3

The Renal Circulations and Glomerular Filtration

Luis Gabriel Navar | David A. Maddox | Karen A. Munger

CHAPTER OUTLINE

KEY POINTS

- Renal blood flow and GFR increase with protein intake in men and women and, at any given level of protein intake, GFR is greater in younger (20–50 years of age) than in older subjects (55–88 years of age) and greater in men than in women, even when factored for body surface area.
- The decline in hydraulic pressure between the renal artery and glomerular capillaries is greatest along the afferent arteriole, whereas 70% of the decline in postglomerular hydraulic pressure occurs at the efferent arteriole. The last 50 to 150 μm of the afferent arteriole and the first early part of the efferent arteriole (first 50–150 μm) provide most of the preglomerular and postglomerular resistances.
- Glomerular capillary blood pressure, P_{GC}, declines only slightly within the capillary network, thus maintaining the transcapillary hydraulic pressure gradient ($\Delta P = P_{GC}$ minus Bowman's space pressure, P_{BS}) at a relatively constant rate. However, glomerular capillary protein oncotic pressure (π_{GC}) rises along the length of the glomerular capillaries as protein-free fluid is filtered into Bowman's space, thus reducing the net filtration pressure, with πg reaching and becoming equal to ΔP in some cases such as hydropenia.
- The barrier to the filtration of fluid and macromolecules includes the glycocalyx lining the endothelial cells, the fenestrations of the endothelial layer of the glomerular capillaries, the layers of the glomerular basement membrane, the filtration slits between the podocytes surrounding the capillaries, and the filtration slit diaphragm extending along the filtration slits to connect adjacent foot processes. Breakdown or injury in any of the restrictive barriers may lead to increased passage of albumin and other proteins.
- The high resistance of the afferent and efferent arterioles leads to a large drop in vascular hydraulic pressure prior to the peritubular capillaries so that peritubular capillary pressure is much lower than glomerular capillary pressure (15–20 mm Hg). The oncotic pressure of fluid entering the peritubular capillaries is elevated because of the filtration of protein-free fluid out of the glomerular capillaries. The net balance between the transcapillary oncotic and hydraulic pressure gradients thus favors entry of the fluid reabsorbed by the tubules into the peritubular capillaries.
- The finding that both renal blood flow and GFR are autoregulated indicates that the principal resistance change due to autoregulatory adjustments is primarily localized to the preglomerular vasculature. The autoregulation mechanisms provide a powerful mechanism to maintain the intrarenal hemodynamic environment in balance with the metabolically determined tubular transport function.

KEY POINTS—cont'd

- The myogenic and tubuloglomerular feedback mechanisms are primarily responsible for the autoregulatory responses. The myogenic mechanism refers to the ability of arterial smooth muscle to contract and relax in response to increases and decreases in vascular wall tension. The tubuloglomerular feedback mechanism provides signals from the macula densa to the afferent arteriole. Paracrine mediators include ATP, adenosine, nitric oxide and arachidonic acid metabolites.
- Overall renal hemodynamic function is regulated by complex interactions between extrinsic and intrinsic mechanisms, including the sympathetic nervous system, circulating vasoactive factors, endothelial nitric oxide, intrarenal angiotensin II, arachidonic acid metabolites, purinergic factors and other paracrine systems, which exert direct effects and modulate the sensitivity of the tubuloglomerular feedback mechanism.

INTRODUCTION

The kidneys are unique in having three distinct microvascular networks identified as the glomerular capillary microcirculation, the cortical peritubular capillary microcirculation, and the unique medullary microcirculation, which nourishes the medullary tissues and maintains the interstitial environment. Each of these circulations has specialized functions that allow filtration of a large volume of fluid at the glomerular capillaries, the consequent reabsorption of most of the filtrate back into the circulation, and the establishment of a medullary environment having a high interstitial osmolality. These renal circulations not only provide the renal cells and tissues with oxygen and nutrients, but also maintain and regulate the hemodynamic environment to achieve their designated functions. Under resting conditions, blood flow to the kidneys represents approximately 20% of cardiac output in humans, even though these organs constitute less than 1% of body mass. This renal blood flow (RBF), approximately 400 mL/100 g of tissue per minute, is significantly greater than that observed in other vascular beds considered to be well perfused, such as the heart, liver, and brain.[1,2] From this enormous blood flow (1.0–1.2 L/min), approximately 20% of the plasma flow is filtered and becomes the glomerular filtrate, but only a small volume of urine, about 1%, is formed from that filtrate. Although the metabolic energy requirements of tubular transport processes are relatively high, the renal arteriovenous O_2 difference reveals that blood flow far exceeds that needed for metabolic demands. In fact, the high blood flow is essential to provide the appropriate hemodynamic environments necessary for the filtration at the glomeruli and the reabsorption into the postglomerular capillaries.[3,4]

In Chapter 2, the gross anatomy of the kidney and arrangement of tubular segments are described in detail. In this chapter, we consider the intrarenal organization of the discrete microcirculatory networks, as shown in Fig. 3.1. We will also consider the differences in regional blood flows and how the structure of the microcirculation contributes to the regulation of the intrarenal hemodynamic environment, thus maintaining appropriate levels of RBF and glomerular filtration rate (GFR).

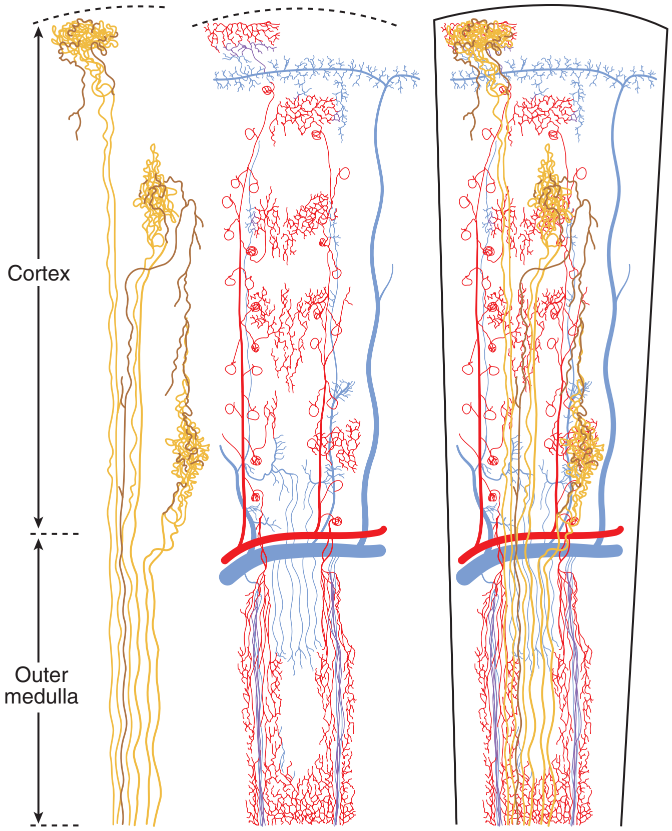

Fig. 3.1 Renal vasculature and tubule organization. *Left,* Three nephrons are shown without accompanying vascular structures. Vascular structures are shown in the *central portion* of the figure. *Right,* Vascular and tubular structures are superimposed. Configurations of tubular segments were generalized from patterns found by silicone rubber injections. For clarity, more distal parts of the nephron are shown in deeper colors. Arterial components of the vascular system are shown in *red*, venous components in *blue*. Only representative venous connections are shown. (From Beeuwkes R III, Bonventre JV. Tubular organization and vascular tubular relations in the dog kidney. *Am J Physiol.* 1975;229:695–713.)

RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

Historically, GFR and renal plasma flow (RPF) have been estimated using the clearance of inulin for the determination of GFR and of *p*-aminohippuric acid (PAH) for the determinations of RPF, from which RBF can be calculated using RPF and hematocrit.[5] Whereas inulin is only filtered across the glomerular capillaries, PAH is both filtered at the glomerulus and actively secreted by the tubules. This results in the renal extraction of 80% to 90% of PAH from the blood. PAH is not completely extracted from the blood because of flow through regions of the kidney, in particular the medulla, that do not perfuse proximal tubule segments, where secretion occurs, and limitations of the secretory process of PAH in cortical regions, along with the presence of a few periglomerular shunts (Fig. 3.2). Thus, PAH clearance is an approximation often termed "effective" or "estimated" renal plasma flow (ERPF) and provides an estimate of RPF without the need for a renal venous blood sample.[5] However, this estimate of RPF is much less accurate in renal disease because extraction is further reduced by damage to proximal tubule segments or rarefaction of the peritubular capillaries involved in PAH secretion.[6] Values taken from a composite of studies in normal human subjects[5] are shown in Table 3.1. As indicated in the data presented, RBF and GFR are lower in women than in men, even when corrected for body surface area. Values for normal subjects vary considerably in different studies, as reflected in Fig. 3.3, which shows more recently obtained data on ERPF and GFR in adult humans from various studies that were not corrected for body surface area. The data also reveal marked differences in GFR and ERPF between obese and lean subjects. These differences are likely the result of increased food intake and, hence, increased protein consumption.[7] Indeed, as shown in Fig. 3.4, GFR increases with protein intake in men and women and, at any given level of protein intake, GFR is higher in young people (20–50 years of age; mean, 31 years) than in older subjects (55–88 years of age; mean, 70 years). Improved methods of RBF

Table 3.1 Renal Blood Flow, Renal Plasma Flow, Glomerular Filtration Rates and Filtration Fractions in Healthy Men and Women[a]

Subject(s)	RBF, mL/min	RPF, mL/min	GFR, mL/min	Filtration Fraction
Men	1166	655	127	0.193
Women	940	600	118	0.197
Combined	1165	634	123	.197

[a]Data for RPF based on clearances of Diodrast as well as *p*-aminohippuric acid (PAH) and for GFR based on clearance of inulin. Clearances are corrected to 1.73 m^2 and suggest lower normalized RBF, RPF, and GFR values in women

GFR, glomerular filtration rate; *RBF,* renal blood flow; *RPF,* renal plasma flow.

Data from Smith HW. *The kidney: Structure and function in health and disease.* New York: Oxford University Press; 1951:544–545.

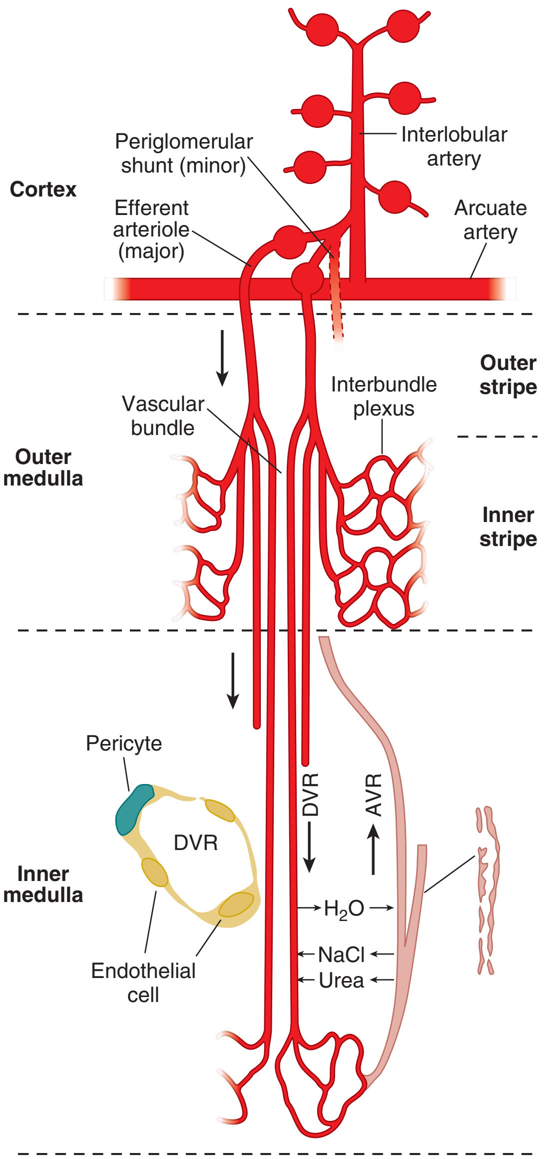

Fig. 3.2 The medullary microcirculation. In the cortex, interlobular arteries arise from the arcuate artery and ascend toward the cortical surface. Cortical and juxtamedullary afferent arterioles leading to glomeruli branch from the interlobular artery. Most of the blood flow reaches the medulla through juxtamedullary efferent arterioles; however, a small fraction may also arise from periglomerular shunt pathways. In the outer medulla, juxtamedullary efferent arterioles in the outer stripe give rise to descending vasa recta *(DVR),* which coalesce to form vascular bundles in the inner stripe. DVR on the periphery of vascular bundles give rise to the interbundle capillary plexus that surrounds nephron segments—thick ascending limb, collecting duct, long looped thin descending limbs (not shown). DVR in the center continue across the inner-outer medullary junction to perfuse the inner medulla. Vascular bundles disappear in the inner medulla, and vasa recta become dispersed with nephron segments. Ascending vasa recta *(AVR)* that arise from the sparse capillary plexus of the inner medulla return to the cortex by passing through outer medullary vascular bundles. *Inset,* DVR have a continuous endothelium. (From Pallone TL, Zhang Z, Rhinehart K. Physiology of the renal medullary microcirculation. *Am J Physiol*. 2003;284:F253–F266, 2003.)

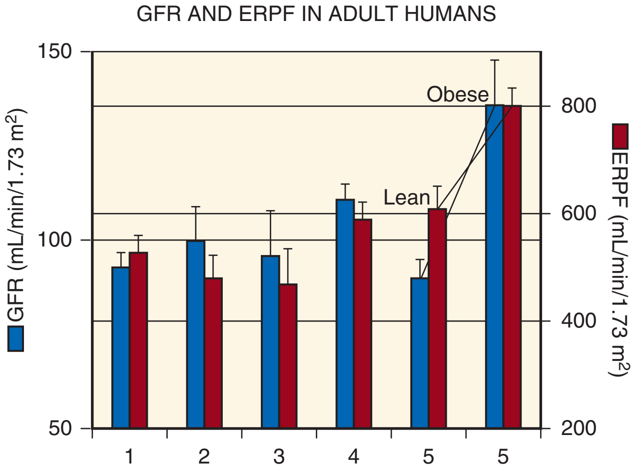

Fig. 3.3 Typical values for glomerular filtration rate *(GFR)* and estimated renal plasma flow *(ERPF)* from five studies in adults. Values from men and women were pooled. Numbers under each set of bars refer to the following studies: 1, Giordano and DeFronzo[517]; 2, Winetz et al[518]; 3, Hostetter[519]; 4, Deen et al[520]; 5, Chagnac et al.[521] For studies 1 through 3 and 5, values were obtained after approximately 12 hours of fasting; subjects in study 4 were allowed food ad lib. For study 5, values from lean subjects (average body mass index [BMI] = 22) were compared with those from obese nondiabetic individuals (BMI >38) after a 10-hour fast but are not corrected for body surface area.

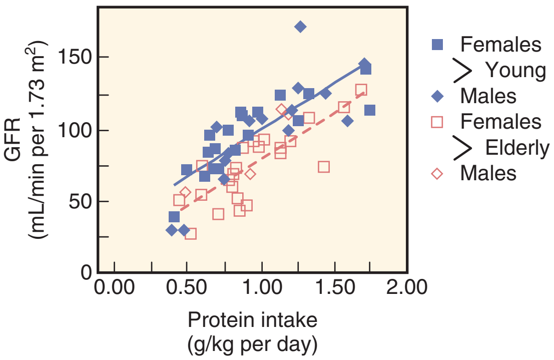

Fig. 3.4 Relationship between protein intake and glomerular filtration rate *(GFR)*. Data from younger (mean age, 31 years; range, 20–50 years) and older (mean age, 70 years; range, 55–88 years) healthy humans. *Closed symbols,* younger subjects; *open symbols,* older subjects; *squares,* women; *diamonds,* men. (From Lew SQ, Bosch P. Effect of diet on creatinine clearance in young and elderly healthy subjects and in patients with renal disease. *J Am Soc Nephrol*. 1991;2:856–865.)

measurement include laser Doppler flowmetry, video microscopy, and imaging techniques such as positron emission tomography (PET), high-speed computed tomography (CT), and magnetic resonance imaging (MRI).[6,8–12] These methods have been especially useful in determining changes in regional blood flow.

MAJOR ARTERIES AND VEINS

Blood supply to each kidney is provided by a renal artery that branches directly from the abdominal aorta. The renal artery typically branches into multiple segmental vessels at a point just before entry into the renal parenchyma and continues to branch in a nonanastomotic manner to supply the glomeruli prior to entering the postglomerular microcirculation (see Fig. 3.1).[3] Therefore, complete obstruction of an arterial segmental vessel results in ischemia and infarction of the tissue in its area of distribution. Ligation of individual segmental arteries has frequently been performed in experimental studies to reduce renal mass and produce the remnant kidney model of chronic renal failure.[13,14] Morphologic studies in this model have revealed the presence of ischemic zones adjacent to the totally infarcted areas. These regions contain viable glomeruli that appear shrunken and crowded together, demonstrating that some portions of the renal cortex may have partial dual perfusion.[15] The anatomic distribution just described is most common; however, other patterns may occur.[16,17] Secondary renal arteries found in 20% to 30% of normal individuals may result from division of the renal artery at the aorta. These vessels, which most often supply the lower pole,[18] may be the sole arterial supply of some part of the kidney.[19]

Within the renal sinus of the human kidney, division of the segmental arteries gives rise to the interlobar arteries. These vessels, in turn, give rise to the arcuate arteries, whose several divisions lie at the border between the cortex and medulla. The interlobular arteries branch from the arcuate arteries more or less sharply, usually as a common trunk that divides two to five times as it extends toward the kidney surface[20,21] (see Fig. 3.1). Afferent arterioles leading to glomeruli arise from the interlobular arteries (see Figs. 3.1 and 3.2). Glomeruli are classified according to their position within the cortex as superficial (i.e., near the kidney surface), midcortical, or juxtamedullary, near the corticomedullary border (see Fig. 3.1). The capillary network of each glomerulus originates from the afferent arteriole as it enters into a manifold-like chamber. The glomerular capillaries coalesce into an efferent chamber leading to an efferent arteriole that delivers blood to the postglomerular capillary circulation, forming both the cortical peritubular capillaries and complex medullary capillaries. The arrangement of the medullary microcirculation plays an important role in the process of concentration of urine.

Venous drainage of the peritubular capillaries from the superficial cortex is via superficial cortical veins.[20,22] In the middle and inner cortices, venous drainage is achieved mainly by the interlobular veins. The dense peritubular capillary network surrounding the interlobular vessels drains directly into the interlobular veins through multiple connections, whereas the less dense, long-meshed network of the medullary rays appears to anastomose with the interlobular network and thus drains laterally (see Fig. 3.1). The medullary circulation also shows two different types of drainage. The outer medullary networks typically extend into the medullary rays before joining interlobular veins, whereas the long vascular bundles of the inner medulla (vasa recta) converge abruptly and join the arcuate veins.

OXYGEN CONSUMPTION

Because of the unique juxtaposition of the arteriolar and venular network, much of the abundant oxygen supply to the kidneys diffuses from the arterioles to the venules.[23] The shunting of oxygen, coupled with the very high rate of oxygen consumption (~4 μmol/min/g), leaves the oxygen tension (pO_2) in the cortex much lower than what would be predicted from the pO_2 in renal venous blood. Tissue pO_2 values in the cortex border on hypoxia, varying from 40 to 45 mm Hg in the outer and mid cortices and even lower (30 mm Hg) in the deep cortex.[13,24,25] As shown in Fig. 3.5, the countercurrent arrangement between the descending and ascending vasa recta permits further shunting of oxygen, leaving pO_2 values in the medullary tissues of 20 mm Hg or lower toward the papillary tip.[23,25–27]

About 75% of the oxygen consumption by the kidneys provides the energy required by the Na^+-K^+-ATPase, which is the major active transport system of the tubules. Changes in oxygen consumption rate vary proportionally with the changes in net Na transport by the tubules. Reduced tissue oxygen levels occur in hypertension, which can compromise renal function.[24,28] The shunting of oxygen from arterioles to venules suggests that diffusible gas molecules that are formed in the kidney, including CO_2, NO, and hydrogen sulfide (H_2S), may also undergo shunting, but from venules to arterioles, thus making it more difficult to wash out unwanted substances such as CO_2 but also accentuating intrarenal retention of protective molecules such as nitric oxide (NO; see Fig. 3.5).[29]

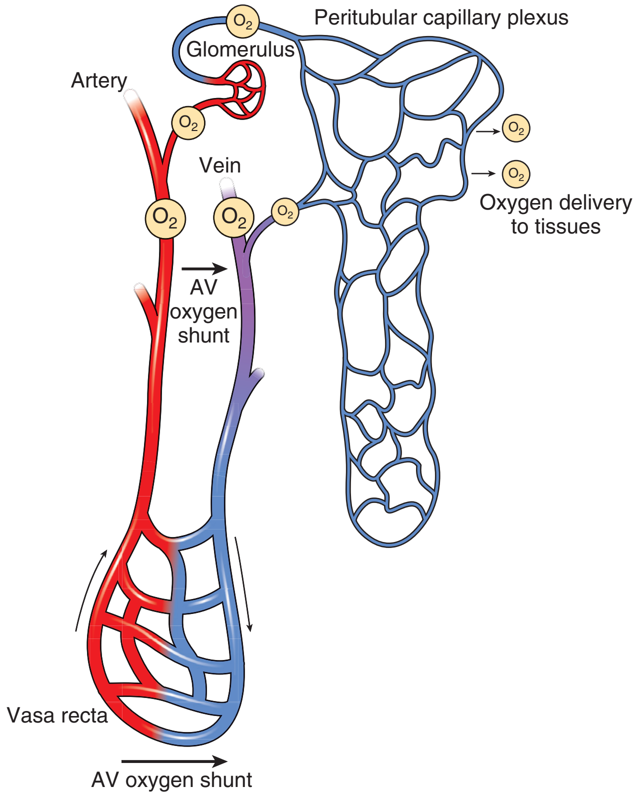

Fig. 3.5 The arterial to venous *(AV)* oxygen shunt. Oxygen is delivered by red blood cells in the artery. Red blood cells release oxygen in the capillaries, which diffuses into the interstitium to reach target cells. Blood with low oxygen tension passes into the vein. The juxtaposition of the arteries and veins present in the kidney facilitates oxygen diffusion from the artery to the vein. Oxygen tensions in the capillaries are relatively low when red blood cells reach the peritubular capillary plexus, which indicate that the kidney is inefficient in extracting oxygen. The relative oxygen tensions are represented by the size of the circles surrounding O_2. (From Mimura I, Nangaku M. The suffocating kidney: tubulointerstitial hypoxia in end-stage renal disease. *Nat Rev Nephrol*. 2010;6:667–678.)

HYDRAULIC PRESSURE PROFILE AND VASCULAR RESISTANCES

The decline in hydraulic pressure between the systemic vasculature and the end of the interlobular artery in both the superficial and juxtamedullary microvasculatures can be as much as 25 mm Hg at normal perfusion pressures, with most of that pressure drop occurring along the interlobular arteries (Fig. 3.6). Based on studies of the vasculature of juxtamedullary nephrons, most of the preglomerular pressure drop between the arcuate artery and the glomerulus occurs along the afferent arteriole.[30,31] Approximately 70% of the postglomerular hydraulic pressure drop occurs along the efferent arterioles. The very late portion of the afferent arteriole (last 50–150 μm) and the early portion of the efferent arteriole (first 50–150 μm) provide the major fraction of the total preglomerular and postglomerular resistance (see Fig. 3.6).[30,31] Multiphoton imaging studies have indicated the presence of an intraglomerular precapillary sphincter at the terminal end of afferent arterioles (Fig. 3.7).[32,33] Collectively, the total resistance (R_T) consists of two major sites, the afferent (Ra) and efferent (Re) arterioles and a minor contribution

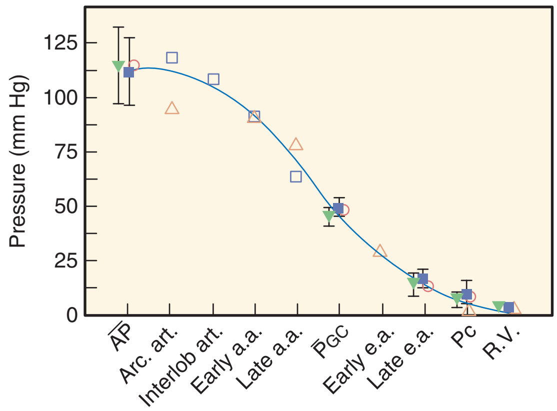

Fig. 3.6 Hydraulic pressure profile in the kidney. Filled squares and triangles denote values (mean ± 2 standard deviation) obtained from euvolemic and hydropenic Munich-Wistar rats. Values from studies in the squirrel monkey[78] are shown as *open diamonds*. *Open inverted triangles* and *open squares* are from studies in the Sprague Dawley rat in juxtamedullary nephrons perfused with whole blood. These nephrons are located inside the cortical surface opposed to the pelvic lining and arcuate veins, in which entire pressure profiles can be obtained from the interlobular artery *(Interlob Art)*, the proximal *(Early a.a.)* and distal *(Late a.a.)* portions of the afferent arteriole, the glomerular capillaries *(P_{GC})*, the proximal *(Early e.a.)* and late *(Late e.a.)* segments of the efferent arteriole, the peritubular capillaries *(P_c)*, and the renal vein *(R.V.)*. *AP,* Arterial pressure; *Arc. art.,* arcuate artery; *P_{GC},* glomerular capillary pressure. (Modified from Maddox DA, Brenner BM. Glomerular ultrafiltration. In *Brenner and Rector's The Kidney*, 7th Edition. Philadelphia: W.B. Saunders Company; 2004, pp 353–412; Casellas D, Navar LG. In vitro perfusion of juxtamedullary nephrons in rats. *Am J Physiol.* 1984;246:F349–F358; Imig JD, Roman RJ. Nitric oxide modulates vascular tone in preglomerular arterioles. *Hypertension.* 1992;19(6 Pt 2):770–774.)

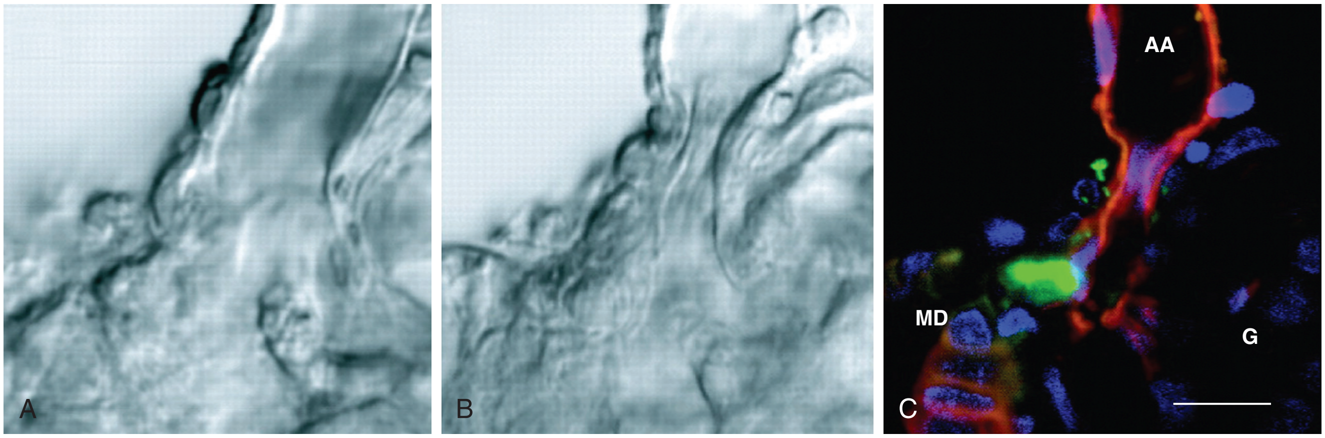

Fig. 3.7 Constriction of the terminal afferent arteriole *(AA)* via an intraglomerular precapillary sphincter in response to elevations in distal tubular NaCl content. (A, B) Transmitted light differential interference contrast (DIC) images. (A) Control, with NaCl concentration at the macula densa at 10 mM. (B) NaCl concentration is increased to 60 mM, resulting in an almost complete closure of the AA. (C) Fluorescence image of the same preparation as shown in B. Vascular endothelium and tubular epithelium are labeled with R18 *(red),* renin granules with quinacrine *(green),* and cell nuclei with Hoechst 33342 *(blue).* Note that renin-positive granular cells constitute the sphincter, demonstrating contractile responses in glomerular cells. *G;*Glomerulus; *MD,* macula densa. Scale bar = 10 µm. (From Peti-Peterdi J. Multiphoton imaging of renal tissues in vitro. *Am J Physiol Renal Physiol.* 2005;288:F1079–F1083.)

from the outflowing venules and veins (Rv). Accordingly, the following relationships describe the intrarenal cortical vascular resistances:

$$R_T = Ra + Re + Rv \quad (1)$$

$$Ra = (AP - P_{GC})/RBF \quad (2)$$

$$Re = (P_{GC} - P_C)/EABF \quad (3)$$

where EABF = RBF – GFR

$$Rv = (P_C - P_V)/RBF \quad (4)$$

where AP = arterial pressure; P_{GC} = glomerular capillary pressure; P_C = peritubular capillary pressure; P_V = renal vein pressure; and EABF = efferent arteriolar blood flow.

INTRARENAL BLOOD FLOW DISTRIBUTION

The cortex accounts for filtration at the glomeruli and most of the reabsorption from proximal and distal nephron cortical tubules, whereas the medulla reabsorbs less than 20% of the total reabsorbate, in keeping with its primary function to maintain a hypertonic interstitial gradient when needed to excrete a concentrated urine. Blood flow to these regions is differentially regulated in response to the differing functions and demands of these two kidney regions.[34] Approximately 80% of the RBF perfuses the cortex and is under the control of numerous intrinsic paracrine vasoactive factors, as well as extrinsic humoral and neural influences. Vasoconstrictors, including angiotensin II (Ang II), endothelin, purinergic agents, and norepinephrine, as well as vasodilators, including bradykinin and nitric oxide, interact to regulate cortical blood flow and medullary blood flow.[34,35] There can be extensive redistribution of blood flow in the kidney under various conditions that may be important in physiologic and pathophysiologic conditions.[36]

Structural differences in vascular components of the cortex and the medulla may account for differences in RBF—namely, the organization of the afferent and efferent arterioles of the cortical and juxtamedullary glomeruli. The cortical afferent arterioles have larger internal diameters than the efferent arterioles, whereas juxtamedullary afferent and efferent arterioles are larger than the outer cortical arterioles, and efferent arterioles of juxtamedullary nephrons are more muscular compared with the cortical arterioles.[34,37] In addition, the cortical peritubular capillaries, derived from efferent arterioles of cortical glomeruli, are about half the diameter of the medullary vasa recta derived from efferent arterioles of the juxtamedullary glomeruli (Fig. 3.8).[34] These features may partially explain the differential control of medullary and cortical blood flows.

VASCULAR-TUBULE RELATIONS

Cortical vascular-tubule relations have been described extensively (see Figs. 3.1 and 3.2).[20,38,39] The efferent peritubular capillary network and the tubules arising from each glomerulus in the outermost region of the cortex are tightly associated, but this relationship becomes dissociated in deeper regions of the cortex. This close association does not mean that each vessel adjacent to a given tubule necessarily arises from the same glomerulus. Although superficial nephron segments and peritubular capillaries arising from the same glomerulus are closely associated, each postglomerular efferent arteriole may serve segments of more than one nephron.[40] However, the loops of Henle of all nephrons, as they descend into the medullary ray, are supplied by postglomerular blood vessels emerging from midcortical and deep nephrons, with some of the branches from deep nephrons descending into the medullary ray, termed the *vasa recta* (see Figs. 3.1 and 3.2). The dissociation between individual tubules and the corresponding postglomerular capillary network is most apparent in the inner cortex. The convoluted tubule segments of these nephrons lie above the glomeruli surrounded by the dense network close to the interlobular vessels or by capillary networks arising from other inner cortical glomeruli. With regard to efferent vessel patterns and vascular-tubule relationships,[39,41] there is a close association between the initial portions of peritubular capillaries and early and late proximal tubule segments of the same glomerulus.[42–44]

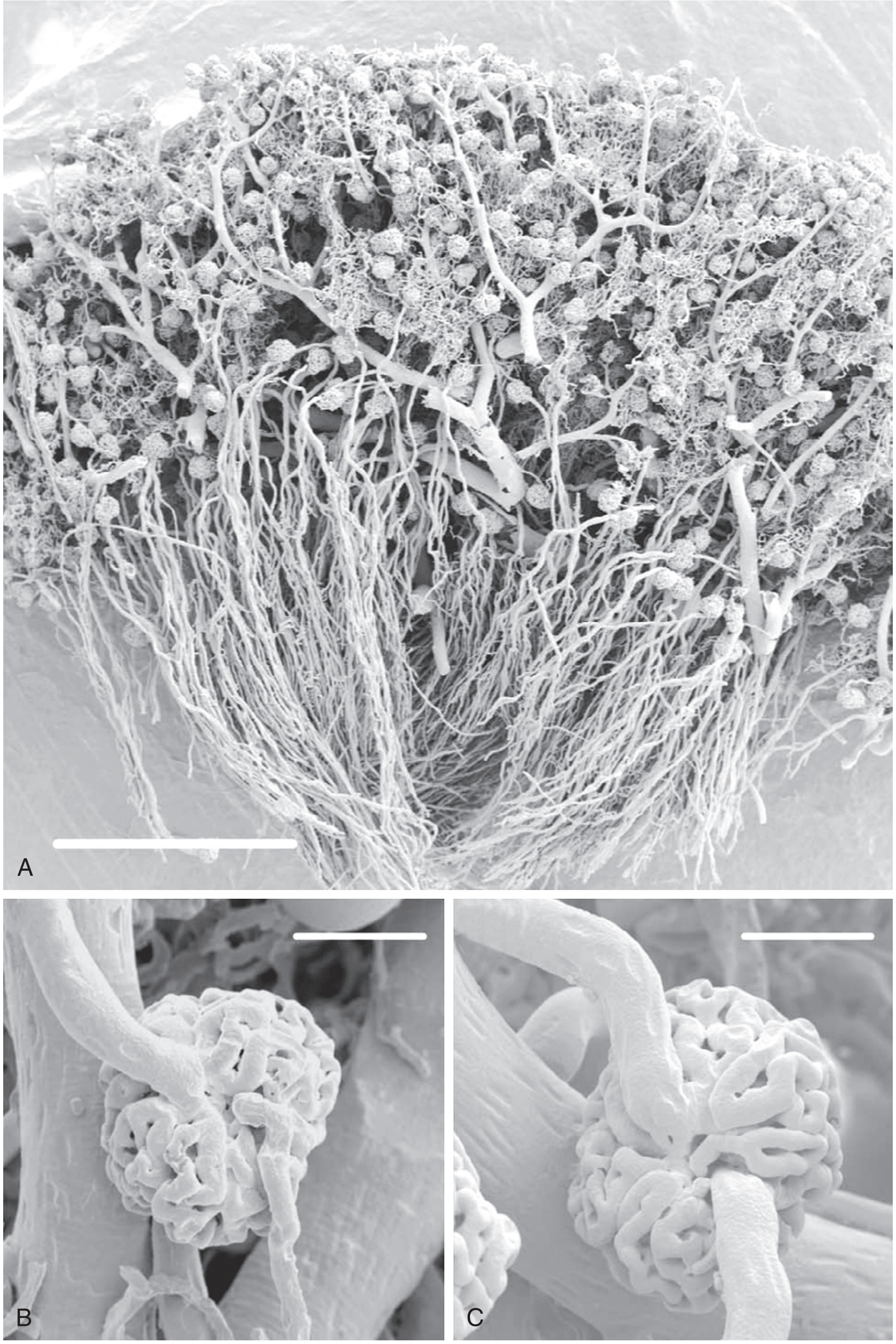

Fig. 3.8 A, Resin casts of renal glomeruli of a rabbit, depicting both cortical and medullary glomeruli. For Panel A, the scale bar represents 1 mm. B, Cortical glomerulus showing afferent (upper vessel) and efferent arterioles and the capillary tuft (scale bar = 60 µm). C, A juxtamedullary glomerulus showing afferent (upper vessel) and efferent arterioles and the capillary tuft (scale bar 60 = µm). Note that the juxtamedullary arterioles are larger in diameter than the cortical glomerular arterioles, particularly the efferent arterioles. (From Evans RG, Eppel GA, Anderson WP, Denton KM: Mechanisms underlying the differential control of blood flow in the renal medulla and cortex. *J Hypertens.* 2004;22:1439–1451.)

STRUCTURAL AND FUNCTIONAL ASPECTS OF THE GLOMERULAR MICROCIRCULATION

Some of the structural relationships of the glomerular microcirculation are seen in Fig. 3.8, which shows a scanning electron micrograph of a resin-filled cast of a kidney. The afferent arteriole branching from the interlobular artery, the many loops of the glomerular capillaries, and the efferent arteriole as it emerges from the glomerular tuft can be seen in B and C. An ultrastructural analysis of the vascular pole of the renal glomerulus has revealed differences in the structure and branching patterns of the afferent and efferent arterioles as these vessels enter and exit the tuft.[45] Afferent arterioles lose their internal elastic layer and smooth muscle cell layer prior to entering the glomerular tuft. Smooth muscle cells are replaced by renin-positive, myosin-negative granular cells that are in close contact with the extraglomerular mesangium (Fig. 3.9).[32,46] On entering Bowman's space, afferent arterioles undergo a transition into a vascular chamber that has a manifold-like structure distributing primary branches along the surface of the glomerular tuft, which branch further into the individual glomerular capillaries.[45,47]

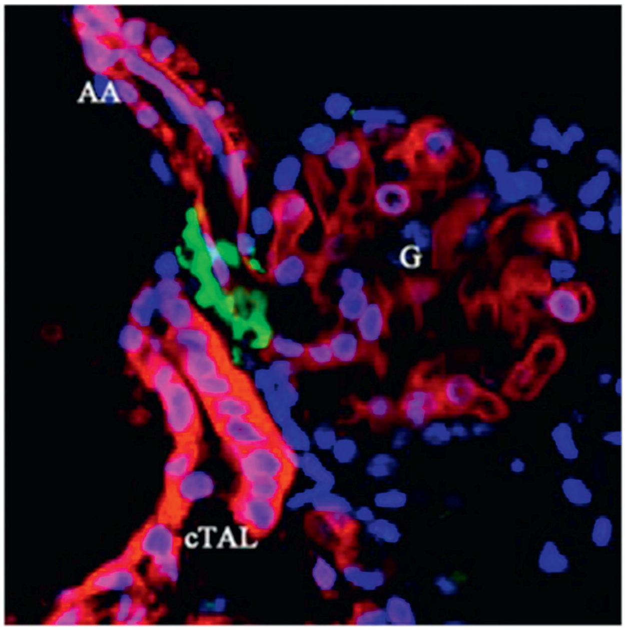

Fig. 3.9 Multicolor labeling of the in vitro microperfused juxtaglomerular apparatus with attached glomerulus. Cell membranes of tubular epithelium (cortical thick ascending limb, *[cTAL]* containing the macula densa), vascular endothelium of the afferent arteriole *(AA),* and glomerulus *(G)* are labeled with R18 *(red),* renin granules with quinacrine *(green),* and cell nuclei with Hoechst 33342 *(blue).* (From Peti-Peterdi J. Multiphoton imaging of renal tissues in vitro. *Am J Physiol Renal Physiol.* 2005;288:F1079–F1083.)

The primary branches have wide lumens and immediately acquire features of glomerular capillaries, including a fenestrated endothelium, characteristic glomerular basement membrane, and epithelial foot processes. In human glomeruli, however, these primary branches serve as conduit vessels that branch into the filtering capillaries (Fig. 3.10).[47] In contrast, the efferent arteriole originates deep within the tuft, from the convergence of capillaries into multiple lobules that exit into an efferent vascular chamber, which narrows into an efferent arteriole. Additional tributaries join the efferent arteriole as it travels toward the vascular pole. The structure of the capillary wall begins to change, even before the vessels coalesce to form the efferent arteriole, losing fenestrae progressively until a smooth epithelial lining is formed. At the arteriole's terminal portion within the tuft, endothelial cells may bulge into the lumen, reducing its internal diameter.[45] Efferent arterioles acquire a smooth muscle cell layer, which is observed distal to the entry point of the final glomerular capillary. The efferent arteriole is also in close contact with the glomerular mesangium as it forms inside the tuft and with the extraglomerular mesangium as it exits the tuft. This precise and close anatomic relationship between the afferent and efferent arterioles and mesangium with the macula densa cells of the ascending loop of Henle provides the structural basis for the presence of an intraglomerular signaling system, known as the "tubuloglomerular feedback mechanism," which participates in the regulation of blood flow and GFR.[32,45,48]

The appearance of the vascular pathways within the glomerulus may change under different physiologic conditions. The glomerular mesangium (Fig. 3.11) has been shown to contain contractile elements[45,49] and to exhibit contractile activity when exposed to Ang II.[50] Mesangial cells, which possess AT1 receptors for Ang II, undergo contraction when exposed to this peptide in vitro.[51] Three-dimensional

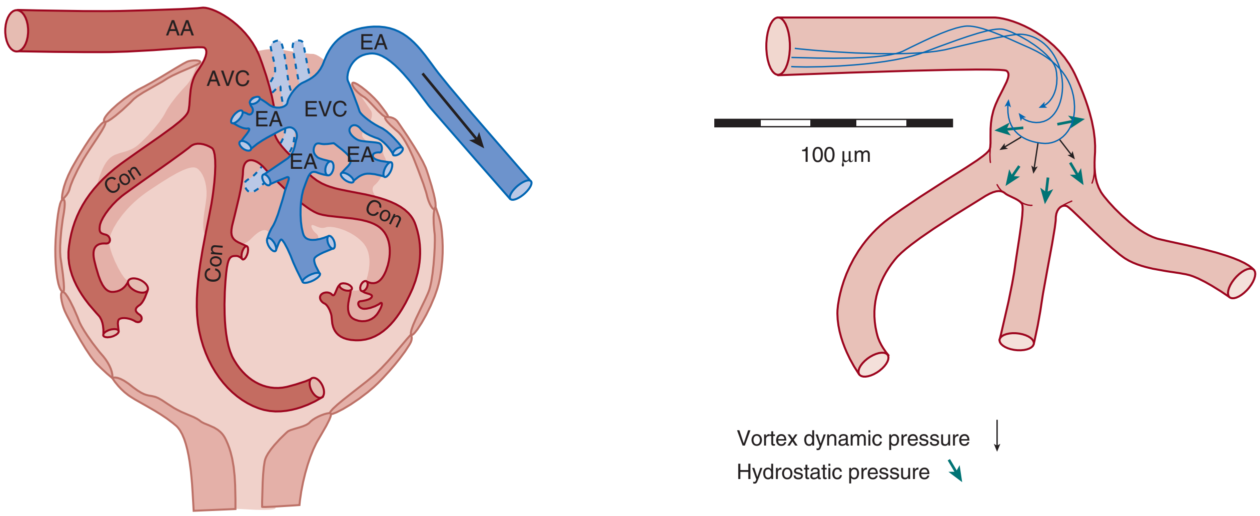

Fig. 3.10 Structure of the human glomerulus. Note the sharp turn made by the afferent arteriole *(AA)* as it enters the afferent vascular chamber *(AVC),* which flows into the conduit afferent capillaries *(Con)*. Efferent first-order vessels (E1) and the efferent vascular chamber *(EVC)* transition into the efferent arteriole *(EA)*. *Right,* Distribution of hydraulic forces in the AVC. (From Neal CR, Arkill K, Bell JS, et al. Novel hemodynamic structures in the human glomerulus. *Am J Physiol Renal Physiol.* 2018;315(5):F1370–F1384; color figures courtesy Dr. Christopher Neal.)

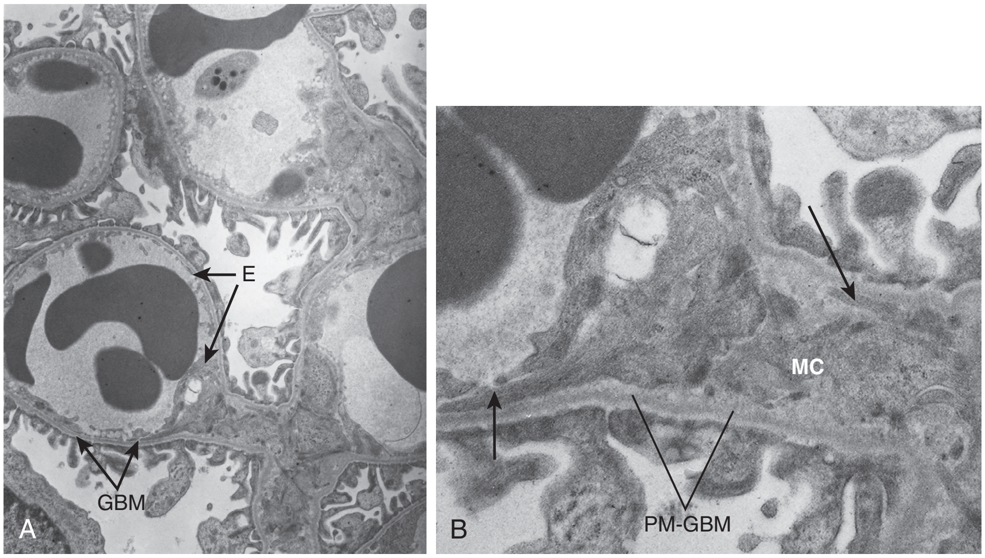

Fig. 3.11 Electron micrographs of glomerular capillaries of a Munich-Wistar rat. (A) Overview of several capillaries (≈×14,500). Most of the glomerular capillary endothelium *(E)* is in contact with the glomerular basement membrane *(GBM)*, with only a small portion in contact with the mesangium (M). At its outer aspect, the GBM is covered by podocyte foot processes. There is no basement membrane separating the endothelium from the mesangium at their interface. (B) Mesangial cell *(MC)* extends outward to meet the glomerular capillary (≈×42,000). Kriz and co-workers have suggested that such cylinder-like stalks appear to be contractile filament bundles (*short arrow*) that attach to the perimesangial glomerular basement membranes *(PM-GBM)* and extend to the GBM at the mesangial angles (*long arrow*). For this preparation the nephron was perfusion-fixed by micropuncture with 1.25% glutaraldehyde through Bowman's space, thereby yielding the fixation of glomerular structures, as well as the red cells in the capillaries. (Modified from Kriz W, Elger M, Mundel P, Lemley KV. Structure-stabilizing forces in the glomerular tuft. *J Am Soc Nephrol.* 1995;5(10):1731–1739; Kriz W, Kaissling B. *Structural organization of the mammalian kidney.* Third Edition ed: Lippincott, Williams & Wilkins; 2000; Drenckhahn D, Schnittler H, Nobiling R, Kriz W. Ultrastructural organization of contractile proteins in rat glomerular mesangial cells. *Am J Pathol.* 1990;137(6):1343–1351.)

reconstruction of the entire mesangium in the rat has suggested that approximately 15% of capillary loops may be entirely enclosed within armlike extensions of mesangial cells that are anchored to the extracellular matrix.[52] Contraction of these cells might alter local blood flow and filtration rate, as well as the intraglomerular distribution of blood flow and total filtration surface area. Many hormones and other vasoactive substances capable of altering glomerular filtration may bring about this adjustment, in part by altering the state of contraction of mesangial cells.

DETERMINANTS OF GLOMERULAR FILTRATION

A critical function of the mechanisms regulating renal hemodynamics is to maintain the blood flow and pressure profile within the glomerular tufts at levels such that the filtration rate allows optimum function of reabsorptive and secretory processes by the tubular network. The exquisite differential regulation of the preglomerular (afferent) and postglomerular (efferent) resistances exert fine control of the intraglomerular hemodynamic environment and thus the GFR. This nesting of the glomerular capillary system between the afferent and efferent arterioles allows for precise regulation of the intraglomerular forces governing filtration. These forces, coupled with the unique restrictive molecular permeability of the glomerular capillary structure, lead to the formation of a nearly protein-free filtrate, from the glomerular capillaries into Bowman's space, as the first step in the process of urine formation.

PERMEABILITY OF THE GLOMERULAR FILTRATION BARRIER

In addition to the unique role of the glomerular capillary wall and surrounding glomerular epithelial cells in regulating fluid movement from the glomerular capillary into Bowman's space, these structures also regulate the movement of macromolecules. Molecules the size of inulin or smaller are normally able to cross the glomerular capillary wall without restriction. However, the transglomerular permeation of molecules of increasing size becomes limited, so that molecules the size of albumin or larger are almost completely prevented from crossing into Bowman's space.

As shown in Fig. 3.12, the composite filtration barrier includes the glycocalyx lining the endothelial cells, the fenestrations of the endothelial layer of the glomerular capillaries, the three layers of the glomerular basement membrane, the filtration slits between adjacent foot processes of the visceral epithelial cells (podocytes) that surround the capillaries, and the filtration slit diaphragm that extends along the filtration slits and connects adjacent foot processes to form the ultimate barrier to filtration[53] (see Fig. 3.12). This complex barrier has a high permeability to small molecules such as water, electrolytes, amino acids, glucose, and other endogenous or exogenous compounds with molecular radii smaller than 20 Å. This allows these compounds to be freely filtered from the blood into Bowman's space while virtually excluding molecules larger than around 50 Å.[54–61] In studies using fractional clearances of neutral

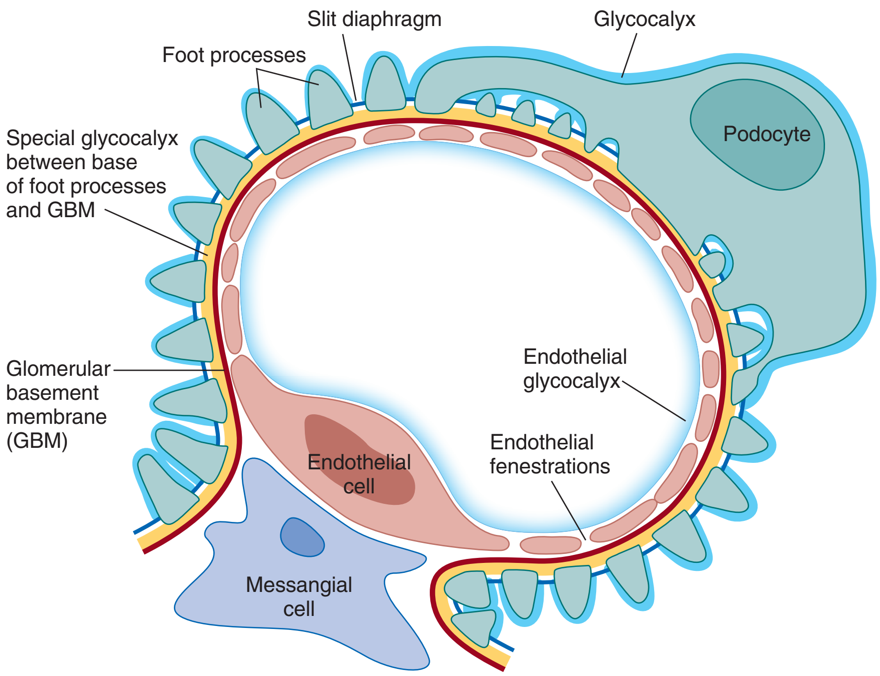

Fig. 3.12 Schematic drawing of a glomerular capillary. The elements considered as part of the glomerular filtration barrier include the following: endothelial glycocalyx, fenestrated endothelium, basement membrane, specialized glycocalyx between the foot processes and basement membrane, and podocyte and slit diaphragm. (From Schlöndorff D, Wyatt CM, Campbell KN. Revisiting the determinants of glomerular filtration barrier: what goes round must come round, *Kidney Int.* 2017;92:533–536.)

dextran, comparable size selectivity[62] was observed in rats,[63] dogs,[64] and humans.[65] As recently emphasized,[53] there has been considerable controversy regarding the component primarily responsible for the exclusion of macromolecules from the filtrate. The size selectivity of the glomerular filtration barrier is determined largely by a combination of the slit diaphragms between podocyte foot processes and the glomerular basement membrane (GBM). Although there is some controversy regarding glomerular charge selectivity, there is good evidence for a role of charge in restricting the transmural movement of negatively charged macromolecules such as albumin at the level of the GBM, which contains negatively charged heparan sulfate proteoglycans, as well as laminin, type IV collagen, and nidogen.[66,67]

Other studies have now demonstrated the presence of a fine meshwork of glycosaminoglycans covering the luminal endothelial layer and bridging the endothelial fenestrations so that the endothelial layer is now considered the initial coarse barrier for macromolecular exclusion.[68] Further studies using negatively charged gold nanoparticles have confirmed that the lamina densa of the basement membrane serves as an exclusion barrier for molecules the size of immunoglobulin G (IgG) and albumin. Small particles permeated into the lamina densa and accumulated upstream, covering the base of the foot processes.[60]

HYDRAULIC AND ONCOTIC FORCES IN THE GLOMERULAR CAPILLARIES AND BOWMAN'S SPACE

The process of filtration of fluid at any given point of the glomerular capillary is governed by the net balance among the transcapillary hydraulic pressure gradient (ΔP), the transcapillary colloid osmotic pressure gradient ($\Delta\pi$), and the hydraulic conductivity of the filtration barrier per surface area unit (Lp), coupled with the surface area (Sf). The product of Lp and Sf is called the "filtration coefficient" (K_f). Fluid flow (J_v) at any given point in the capillary is determined by the Starling equation:

$$J_v = Lp\,[(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})] \quad (5)$$

where P_{GC} and P_{BS} are the hydraulic pressures in the glomerular capillaries and Bowman's space, respectively, and π_{GC} and π_{BS} are the corresponding colloid osmotic pressures at any given point. Because the protein concentration of the fluid in Bowman's space is very low, π_{BS} approaches zero and can be disregarded. The total GFR of fluid for a single nephron (SNGFR) is equal to the product of the surface area for filtration (S_f) and the hydraulic conductivity (Lp), defined as the filtration coefficient (K_f) and the values of the right-hand terms in Eq.5 averaged along the length of the glomerular capillaries yielding the expression:

$$SNGFR = LpS_f\,[(P_{GC} - P_{BS}) - (\pi_{GC})] \; or \; SNGFR = K_f\,(\Delta P - \pi_{GC}) \quad (6)$$

Thus,

$$SNGFR = K_f \times P_{UF} \quad (7)$$

where P_{UF} = the mean filtration pressure, the difference between the mean transcapillary hydraulic and colloid osmotic pressure gradients, ΔP and π_{GC}, respectively.

It is important to emphasize the difference between Lp, the hydraulic conductivity, and the macromolecular permeability. These are regulated differently and are not closely coupled. Based on known ultrastructural detail and the hydrodynamic properties of the individual components of the filtration barrier, mathematical modeling suggests that only around 2% of the total hydraulic resistance is accounted for by the fenestrated capillary endothelium, whereas the basement membrane accounts for nearly 50%.[66,69,70] The remaining hydraulic resistance resides in the diaphragm of the filtration slits, which are complex structures containing numerous proteins, including nephrin and podocin.[66,70–72] Disruption of these slit diaphragm proteins leads to substantial proteinuria.[72] A reduction in the frequency of intact filtration slits is an important factor in the deterioration of filtration in some disease states.[66,73] In the case of macromolecule permeability, mathematical modeling efforts contend that the sieving efficiency of the layers of the glomerular filtration barrier are interdependent. Moreover, whereas hydraulic resistances are additive, macromolecule sieving coefficients are multiplicative; thus, a small change in the macromolecule permeability of one layer can significantly change the overall permeability of the filtration barrier.[62,66]

Because surface glomeruli are not present in most experimental animals, indirect approaches to measure glomerular pressure have been used in many experimental studies to evaluate the responses of glomerular pressure. The stop-flow technique has been used by many investigators to estimate P_{GC}.[2,35,74,75] When fluid movement in the early proximal tubule is blocked, intratubular pressure upstream from the block increases until net filtration at the glomerulus ceases.[76] At that point, the sum of this hydrostatic pressure in the early proximal tubule plus the systemic colloid oncotic pressure is equal to the pressure in the glomerular capillaries (P_{GCSF}). The stop-flow technique has been used in different strains of rats, as well as in dogs and mice, with the P_{GCSF} averaging 55 to 60 mm Hg in the dog and about 50 mm Hg in the rat.[4,55] Glomerular capillary pressures calculated using this stop-flow have been compared with values obtained by direct micropuncture of glomerular capillaries in a number of studies; these indicate that P_{GCSF}, measured at normal APs, provides a close estimate of P_{GC} measured directly.[35,55,74,75]

Direct measurements of glomerular capillary hydraulic pressure in vertebrates (P_{GC}) became possible when it was discovered that a mutant strain of the Munich-Wistar rat had surface glomeruli that allowed direct puncture of glomerular capillaries.[77] Subsequent studies have confirmed the original observations, demonstrating that values for P_{GC} in surface glomeruli average 43 to 49 mm Hg in this strain of rats (Fig. 3.13), and similar values were found in the squirrel monkey, which also has some superficial glomeruli.[78] Because the glomerular capillaries are nested between the afferent and efferent arterioles, P_{GC} is nearly constant along the length

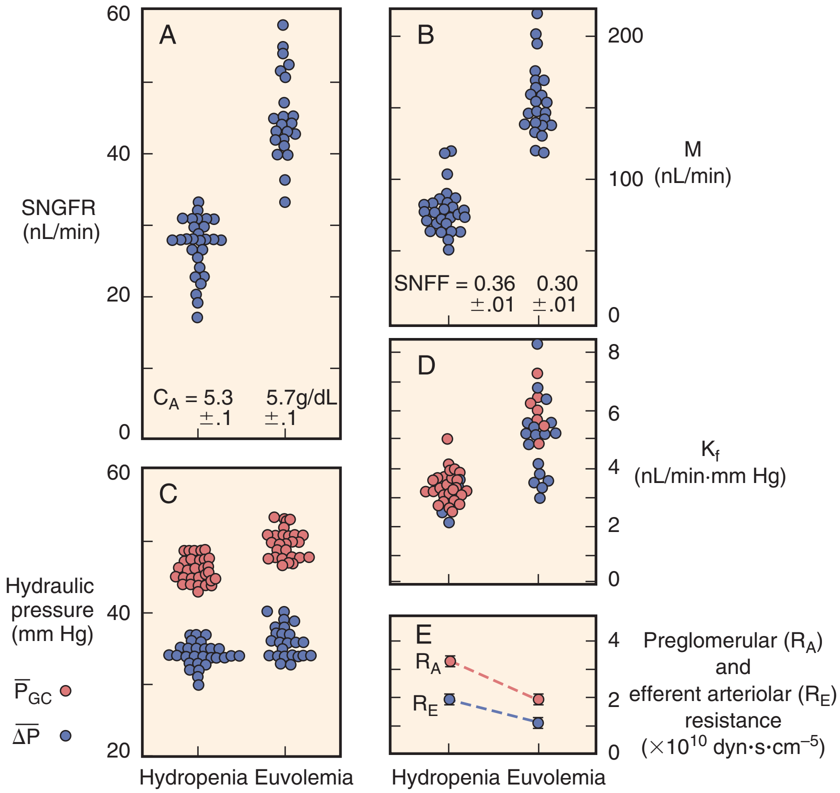

Fig. 3.13 Glomerular filtration in the Munich-Wistar rat. (A–E) Each point represents the mean value reported for studies in hydropenic and euvolemic rats provided food and water ad lib until the time of study. Data from euvolemic rats are thought to be representative of nonanesthesia conditions. Only data from studies using male or a mix of male and female rats are shown. Values of the ultrafiltration coefficient, K_f (*red circles* in D), denote minimum values because the animals were in filtration pressure equilibrium. *Blue circles* represent unique values of K_f calculated under conditions of filtration pressure disequilibrium ($\pi_E/\Delta P \leq 0.95$). *C_A*, Concentration in the afferent arteriole; *ΔP*, pressure gradient; *P_{GC}*, pressure in the glomerular capillaries; *Q_A*, glomerular plasma flow rate; *SNFF,* single-nephron filtration fraction; *SNGFR*, single-nephron glomerular filtration. (From Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. *Handbook of physiology: Section 8; Renal physiology* Vol 1. New York: Oxford University Press; 1992, pp. 545–638.)

GLOMERULAR PRESSURES IN THE MUNICH-WISTAR RAT

	Hydropenia		Euvolemia	
	Afferent end	Efferent end	Afferent end	Efferent end
P_{GC}	46	46	50	50
P_{BS}	12	12	14	14
π_{GC}	17	34	19	33
P_{UF}	17 mm Hg	0 mm Hg	17 mm Hg	3 mm Hg

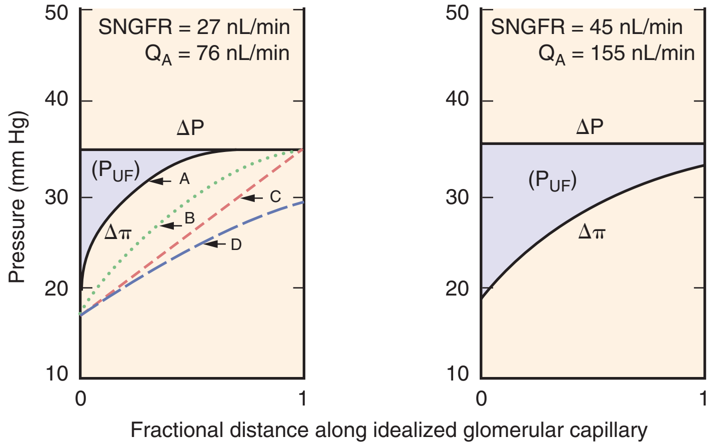

Fig. 3.14 Hydraulic and colloid osmotic pressure profiles along idealized glomerular capillaries in hydropenic and euvolemic rats. Values shown are mean values derived from the studies shown in Fig. 3.13. The transcapillary hydraulic pressure gradient, ΔP, is equal to P_{GC} – PT, and the transcapillary colloid osmotic pressure gradient, Δπ, is equal to $\pi_{GC} - \pi_{BS}$, where P_{GC} and P_{BS} are the hydraulic pressures in the glomerular capillary and Bowman's space, respectively, and π_{GC} and π_{BS} are the corresponding colloid osmotic pressures. Because the value of π_{BS} is negligible, Δπ essentially equals π_{GC}. P_{UF} is the ultrafiltration pressure at any point. The area between the ΔP and Δπ curves represents the net ultrafiltration pressure, P_{UF}. *Left,* Lines A and B represent two of the many possible profiles under conditions of filtration pressure equilibrium; line D represents disequilibrium and line C represents the hypothetic linear Δπ profile. *Right,* Pressure profile after correction for surgical induced loss of plasma volume depicts a small positive ΔP at the efferent level indicative of disequilibrium conditions. Q_A, Glomerular plasma flow; *SNGFR,* single-nephron glomerular filtration rate.

of the capillary bed, resulting in a transcapillary hydraulic pressure gradient that averages 34 mm Hg in hydropenic Munich-Wistar rats (see Fig. 3.13). Coupling these hydraulic pressure measurements with determinations of systemic plasma protein concentration and efferent arteriolar protein concentrations of superficial nephrons has permitted an opportunity to determine the hydraulic and oncotic pressures that govern glomerular filtration at the beginning and end of the capillary network.

The early direct measurements of P_{GC} made in hydropenic rats were under conditions of surgically induced reductions in plasma volume and GFR. Subsequent studies, in which plasma volume was restored to a euvolemic state, equal to that of the awake animal,[79] by infusion of iso-oncotic plasma, yielded SNGFR values substantially higher than in hydropenic rats. This is primarily as a consequence of increases in glomerular plasma flow (Q_A) associated with a fall in preglomerular (R_A) and efferent arteriolar (R_E) resistance values (see Fig. 3.13), indicating that the early studies in hydropenic rats were performed under conditions of elevated activity of the sympathetic nervous system. Collectively, the experimental studies in Munich-Wistar rats (see Fig. 3.13) and several other strains of rats studied under euvolemic conditions have indicated that glomerular pressure in rats is in the range of 50 mm Hg, leading to higher transglomerular capillary pressures.

Glomerular capillary hydraulic and oncotic pressure profiles for rats under hydropenic and euvolemic conditions are shown in Fig. 3.14, using the mean values determined from the studies shown in Fig. 3.13. In hydropenic animals, by the time the blood reaches the efferent end of the glomerular capillaries, plasma oncotic pressure (π_E) rises to a value that, on average, equals ΔP. As a consequence, the net local filtration pressure, P_{UF} ($P_{GC} - [P_T + \pi_{GC}]$), is reduced from approximately 17 mm Hg at the afferent end of the glomerular capillary network to essentially zero by the efferent end (referred to as "filtration pressure equilibrium"). The value of ΔP is nearly constant along the glomerular capillaries, and the decline in P_{UF} along the capillary network is due to a rise in π_{GC} (see Fig. 3.14). An exact profile of the Δπ curve cannot be ascertained under conditions of filtration pressure equilibrium, and hence only maximum estimates of P_{UF} and minimum estimates of K_f can be obtained by assuming a linear rise in the Δπ curve. To obtain accurate measurements of the Δπ curve, Deen and colleagues plasma-expanded rats to increase plasma flow to obtain filtration pressure disequilibrium (Fig. 3.14, curve D), permitting an exact determination of P_{UF} and hence K_f.[80] Filtration pressure disequilibrium is present in about 60% of the studies in the euvolemic Munich-Wistar rat (see Fig. 3.13) and in most of the studies in Sprague-Dawley rats and in dogs,[4,74,75,81] permitting exact determinations of P_{UF} along the entire length of the glomerular capillaries and hence an accurate K_f reflecting the total surface area.

DETERMINATION OF THE FILTRATION COEFFICIENT

As shown in Eq. 7, SNGFR equals the filtration coefficient (K_f) times the net driving force for filtration averaged over the length of the glomerular capillaries (P_{UF}). The values of K_f from many studies in euvolemic Munich-Wistar rats studied under conditions of filtration pressure disequilibrium (see Fig. 3.13) averaged 5.0 ± 0.3 nL/(min·mm Hg). These are values similar to those found in other rat strains and in dogs (3–5 nL/[min·mm Hg]).[4,55,74,75] Because this value remains essentially unchanged over a twofold range of changes in Q_A, the data suggest that changes in Q_A per se do not affect K_f.[80]

In the rat, total capillary basement membrane surface area per glomerulus (A_s) has been determined to be around 0.003 cm^2 in superficial nephrons and 0.004 cm^2 in deep nephrons.[82] A large portion of the capillary surface area faces the mesangium and, as a consequence, only the peripheral area of the capillaries surrounded by podocytes participates in filtration. This peripheral area available for filtration (A_p) is only about half that of A_s (~0.0016–0.0018; 0.0019–0.0022 cm^2 in the superficial and deep glomeruli, respectively).[82] Using a value of K_f of around 5 nL/min mm Hg, as determined by micropuncture techniques, with these estimates of A_p, yields a hydraulic conductivity (Lp) of 45 to 48 nL/(sec·mm Hg·cm^2). These estimates of k for the rat glomerulus are all one or two orders in magnitude higher than those reported for capillary networks in mesentery, skeletal muscle, omentum or peritubular capillaries of the kidney,[55,83] thus supporting the premise of very high glomerular hydraulic permeability of the glomerular capillaries.

DETERMINANTS OF GLOMERULAR FILTRATION COEFFICIENT IN HUMAN SUBJECTS

Hydraulic pressure in the glomerular capillaries of human kidneys cannot be measured using micropuncture of glomerular capillaries or measurements of stop-flow pressures in proximal tubules or free-flow pressures in Bowman's space. From determinations of plasma protein concentrations, and thus the afferent arteriolar oncotic pressure together with the whole-kidney filtration fraction, efferent arteriolar oncotic pressure can be calculated, generally yielding values around 37 mm Hg. In addition, peritubular capillary pressure has been estimated from intrarenal venous pressure measurements, yielding estimates of proximal tubule hydraulic pressure of 20 to 25 mm Hg.[84,85] Coupled with an efferent oncotic pressure of 37 mm Hg, this indicates that the minimal value for glomerular capillary pressure in humans is in the range of 57 to 62 mm Hg. Glomerular volumes and diameters of human kidneys are larger than in the experimental species, suggesting that the single-nephron K_f is greater than in the experimental animals. It is thought that in humans, filtration pressure disequilibrium is normally present, and GFR exhibits less plasma flow dependency than in rats.[2] The molecular sieving approach has been used as an alternative noninvasive means for the evaluation of the hydraulic conductivity characteristics and filtration coefficient in studies of glomerular dynamics in humans. By using uncharged macromolecules with varying molecular radii, which are partially restricted, the sieving coefficients for molecules of different sizes can be obtained.[63,86,87] Combining the sieving data with mathematical models, estimated data for K_f values can be derived. Values for single-nephron K_f in normal human subjects have varied from 3.6 to 9.4 nL/min mm Hg, with an average value of about 6 to 7 nL/min mm Hg.[2,4,75,88,89]

SELECTIVE ALTERATIONS IN THE PRIMARY DETERMINANTS OF GLOMERULAR FILTRATION

The four primary determinants of filtration are Q_A, ΔP, K_f, and π_A, and alterations in each of these will affect the GFR. The degree to which such alterations will modify SNGFR has been examined by mathematical modeling[80] and compared with values obtained experimentally (see Arendshorst and Navar,[2] Navar et al.,[4] and Lowenstein et al.[84]).

Glomerular Plasma Flow (Q_A)

Because protein is normally excluded from the glomerular filtrate, the total amount of protein entering the glomerular capillary network from the afferent arteriole is maintained, leading to progressively increasing protein concentration as the plasma traverses the glomerular capillaries to the efferent arteriole. Thus, there is a substantial increase in the plasma colloid osmotic pressure (πg) in the glomerular capillaries as the plasma traverses from the inlet to the outlet. This increase in πg counteracts the net hydraulic filtration pressure and may completely offset the ΔP, so that $\pi g = \Delta P$ if equilibrium is reached before the plasma reaches the efferent arteriole, preventing further net filtration of fluid into Bowman's space. Under this condition of filtration pressure equilibrium, where $\Delta P = \pi_E$, SNGFR will vary directly with changes in Q_A as a greater filtering surface area is recruited. Once Q_A increases enough to produce disequilibrium, π_E becomes less than ΔP, and the SNGFR will no longer vary linearly with Q_A.[90] However, there is still an effect of increases in plasma flow to increase GFR, although to a lesser extent, because the filtration fraction will decrease and there will be a lesser overall increase in πg. As shown in Fig. 3.15, increases in plasma flow are associated with increases in GFR in a number of studies in rats, dogs, nonhuman primates, and humans.

Transcapillary Hydraulic Pressure Difference (ΔP)

Mathematical modeling indicates that isolated changes in the glomerular transcapillary hydraulic pressure gradient exert strong effects on SNGFR.[2,4,80] In particular, when ΔP exceeds the colloid osmotic pressure at the efferent end of the glomerular capillary, filtration occurs throughout the glomerular capillary network, and SNGFR increases as ΔP increases. The relationship between SNGFR and ΔP is nonlinear, however, because the rise in SNGFR at any given fixed value of Q_A results in a concurrent increase in $\Delta\pi$. Because net effective filtration pressure is a small fraction of P_{GC}, small isolated changes in P_{GC} can cause large percentile changes in net filtration pressure.

Glomerular Capillary Filtration Coefficient (K_f)

Glomerular damage from a variety of kidney diseases and various hormonal and pharmacologic influences can result in alterations in the glomerular filtration coefficient (K_f) due to reductions in surface area available for filtration and/or to reductions in the hydraulic conductivity because of thickening of the basement membrane or other derangements.

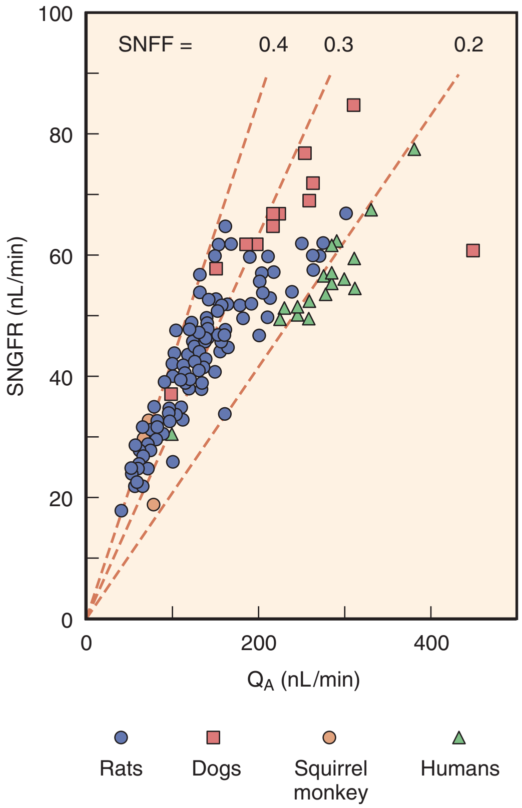

Fig. 3.15 Association between single-nephron glomerular filtration rate *(SNGFR)* and glomerular plasma flow (Q_A); data are from rats, dogs, squirrel monkeys, and humans. The values for SNGFR and Q_A for humans were calculated by dividing the whole-kidney GFR and renal plasma flow by the estimated total number of nephrons/kidney (1 million). Each point represents the mean value for a given study. *SNFF,* Single-nephron filtration fraction. (Data from Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. *Handbook of physiology: Section 8; Renal physiology* Vol 1. New York: Oxford University Press; 1992.)

Hydraulic conductivity of the glomerular basement membrane demonstrates an inverse relationship to ΔP, indicating that K_f may be directly influenced by ΔP.[91] K_f is also affected by the plasma protein concentration.[81,92] Under conditions of filtration pressure equilibrium, reductions in K_f do not affect SNGFR until K_f is reduced enough to produce filtration pressure disequilibrium. However, increases in K_f above normal will increase SNGFR until equilibrium conditions occur.[80,83] When plasma flow is high, and disequilibrium exists, there is a more direct relationship between K_f and SNGFR.[80,83]

Colloid Osmotic Pressure (π_A)

SNGFR and the single-nephron filtration fraction (SNFF) are each theoretically predicted to vary reciprocally as a function of π_A.[80] If Q_A, ΔP, and K_f are held constant, reductions in π_A are predicted to increase P_{UF}, leading to an increase in SNGFR. An increase in π_A should produce a decrease in SNGFR until π_A equals ΔP (normally, ~35 mm Hg), at which point filtration stops. In contrast to theoretic predictions, experimentally induced reductions in π_A do not lead to a rise in SNGFR because there are changes in P_{BS} and K_f so that a reduction in π_A results in a reduction in K_f, thereby offsetting variations in P_{UF} that occur with changes in π_A.[83] Studies in isolated glomeruli have indicated that extremely low concentrations of albumin produce an increase in K_f, whereas extremely high concentrations of albumin result in a decrease in K_f.[83] However, in vivo studies have shown that increases in plasma π will increase K_f in both rats[92] and dogs.[81] These divergent results of the effects of protein concentration or π_A on K_f can be partially explained by the results from studies of isolated glomerular basement membranes, which have shown a biphasic relationship between albumin concentration and hydraulic permeability.[91] There were lower values of hydraulic permeability at an albumin concentration of 4 g/dL than at either 0 or 8 g/dL.

POSTGLOMERULAR CIRCULATION

PERITUBULAR CAPILLARY DYNAMICS

The same Starling forces that control fluid movement across all capillary beds govern the rate of fluid movement across the peritubular capillary walls of the renal cortex. Owing to the high resistance of the afferent and efferent arterioles, a large drop in hydraulic pressure occurs prior to the peritubular capillaries so that peritubular capillary pressure is 15 to 20 mm Hg. In addition, because protein-free fluid is filtered out of the glomerular capillaries and into Bowman's space, the plasma proteins become concentrated, yielding an elevated oncotic pressure of blood flowing into the peritubular capillaries. As a consequence, the balance between the transcapillary oncotic and hydraulic pressure gradients favors movement of the tubular reabsorbate into the capillaries. However, variations in these forces have significant effects on net proximal reabsorption.[35,54,93] The absolute amount of movement resulting from this driving force also depends on the peritubular capillary surface area available for fluid uptake and the hydraulic conductivity of the peritubular capillary wall. Values for the hydraulic conductivity of the peritubular capillaries are not as great as those for the glomerular capillaries, but this difference is offset by the much greater total surface area of the peritubular capillary network.

The peritubular capillary surface contains fenestrations that are bridged by a thin diaphragm and glycocalyx that is negatively charged.[3,94] Beneath the fenestrae of the endothelial cells lies a thin basement membrane that surrounds the capillary. The peritubular capillaries are closely opposed to cortical tubules (Fig. 3.16), so that the extracellular space between the tubules and capillaries constitutes only about 5% of the cortical volume.[95] The tubular epithelial cells are surrounded by the tubular basement membrane, which is distinct from and wider than the capillary basement membrane (see Fig. 3.16). Numerous microfibrils connect the tubular and capillary basement membranes, a feature that may help limit expansion of the interstitium and maintain close contact between tubular epithelial cells and the peritubular capillaries during periods of high fluid flux.[96] Thus, the pathway for fluid reabsorption from the tubular lumen to the peritubular capillary is composed, in series, from the epithelial cell, lateral spaces, tubular basement membrane, a narrow interstitial region containing microfibrils, the capillary basement membrane, and the thin membrane bridging the endothelial fenestrae.[96]

Like the endothelial cells, the basement membrane of the peritubular capillaries possesses anionic sites.[94,97] The

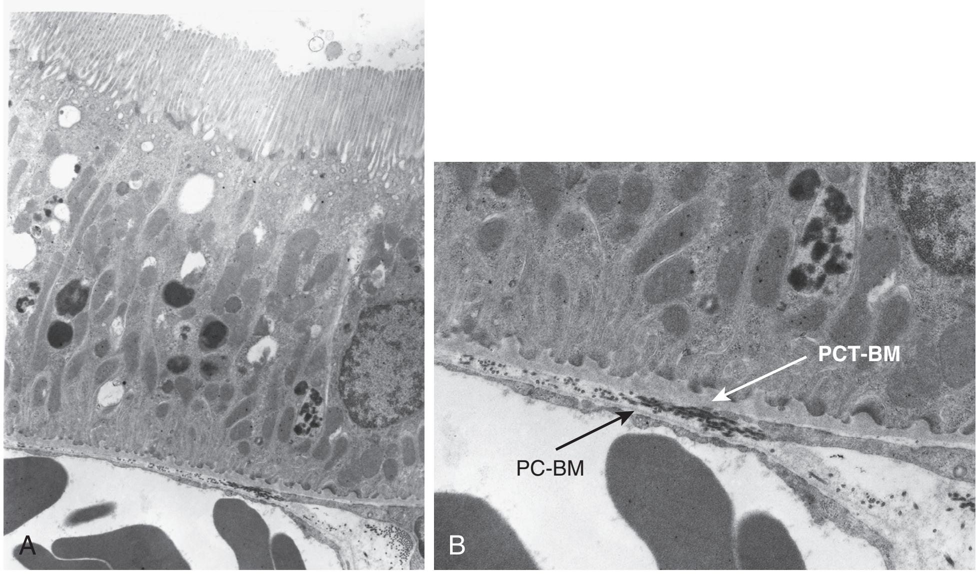

Fig. 3.16 Apposition of peritubular capillaries with basolateral tubular membranes. Shown are electron micrographs of a proximal tubule of a Munich-Wistar rat. The tubule was perfusion-fixed with 1.25% glutaraldehyde, thereby also fixing red cells in adjacent capillaries. (A) The apposition of the basolateral surface of the tubular cells with the adjacent peritubular capillaries is close, leaving little interstitial space where the two come in contact (magnification ≈×13,000). (B) The proximal tubule basement membrane *(PCT-BM)* is relatively thick in comparison with the peritubular capillary endothelial basement membrane (*PC-BM*; magnification ≈×25,000). (Courtesy D. Maddox.)

electronegative charge density of the peritubular capillary basement membrane is significantly greater than that observed in the unfenestrated capillaries of skeletal muscle and similar to that observed in the glomerular capillary bed. These anionic sites in the peritubular capillaries compensate for the greater permeability of fenestrated capillaries, allowing free exchange of water and small molecules while restricting anionic plasma proteins to the circulation. The renal peritubular capillaries are reported to be more permeable to both small and large molecules than are other beds,[98] but this may be an artifact of the experimental conditions used. Indeed, other studies have indicated that the permeability of the peritubular vessels to dextrans and albumin is extremely low.[97,99]

MEDULLARY MICROCIRCULATION

Similar to cortical peritubular vessels, the functional role of the medullary peritubular vasculature is to supply the metabolic needs of nearby tissues, but this unique vasculature is also responsible for the uptake and removal of water extracted from collecting ducts during the process of urine concentration. Because the urinary concentration process requires the development and maintenance of a hypertonic interstitium, the countercurrent arrangement of vasa recta plays a vital role in maintaining the medullary solute gradient through passive countercurrent exchange.

Medullary blood flow constitutes only about 10% to 15% of total RBF[3,6,100,101] and is derived entirely from efferent arterioles of the juxtamedullary nephrons (see Figs. 3.1 and 3.2).[22,37,102–105] Depending on the species and the method of evaluation, from 7% to 18% of glomeruli give rise to efferent arterioles that supply the medulla.[22,106] Efferent arterioles of juxtamedullary nephrons are larger in diameter, possess thicker endothelium, and have more prominent smooth muscle layers than efferent arterioles originating from superficial glomeruli.[34,107]

Despite the fact that medullary flow is less than 20% of the cortical flow, it is still relatively high compared with other tissues per gram of tissue; outer medullary flow exceeds that of liver, and inner medullary flow is comparable to that of resting muscle or brain.[108] The high efficiency of countercurrent mechanisms in this area permits the existence and maintenance of the inner medullary solute concentration gradients in the presence of such large flows. The descending vasa recta have a continuous endothelium, in which water moves across water channels, and urea moves through endothelial carriers.[109,110] The ascending vasa recta are fenestrated, with a high hydraulic conductivity, and water movement is governed by transcapillary hydraulic and oncotic pressure gradients.[110] Medullary blood flow is highest under conditions of water diuresis and declines during antidiuresis.[100] A direct vasoconstrictive effect of vasopressin on the medullary microcirculation contributes to this decrease during antidiuresis.[111] Vasodilatory factors act to preserve medullary blood flow and prevent ischemia. Acetylcholine,[112] vasodilator prostaglandins,[113] kinins,[114] adenosine,[115] atrial peptides,[116] bradykinin,[9] and nitric oxide[117] increase medullary RBF. In contrast to their vasoconstrictor effects in the renal cortex, Ang II[118–121] and endothelin[118] increase medullary blood flow, effects mediated in part by vasodilatory prostaglandins,[119,120] whereas vasopressin decreases medullary blood flow.[111,122] Alterations in medullary blood flow may be a key determinant of medullary fluid tonicity and, thereby, of solute transport in the loops of Henle and the control of sodium excretion and blood pressure.[123] During hemorrhage, there is primarily cortical ischemia, with maintained blood flow through the medulla.[124]

The precise location of the boundary between the renal cortex and medulla is difficult to discern because the medullary rays of the cortex merge imperceptibly with the medulla. In general, the arcuate arteries or sites at which the interlobular arteries branch into arcuate arteries mark this boundary. When considering the medullary circulation, most studies have focused on its relation to the countercurrent mechanism, as facilitated by the parallel array of descending and ascending vasa recta. This configuration is characteristic of the inner medulla, but the medulla also contains an outer zone consisting of two morphologically distinct regions, the outer and inner stripes of the outer medulla (see Fig. 3.2). The boundary between the outer medulla and inner medulla is defined by the beginning of the thick ascending limbs of Henle (see Fig. 3.1). In addition to the thick ascending limbs, the outer medulla contains descending straight segments of proximal tubules (pars recta), descending thin limbs, and collecting ducts. The nephron segments of the inner stripe of the outer medulla include thick ascending limbs, thin descending limbs, and collecting ducts. Each of these morphologically distinct medullary regions is supplied and drained by a specific vascular system.

Both the outer and inner stripes contain two distinct circulatory regions—the vascular bundles, formed by the coalescence of the descending and ascending vasa recta, and the interbundle capillary plexus. Vascular bundles of the descending and ascending vasa recta arise from the efferent arterioles of juxtamedullary glomeruli and descend through the outer stripe of the outer medulla to supply the inner stripe of the outer medulla and inner medulla (see Fig. 3.2). Within the outer stripe, nutrient flow is provided by the ascending vasa recta rising from the inner stripe. This notion is supported by the large area of contact between the ascending vasa recta and descending proximal straight tubules within this zone.[103,106,125]

The outer medulla includes the metabolically active thick ascending limbs. Nutrients and O_2 to this energy-demanding tissue in the inner stripe are delivered by a dense capillary plexus arising from a few descending vasa recta at the periphery of the bundles. Of the 10% to 15% of total RBF directed to the medulla, the largest portion perfuses this inner stripe capillary plexus. The descending vasa recta possess a contractile layer composed of smooth muscle cells in the early segments that evolve into pericytes by the more distal portions of the vessels. These pericytes contain smooth muscle α-actin, suggesting that they serve as contractile elements and participate in the regulation of medullary blood flow,[126] as well as in vascular-tubular crosstalk.[127] Each of these vessels also displays a continuous endothelium that persists until the hairpin turn is reached, and the vessels divide to form the medullary capillaries. In contrast, ascending vasa recta, like true capillaries, lack a contractile layer and are characterized by a highly fenestrated endothelium.[128,129] The smooth muscle cells of the descending vasa recta are replaced by pericytes surrounding the endothelium, with subsequent loss of the pericytes and transformation into medullary capillaries accompanied by endothelial fenestrations.[102,125]

The rich capillary network of the inner stripe drains into numerous veins, which, for the most part, do not join the vascular bundles but ascend directly to the outer stripe. These veins subsequently rise to the cortical-medullary junction and join with cortical veins at the level of the inner cortex.[130] A few veins may extend within the medullary rays to regions near the kidney surface.[20,104,130] Thus, the capillary network of the inner stripe makes no contact with the vessels draining the inner medulla.

The inner medulla contains the thin descending and thin ascending limbs of Henle, together with collecting ducts (see Fig. 3.2). Within this region, the straight, unbranching vasa recta descend in bundles, with individual vessels leaving at every level to divide into a simple capillary network characterized by elongated links (see Figs. 3.1 and 3.2).[102,106,130] These capillaries converge to form the venous vasa recta. Within the inner medulla, the descending and ascending vascular pathways remain in close apposition, although distinct vascular regions can no longer be clearly discerned. The venous vasa recta rise toward the outer medulla in parallel with the supply vessels to join the vascular bundles. Within the outer stripe of the outer medulla, the vascular bundles spread out and traverse the outer stripe as wide tortuous channels that lie in close apposition to the tubules, eventually emptying into arcuate or deep interlobular veins.[106] The venous pathways in the bundles are both larger and more numerous than the arterial vessels, suggesting lower flow velocities in the ascending (venous) than in the descending (arterial) vessels.[131] The close apposition of the arterial and venous pathways in the vascular bundles is important for maintaining the hypertonicity of the inner medulla.

The mechanism of urine concentration requires coordinated function of the vascular and tubular components of the medulla. In species capable of marked concentrating ability, medullary vascular-tubular relationships show a high degree of organization favoring particular exchange processes by the juxtaposition of specific tubular segments and blood vessels.[130,132] In addition to anatomic proximity, the absolute magnitude of these exchanges is greatly influenced by the permeability characteristics of the structures involved, which may vary significantly among species.[133]

PARACRINE AND ENDOCRINE FACTORS REGULATING RENAL HEMODYNAMICS AND GLOMERULAR FILTRATION RATE

EXTRINSIC AND INTRINSIC REGULATION OF THE RENAL MICROCIRCULATIONS

A variety of hormonal, neural, and paracrine factors exert regulating influences on RBF and GFR.[35,55,83] Renal blood vessels from the arcuate arteries and interlobular arteries to the afferent and efferent arterioles are influenced to a greater or lesser extent by these intrinsic and extrinsic influences. As a result, the vascular tones of preglomerular and postglomerular resistance vessels are regulated to control RBF, glomerular hydraulic pressure, and the transcapillary hydraulic pressure gradient. The glomerular mesangium is the site of action and production of many such substances. Vasoactive compounds may elicit acute alterations in K_f by changing the effective surface area for filtration through contraction of mesangial cells, causing shunting of blood to fewer capillary loops.[35,134,135] In addition, contraction of glomerular epithelial cells (podocytes), which contain filamentous actin molecules, may decrease the size of the filtration slit pores, thereby altering hydraulic conductivity of the filtration pathway and reducing K_f.[136] Various growth factors influence chronic changes in renal hemodynamics by promoting mesangial cell proliferation and expansion of the extracellular matrix, leading to obliteration of capillary loops and a reduction in the filtration coefficient.

Our understanding of afferent and efferent arteriolar vascular responses to neural, paracrine hormonal, and vasoactive substances have, to a large extent, come from micropuncture studies of glomerular hemodynamics. Various other methods have been used to examine the effects of vasoactive substances on the preglomerular and postglomerular vasculature.[35,55,83,137–144] The results using these different techniques have provided important insights into the vasoactive properties of the preglomerular and postglomerular vasculature that control renal hemodynamics and glomerular filtration rate.

The renal vasculature and glomerular mesangium respond to numerous endogenous hormones and vasoactive peptides, such as Ang II, by vasoconstriction, reductions in RBF and GFR, and reductions in the glomerular capillary filtration coefficient. Among the vasoconstrictors are Ang II, norepinephrine, leukotrienes C4 and D4, platelet-activating factor (PAF), adenosine 5'-triphosphate (ATP), endothelin, vasopressin, serotonin, and epidermal growth factor.[35,55,83] Similarly, vasodilatory substances, such as NO and prostaglandin E2 (PGE2), and PGI2, histamine, bradykinin, acetylcholine, insulin, insulin-like growth factor, calcitonin gene-related peptide, cyclic adenosine monophosphate, and relaxin can increase RBF and GFR.[35,55,83] However, in addition to having their own direct effects on RBF and GFR, a number of these complex vasoactive systems, such as the renin-angiotensin-aldosterone system (RAAS) and arachidonic acid metabolites, produce both vasoconstriction and vasodilator effects and can also stimulate production and release of other factors thus, masking their primary effect. Furthermore, vasoconstrictor agents such as Ang II may result in a feedback stimulation of vasodilatory compensatory factors yielding a complex interactive balance regulating renal hemodynamics.

INTRINSIC MECHANISMS: RENAL AUTOREGULATION

Renal autoregulation refers to the intrinsic ability of the kidney to respond to a perturbation that elicits a vasoactive response, which alters renal vascular resistance in the direction that maintains RBF and GFR. Changes in perfusion pressure are the manipulation most commonly used to demonstrate autoregulatory efficiency. Although the efficiency with which blood flow is maintained differs from organ to organ (being most efficient in brain and kidney), all organs and tissues exhibit autoregulation. As shown in Fig. 3.17, the kidney autoregulates renal blood flow over a wide range of renal perfusion pressures. Autoregulation of blood flow in response to changes in perfusion pressure requires parallel changes in resistance.

The finding that both RBF and GFR are autoregulated with a high efficiency indicates that the principal resistance change due to autoregulatory adjustments is localized to the preglomerular vasculature. Studies of single-nephron function of superficial nephrons have demonstrated that SNGFR also exhibits efficient autoregulation, as long as the tubular fluid collections do not block flow to the macula densa. Furthermore, direct measurements of glomerular pressures in the Munich-Wistar rat, which has glomeruli on the renal cortical surface that is accessible to micropuncture, have demonstrated autoregulation of glomerular pressure in response to variations in renal arterial perfusion pressure. Fig. 3.18 summarizes the effects of graded reductions in renal perfusion pressure

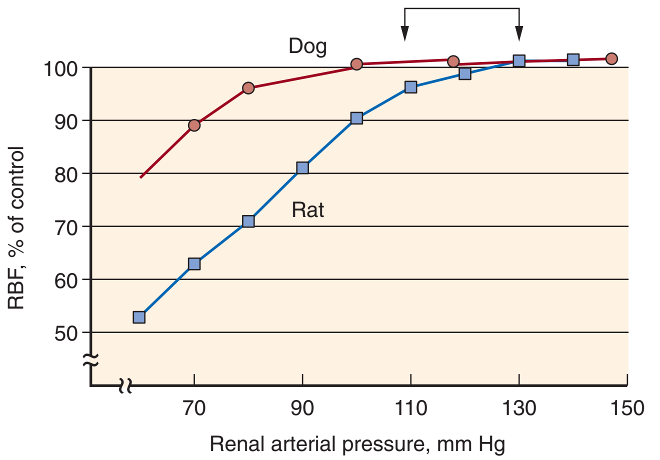

Fig. 3.17 Autoregulatory response of total renal blood flow *(RBF)* to changes in renal perfusion pressure in the dog and rat. In general, the normal anesthetized dog exhibits greater autoregulatory capability to maintain RBF and glomerular filtration rate to lower arterial pressures than the rat. (From Navar LG, Bell PD, Burke TJ: Role of a macula densa feedback mechanism as a mediator of renal autoregulation. *Kidney Int.* 1982;22:S157–S164.)

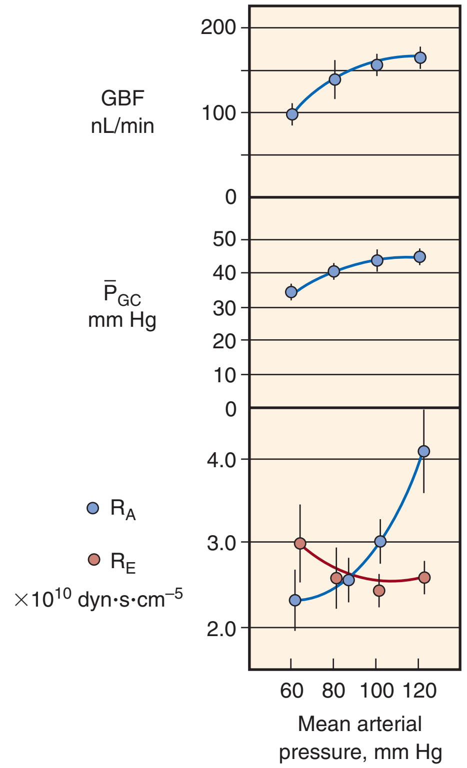

Fig. 3.18 Glomerular dynamics in response to reductions of renal arterial pressure in the normal hydropenic rat. As can be seen, glomerular blood flow *(GBF)* and glomerular capillary hydraulic pressure (P_{GC}) remained relatively constant as blood pressure was lowered from ≈120 to ≈80 mm Hg over the range of perfusion pressure examined, primarily as a result of reductions in afferent arteriolar resistance *(R_A)*. Efferent arteriolar resistance *(R_E)* was relatively constant but increased slightly at lower pressures. (Modified from Robertson CR, Deen WM, Troy JL, Brenner BM: Dynamics of glomerular ultrafiltration in the rat. III: hemodynamics and autoregulation. *Am J Physiol.* 1972;223:1191, 1972.)

on P_{GC} and preglomerular (R_A) and efferent arteriolar (R_E) resistance.[145] Graded reductions in renal perfusion pressure from 120 to 80 mm Hg result in only a modest decline in glomerular capillary blood flow, whereas a further reduction in perfusion pressure to 60 mm Hg leads to a more pronounced decline (see Fig. 3.18).

Autoregulation of glomerular capillary blood flow and P_{GC} as perfusion pressure decreased from 120 to 80 mm Hg is the result primarily of a pronounced decrease in R_A, with little or no change in R_E. Over the range of renal perfusion pressures from 120 to 60 mm Hg, R_E tended to increase slightly at the lower perfusion pressure. Under conditions of modest plasma volume expansion, R_A declines while R_E increases slightly as renal perfusion pressure is lowered so that P_{GC} and ΔP are virtually unchanged over the entire range of renal perfusion pressures.[145] The mean glomerular transcapillary hydraulic pressure difference (ΔP) exhibits almost perfect autoregulation over the entire range of perfusion pressures.[145] These results indicate that autoregulation of GFR is the consequence of the autoregulation of glomerular blood flow and glomerular capillary pressure. Similar results have been obtained in other rat strains and in dogs, where proximal and distal tubular pressures and peritubular capillary pressure also demonstrated autoregulation.[146]

Although more controversial, autoregulation also occurs in the medullary circulation,[147–149] an effect that may be influenced by the volume status of the animal.[148] In the split hydronephrotic rat kidney preparation,[150] reductions in perfusion pressure from 120 to 95 mm Hg elicited dilation of all preglomerular vessels, including the arcuate and interlobular arteries. The large preglomerular arterioles, including the interlobular arteries, contribute to the constancy of outer cortical blood flow in the upper autoregulatory range.[151] These responses notwithstanding, most evidence has indicated that the major preglomerular resistance components are the afferent arterioles.[33,152–154] Direct observations of perfused juxtamedullary nephrons have revealed parallel reductions in the luminal diameters of arcuate, interlobular, and afferent arterioles in response to elevations in perfusion pressure. However, because quantitatively similar reductions in vessel diameter produce much greater elevations in resistance in smaller than in larger vessels, the predominant effect of these changes is an increase in afferent arteriolar resistance.[30,152]

Under conditions of substantial plasma volume expansion, medullary blood flow autoregulation efficiency is diminished, whereas cortical blood flow autoregulatory responses are maintained. This loss of medullary blood flow autoregulation is thought to contribute to the exaggerated pressure natriuresis during plasma volume expansion.[36,155]

Cellular Mechanisms Involved in Renal Autoregulation

Autoregulation of the afferent arteriole and interlobular artery is blocked by administration of L-type calcium channel blockers, inhibition of mechanosensitive cation channels, and a calcium-free perfusate.[156–159] Thus, the autoregulatory response involves gating of mechanosensitive channels, which produces membrane depolarization and activation of voltage-dependent calcium channels and leads to an increase in intracellular calcium concentration and vasoconstriction.[156,160,161] Indeed, calcium channel blockade almost completely blocks the autoregulation of RBF.[162,163] Intrinsic metabolites of the cytochrome P450 epoxygenase pathway attenuate the autoregulatory capacity of the afferent arteriole, whereas metabolites of the cytochrome P450 hydroxylase pathway enhance autoregulatory responsiveness.[164]

Inhibition of nitric oxide (NO) does not prevent autoregulation of GFR and RBF, but values for RBF are reduced at any given renal perfusion pressure as compared with control values.[165–168] In the isolated, perfused, juxtamedullary afferent arteriole, the initial vasodilatation observed when pressure was increased was of shorter duration when endogenous NO formation was blocked, but the autoregulatory response was unaffected.[161] Cortical and juxtamedullary preglomerular vessels in the split hydronephrotic kidney also autoregulate in the presence of NO inhibition.[169] Thus, the evidence indicates that NO is not essential for the manifestation of renal autoregulation, although it does greatly influence the plateau of the autoregulatory response. Furthermore, NO plays a role in tubuloglomerular feedback, as will be discussed.[35,170]

Myogenic and Tubuloglomerular Feedback Mechanisms

There is general consensus that both myogenic and tubuloglomerular feedback mechanisms contribute to autoregulatory responses. The myogenic mechanism refers to the ability of arterial smooth muscle to contract and relax in response to increases and decreases in vascular wall tension.[153,156,171,172] Thus, an increase in perfusion pressure, which initially distends the vascular wall, is followed by contraction of a resistance vessel, resulting in a recovery of blood flow from an initial elevation to a value comparable to the control level. Evidence that the renal vasculature is intrinsically responsive to changes in the transmural hydraulic pressure difference and exhibits myogenic responses has been obtained in isolated afferent arterioles. Myogenic control of renal vascular resistance has been estimated to contribute up to 50% of the total autoregulatory response.[153,173]

Autoregulation of renal blood flow is observed, even when tubuloglomerular feedback is inhibited by furosemide, suggesting an important role for a myogenic mechanism.[174] This myogenic mechanism of autoregulation occurs very rapidly, reaching a full response in 3 to 10 seconds.[174,175] Autoregulation occurs in all the preglomerular resistance vessels of the in vitro, blood-perfused, juxtamedullary nephron preparation.[152,153,164,175–177] Of note, the afferent arterioles in this preparation constricted in response to rapid increases in perfusion pressure, even when flow to the macula densa was prevented by resection of the papilla, indicating a myogenic response.[175] Isolated perfused rabbit afferent arterioles respond to step increases of intraluminal pressure with a decrease in luminal diameter.[143] In contrast, efferent arteriolar segments showed vasodilation when submitted to the same procedure, probably reflecting simple passive physical properties.

Autoregulation is also observed in the afferent arteriole and arcuate and interlobular arteries of the nonfiltering hydronephrotic kidney preparation but, again, the efferent arteriole does not autoregulate in this model.[156,158,159,169,178] However, it should be noted that efferent arteriolar resistance in vivo may increase in response to prolonged reductions in AP.[145,179] This may result from increased activity of the intrarenal renin-angiotensin system (RAS). These data may also explain why autoregulation of GFR is more efficient than autoregulation of RBF.

The autoregulatory threshold can be reset in response to a variety of perturbations. Autoregulation in the afferent arteriole is attenuated in diabetic kidneys and may contribute to the hyperfiltration seen early in this disease.[178] Autoregulation is partially restored by insulin treatment and/or by inhibition of endogenous prostaglandin production.[178] Autoregulation in the remnant kidney is markedly attenuated 24 hours after the reduction in renal mass but is restored by cyclooxygenase inhibition, suggesting that release of vasodilatory prostaglandins may be involved in the initial response to increased SNGFR in the remaining nephrons after an acute partial nephrectomy.[180] Much higher pressures than normal are required to evoke a vasoconstrictor response in the afferent arteriole during the development of spontaneous hypertension.[181] Both the afferent arterioles and the interlobular arteries of Dahl salt-sensitive hypertensive rats exhibit reduced myogenic responsiveness to increases in perfusion pressure when fed a high-salt diet.[182] Thus, alterations in autoregulatory responses of the renal vasculature occur in a variety of disease states and may influence the kidney's ability to alter excretory responses to increased plasma volume expansion.

During development of the nephron, the tubule develops a segment that descends from the cortex into the medulla, but the connection with the originating glomerulus remains throughout the developmental process and provides the structural basis for the regulatory mechanism known as "tubuloglomerular feedback" (TGF; Fig. 3.19). A specialized nephron segment macula densa, at the end of the thick ascending limb of the loop of Henle, has distinct morphologic characteristics, including the presence of a primary cilium.[183] Macula densa cells are adjacent to the cells of the glomerulus and connect with the extraglomerular mesangium and afferent and efferent arterioles of the glomerulus (see Fig. 3.19). This anatomic arrangement of macula densa cells, extraglomerular mesangial cells, arteriolar smooth muscle cells, and renin-secreting cells of the afferent arteriole is known as the "juxtaglomerular apparatus" (JGA).[184]

The JGA is ideally suited to serve as a feedback system whereby a physicochemical stimulus in the tubular fluid activates the macula densa cells, which in turn transmit signals to the arterioles to alter the degree of contraction, thus regulating afferent arteriolar resistance. Changes in the volume flow and composition of the fluid flowing past the macula densa elicit rapid alterations in afferent arteriolar resistance and glomerular filtration, with increases in delivery of fluid resulting in decreases in SNGFR and P_{GC} of the same nephron.[35,185,186] The TGF system senses delivery of fluid to the macula densa and "feeds back" signals to control filtration rate, thus providing a powerful feedback mechanism to regulate the pressures and flows that govern GFR in response to acute perturbations in delivery of fluid to the macula densa. The TGF mechanism is thus another important mechanism that helps explain the very high efficiency of autoregulation of RBF and GFR. Increased RBF or glomerular capillary pressure leads to increased GFR and, therefore, greater delivery of volume and solute to the distal tubule. Increased distal delivery is sensed by the macula densa, which activates effector mechanisms that increase preglomerular resistance, reducing RBF, glomerular pressure, and GFR.[48]

Increased perfusion of the late proximal tubule into the distal tubule causes a reduction in glomerular blood flow, glomerular pressure, and GFR.[187] Furthermore, experimental maneuvers that decrease distal tubule fluid flow induce afferent arteriolar vasodilation and interfere with the normal autoregulatory response.[35,170,188] In addition, perfusion with furosemide-containing solutions into the macula densa segment abrogate the normal constrictor response of afferent arterioles to increased perfusion pressure,[189] presumably by blocking the Na^+-K^+-$2Cl^-$ transporter on the luminal membrane of the macula densa cells.[153,190] These studies have suggested that the autoregulatory response in juxtamedullary nephrons is also highly dependent on the TGF mechanism. Moreover, deletion of the A_1 adenosine receptor gene in mice to block TGF results in less efficient autoregulation, again indicating the role for TGF in the autoregulatory response.[191]

To examine the role of TGF in autoregulation, investigators have studied spontaneous oscillations in proximal tubule pressure and RBF and the response of the renal circulation to high-frequency oscillations in tubule flow or renal perfusion pressure.[192] Oscillations in tubule pressure have been observed in anesthetized rats at a rate of about three cycles/min that are sensitive to small changes in delivery of fluid to the macula densa.[193] These spontaneous oscillations are eliminated by loop diuretics.[194] To examine this hypothesis,[192] sinusoidal oscillations were induced in distal tubule flow in rats at a frequency similar to that of the spontaneous fluctuations in tubule pressure. Varying distal delivery at this rate caused parallel fluctuations in stop-flow pressure (an index of glomerular capillary pressure), probably mediated by alterations in afferent resistance, again consistent with dynamic regulation of glomerular blood flow by the TGF system. To investigate the role of this system in autoregulation,[195] the effects of sinusoidal variations in AP at varying frequencies on renal blood flow were examined. Two separate components of autoregulation were identified, one operating at about the same frequency as the spontaneous fluctuations in tubule pressure, the TGF component, and one operating at a much higher frequency consistent with spontaneous fluctuations in vascular smooth muscle tone by the myogenic component.[196] These data have suggested that slow pressure changes elicit a predominant TGF response, whereas the rapid response reflects the myogenic mechanism.

The TGF mechanism stabilizes delivery of volume and solute to the distal nephron. Under normal conditions, flow-related changes in the tubular fluid composition at the macula densa are sensed, and signals are transmitted to the afferent arterioles to regulate the filtered load. Early distal tubular fluid is hypotonic (~120 mOsm/kg H_2O), and its composition is closely coupled to fluid flow along the ascending loop of Henle, so that increases in flow cause increases in tubular fluid osmolality and NaCl concentration at the macula densa, which lead to vasoconstriction of the afferent arteriole. At the cellular level, increases in tubular fluid osmolality elicit increases in cytosolic [Ca^{2+}] in macula densa cells, which result in release of a vasoconstrictive factor from these cells.[197] As depicted in Fig. 3.20, suggested mediators of TGF include purinergic compounds, such as adenosine and ATP, and one or more of the eicosanoids, such as prostaglandin E2 (PGE2) or 20-hydroxyeicosatetraenoic acid (20-HETE). The factor mediating TGF responses vasoconstricts

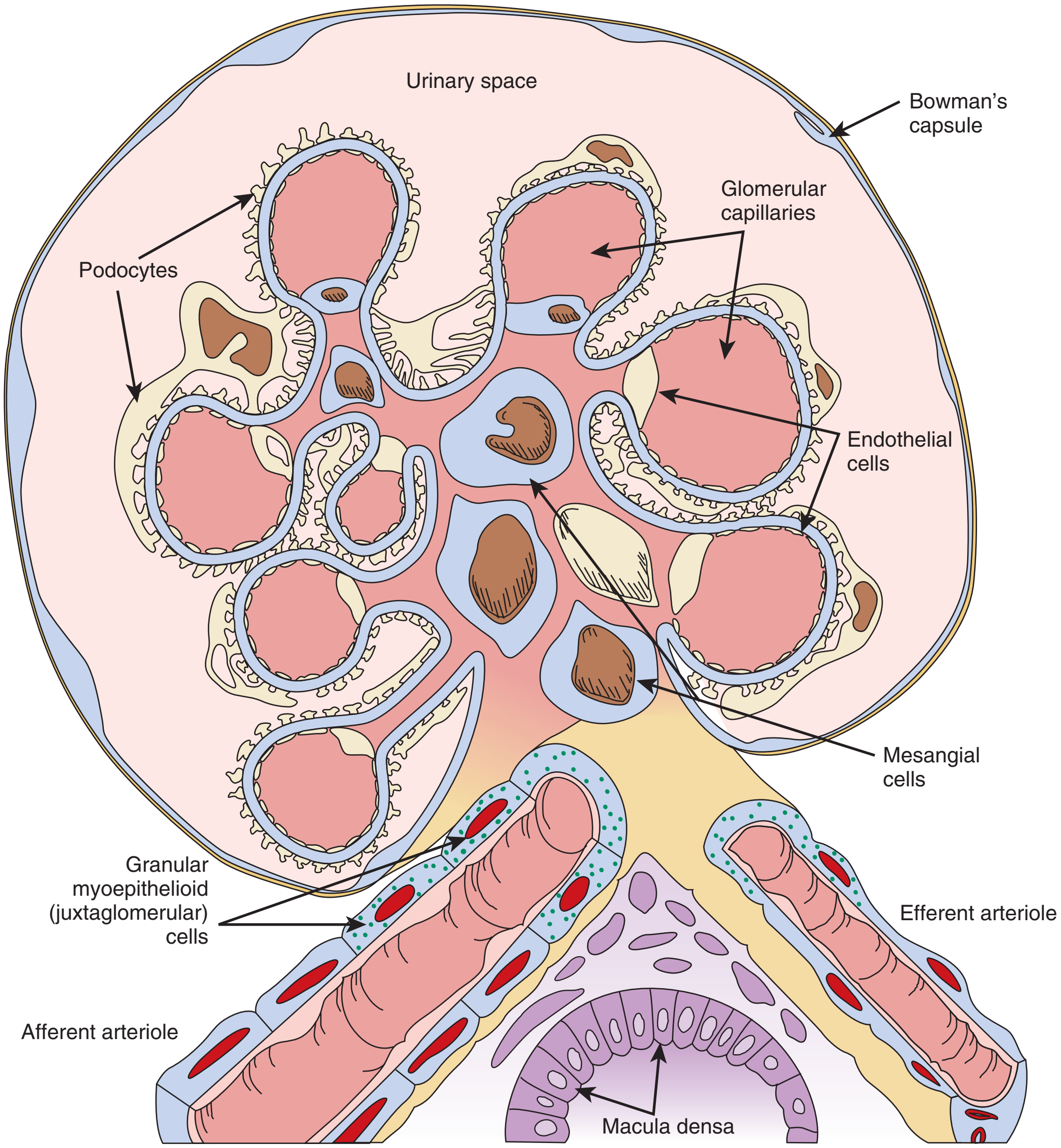

Fig. 3.19. Schematic drawing of a cross-section of a glomerulus, vascular pole, and macula densa cells forming the juxtaglomerular apparatus. As the afferent arteriole enters the glomerular tuft (large vessel, *lower left*), it breaks into a capillary network, and blood leaves the glomerular tuft via the efferent arteriole (large vessel, *lower right*). The glomerular capillaries have a fenestrated endothelium. The capillary network and mesangium located between the capillaries are bound together by a common basement membrane (*blue line* between the podocytes and capillaries). The basement membrane is absent between the capillary lumen and mesangial cells. The outer side of the basement membrane is surrounded by interdigitating visceral epithelial cells known as podocytes. Kriz and coworkers[184] have pointed out that the glomerular mesangium is continuous with the extraglomerular mesangium (consisting of extraglomerular mesangial cells and matrix) at the vascular pole. The extraglomerular mesangium, along with the macula densa cells of the distal tubule and the afferent arteriole, form the juxtaglomerular apparatus. (Courtesy D.A. Maddox.)

afferent arteriolar vessels through the opening of voltage-gated Ca^{2+} channels in vascular smooth muscle cells.[35]

The sensitivity of the TGF mechanism can be modulated by many agents and circumstances. TGF sensitivity is diminished during volume expansion, thus allowing a greater delivery of fluid and electrolytes to the distal nephron for any given level of GFR. Reductions in TGF sensitivity allow correction of volume expansion. In contrast, contraction of extracellular fluid and blood volume is associated with an enhanced sensitivity of the TGF mechanism, which together with an augmented proximal reabsorption helps conserve fluid and electrolytes. A major regulator of TGF sensitivity is Ang II. In states of low Ang II activity (e.g., extracellular volume expansion, salt loading), the TGF mechanism is less responsive, whereas feedback sensitivity is enhanced during conditions of high Ang II activity, such as occurs during dehydration, hypotension, or hypovolemia.

The interactions between the myogenic mechanism and the TGF are complex and not simply additive. Contributions from other systems add additional complexity. For example, glomerulotubular balance, whereby proximal tubule reabsorption increases as GFR rises, blunts the effects of alterations in GFR on distal delivery. In addition, the persistence of some autoregulatory behavior in nonfiltering kidneys[198] and

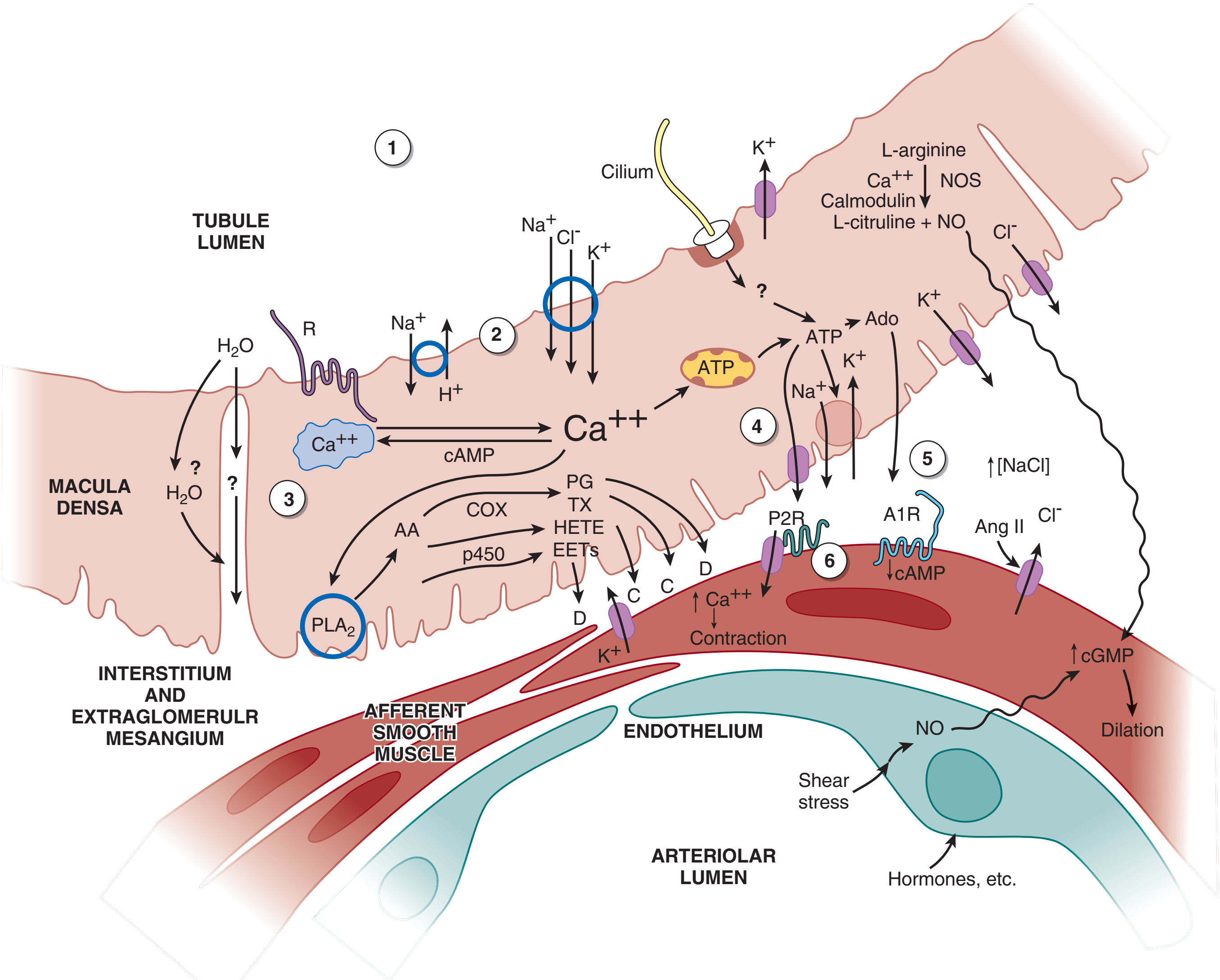

Fig. 3.20 Proposed macula densa tubuloglomerular feedback (TGF) signaling mechanisms. Numbers in circles refer to the following sequence of events: 1, Flow-dependent changes in tubular fluid composition, including Na^+, Cl^-, osmolality, signals from intratubular paracrine agents, and cilia disturbance; 2, membrane activation step, including membrane depolarization, enhanced Na^+, Cl^-, K^+ uptake, or other sensing mechanism; 3, transmission from membrane to intracellular signal mobilization; 4, formation and release of TGF mediators, including ATP and adenosine *(Ado),* arachidonic acid *(AA)* metabolites, and nitric oxide *(NO)*; 5, receptor activation by released agents, membrane depolarization, and activation of Ca^{2+} channels in vascular smooth muscle cells; 6, afferent arteriolar vasoconstriction partially countered by NO-stimulated increases in cGMP and other released mediators—local angiotensin II (Ang II) and neuronal nitric oxide synthase (nNOS) activity modulate the response. *ATP,* Adenosine triphosphate; *A1R,* adenosine A1 receptor; *C,* constriction; *cGMP*, cyclic guanosine monophosphate; *COX,* cyclooxygenase; *D,* dilation; *EET,* epoxyeicosatrienoic acid; *HETE,* hydroxyeicosatetraenoic acid; *P2R,* P2 purinergic receptors; *p450,* cytochrome P 450; *PG,* prostaglandins; *PLA2,* phospholipase A2; *TX,* thromboxanes. (Modified from Navar LG, Bell PD, Burke TJ: Role of a macula densa feedback mechanism as a mediator of renal autoregulation. *Kidney Int.* 1982;22:S157–S164.)

in isolated blood vessels has suggested that the delivery of filtrate to the distal tubule is not absolutely required for constancy of blood flow. Nevertheless, the myogenic and TGF mechanisms are not mutually exclusive, and various models of renal autoregulation incorporate both systems.[172,199] Because the myogenic and TGF responses share the same effector site, the afferent arteriole, interactions between these two systems are unavoidable, and each response is capable of modulating the other. The prevailing view is that these two mechanisms act in concert to accomplish the same end, a stabilization of renal function when blood pressure is altered.[35,173,200] Furthermore, the time constraints are different; the myogenic component of autoregulation requires less than 10 seconds for completion and normally follows first-order kinetics without rate-sensitive components.[173] The response time for the tubuloglomerular feedback may occur as rapidly as 5 seconds,[154] although others have suggested that it takes 30 to 60 seconds and shows spontaneous oscillations at 0.025 to 0.033 Hz.[173] The myogenic and tubuloglomerular feedback mechanisms account for most of the autoregulatory responses, but additional systems have been suggested.[173] Furthermore, the nature of their interaction may be complex, with the TGF primarily influencing the sensitivity of the myogenic mechanism.[35]

Mechanisms of Tubuloglomerular Feedback Control of Renal Blood Flow and Glomerular Filtration Rate

There appear to be several factors that have been identified as tubular signals for TGF.[201] Changes in delivery of Na^+, Cl^-, and K^+ are thought to be sensed by the macula densa through the Na^+-K^+-$2Cl^-$ cotransporter on the luminal cell membrane of the macula densa cells.[202] Alterations in Na^+, K^+, and Cl^- reabsorption result in inverse changes in SNGFR and renal vascular resistance, primarily due to changes in preglomerular resistance. For example, when salt concentration increases at the macula densa, the feedback mechanism increases afferent arteriolar resistance, thus decreasing glomerular pressure and SNGFR. Agents such as furosemide that interfere with the Na^+-K^+-$2Cl^-$ cotransporter in the macula densa cells[190] inhibit the feedback response.[203]

Additional studies have been performed in which macula densa segments were perfused with solutions that contained minimal concentrations of essential ions needed to maintain the integrity of the Na^+-K^+-$2Cl^-$ cotransporter, with the remaining solute being deficient in Na^+ (choline chloride) or deficient in Cl^- (Na isocyanate). These solutions clearly elicited normal TGF responses.[204] Furthermore, orthograde perfusion with nonelectrolyte solutes also elicited TGF responses.[35,205] Collectively, these results indicate that the integrity of the Na^+-K^+-$2Cl^-$ cotransporter must be maintained for the sensing mechanism to function normally. However, the actual sensing mechanism may be activated by changes in total solute concentration. Furthermore, studies evaluating the possible role of the primary cilium in macula densa cells have suggested that flow-dependent signals may stimulate the cilium and alter the magnitude of the TGF response.[183,206]

Another area of uncertainty involves the intracellular signaling cascade to generate a vasoactive agent that is secreted. Some studies have suggested that the luminal signal activates release of Ca^{2+} from the intracellular stores that then lead to the formation of ATP, which is secreted and plays a central role in mediating the signal from the luminal cell membrane of the macula densa. This is illustrated in Fig. 3.20, which is derived from based on several sources.[35,186] According to this scheme, increased delivery of solute to the macula densa results in concentration-dependent increases in solute uptake by the Na^+-K^+-$2Cl^-$ cotransporter and may separately generate a signal to initiate the cascade. This, in turn, stimulates mitochondrial activity in the macula densa cells, leading to the formation of ATP. Macula densa cells respond to an increase in luminal [NaCl] or total solute concentration by releasing ATP at the basolateral cell membrane through ATP-permeable, large-conductance anion channels, possibly providing a communication link between macula densa cells and adjacent mesangial cells via purinoceptors receptors on the latter.[207] The ATP can exert direct actions on the vascular smooth muscle cells and is also metabolized further, with ultimate degradation to the metabolites, adenosine diphosphate (ADP) and adenosine monophosphate (AMP). Activity of cytosolic 5'-nucleotidase or endo-5'-nucleotidase bound to the cell membrane results in the formation of adenosine.[186] In addition to the ATP metabolites, the macula densa cells also produce arachidonic acid metabolites, including PGE2 and PGI2, and nitric oxide and reactive oxygen species. Thus, there are several vasoactive substances that are secreted and may alter afferent arteriolar vascular tone. Although ATP and adenosine are formed in macula densa cells or in the adjacent interstitium, ATP interacts with purinergic (P2) receptors on the extraglomerular mesangial and vascular cells, resulting in an increase in $[Ca^{2+}]_i$.[208] The increase in $[Ca^{2+}]_i$ may occur, in part, via basolateral membrane depolarization through receptor operated channels, followed by a further increase in Ca^{2+} entry into the cells via voltage-gated Ca^{2+} channels.[209] As indicated in Fig. 3.20, gap junctions then transmit the calcium transient to the adjacent afferent arteriole, or ATP can exert similar effects directly on vascular smooth muscle cells, thus leading to vasoconstriction. Some of the ATP elicited by macula densa cells is metabolized to adenosine, which also can directly constrict the afferent arteriole through activation of purinergic P1 receptors.[210]

Although there is general consensus that ATP is secreted by macula densa cells, some investigators have suggested that the ATP metabolite, adenosine, is primarily responsible for mediating tubuloglomerular feedback. Intraluminal administration of an adenosine A1 receptor agonist enhances the TGF response.[211] In addition, TGF is attenuated in adenosine A1 receptor-deficient mice.[212,213] Blocking adenosine A1 receptors, or inhibition of adenosine synthesis via inhibition of 5'-nucleotidase, reduces TGF efficiency.[214] Addition of adenosine to the afferent arteriole causes vasoconstriction via activation of the adenosine A1 receptor, and addition of an A1 receptor antagonist blocks both the effects of adenosine and of high macula densa [NaCl].[215] These results are consistent with the hypothesis that adenosine is also a mediator of TGF responses invoking an effect of Na^+/K^+-ATPase activity and leading to increased adenosine synthesis.[215] However, adenosine also activates adenosine A2 receptors, which cause afferent arteriolar dilation and apparently abrogate the actions of A1 receptors.[216,217]

Efferent arterioles also respond to adenosine but they vasodilate in response to an increase in NaCl concentration at the macula densa or to direct application due to actions of the adenosine A2 receptors, which antagonize the effects of A1 receptors.[217,218] The changes in efferent arteriolar resistance tend to be in an opposite direction to that of the afferent arterioles, which vasoconstrict in response to increased NaCl at the macula densa.[215,219] The net result, however, is decreased glomerular blood flow, decreased glomerular hydraulic pressure, and a reduction in SNGFR.

There are many additional paracrine agents produced and secreted by macula densa cells, as shown in Fig. 3.20. These include metabolites of the arachidonic acid cascade, including prostaglandins PGE2 and PGI2, and other products of the cyclooxygenase pathway, products of the cytochrome P450 pathway, including epoxygenases and the cytochrome P450 4A HETEs.[35,164,220] Another very important regulating paracrine regulator is NO, which exerts vasodilatory responses when released by the macula densa cells. Under normal circumstances, when the NaCl concentration of tubular fluid is increased, there are increases in ATP release coupled with reductions in PGE2 formation until the ATP release reaches a plateau, and the PGE_2 release is markedly reduced. With further increases in [NaCl], NO release is augmented to counteract the effects of increased ATP.[35]

In addition to the paracrine factors released by macula densa cells, there are also many modulatory agents that influence the sensitivity of TGF responses. Ang II is one of the more important factors. TGF is blunted by Ang II antagonists and Ang II synthesis inhibitors, and TGF is markedly reduced in knockout mice lacking the AT1A Ang II receptors or angiotensin-converting enzyme (ACE).[221,222] Furthermore, systemic infusion of Ang II in ACE knockout mice restores TGF.[221,223–228] Ang II also enhances TGF via activation of AT1 receptors on the luminal membrane of the macula densa.[229] Acute inhibition of the AT1 receptor in normal mice reduces TGF responses and reduces autoregulatory efficiency.[224] Several studies have shown the interactions between adenosine and Ang in the TGF mechanism. In these studies, adenosine A_1 receptor antagonist administration results in decreased afferent arteriolar resistance and increased transcapillary hydraulic pressure differences (ΔP), whereas pretreatment with an angiotensin AT1 receptor antagonist prevented these changes.[230] Although it is known that Ang II is not the primary regulator of TGF, these results indicate that Ang II plays a prominent role in modulating tubuloglomerular feedback sensitivity, and that this response is mediated through the AT1 receptor.

Neuronal NO synthase (nNOS or NOS I) is present in macula densa cells.[231] NO derived from nNOS in the macula densa provides a vasodilatory influence on tubuloglomerular feedback, decreasing the amount of vasoconstriction of the afferent arteriole that otherwise would occur.[231,232] Increased distal sodium chloride delivery to the macula densa stimulates nNOS activity and also increases activity of the inducible form of cyclooxygenase (COX-2), which forms PGE2 and counteracts TGF-mediated constriction of the afferent arteriole.[231,232] Macula densa cell pH increases in response to increased luminal sodium concentration and may be related to the stimulation of nNOS.[233] Inhibition of macula densa guanylate cyclase increases the TGF response to high luminal [NaCl], further indicating the importance of NO in modulating TGF.[219] In an isolated perfused JGA preparation, microperfusion of the macula densa with an inhibitor of NO production led to constriction of the adjacent afferent arteriole.[234] When the macula densa was perfused with a solution low in Na concentration, however, the response was blocked, indicating that Na reabsorption is required.[234] Microperfusion of the macula densa with the precursor of NO, L-arginine, blunts TGF responses, especially in salt-depleted animals.[235–237] These results indicate that NO released from macula densa cells or endothelial cells causes afferent arteriolar vasodilation acutely or may blunt TGF responses. An increase in NO production may also inhibit renin release by increasing cyclic guanosine monophosphate (cGMP) in the granular cells of the afferent arteriole,[238] thereby accentuating its vasodilatory effects. When NO production was chronically blocked in knockout mice lacking nNOS, TGF in response to acute perturbations in distal sodium delivery was normal.[225] However, the presence of intact nNOS in the JGA is required for sodium chloride–dependent renin secretion.[225] The TGF system, which elicits vasoconstriction and a reduction in SNGFR in response to acute increases in sodium and solute delivery to the macula densa, appears to activate a vasodilatory response secondarily via NO release.[239] Stimulation of NO production in response to increased distal salt delivery under conditions of volume expansion would be advantageous by resetting TGF and limiting TGF-mediated vasoconstrictor responses.

The dynamics of the TGF mechanism can be temporally divided into two or more responses with different time constants. The initial rapid response occurs within a few seconds and elicits a rapid vasoconstriction and decrease in GFR and P_{GC} when sodium delivery to the macula densa cells is acutely increased. A second vasoconstrictor response occurs in seconds to minutes and changes the slope of the response to a slower time constant. This may be due to modulation of the initial response by some of the modulating agents mentioned. The rapid TGF system prevents large changes in GFR under conditions such as spontaneous fluctuations in blood pressure, thereby maintaining tight control of distal sodium delivery in the short term. Over the long term, renin secretion, controlled by the JGA in accordance with the requirements for sodium balance and the TGF system, resets to a new sodium delivery rate.[225] In the uninephrectomized rat with sustained elevations of the GFR, the TGF system appears to be reset.[240]

Connecting Tubule Glomerular Feedback Mechanism

There is also growing interest in a second feedback loop that links the connecting tubule, which has been shown to be in apposition to its own glomerulus and in close contact with the afferent arteriole.[241] In vitro perfusion of the connecting tubule has shown that increases in luminal NaCl elicit vasodilation of preconstricted afferent arterioles.[242] This action is opposite to the effect of macula densa TGF signaling, in which increases in NaCl cause vasoconstriction, raising the question of how these two opposing systems interact. Additional studies demonstrating that the addition of amiloride to the perfusion solution prevents the action of increased luminal Na^+, have suggested that the epithelial sodium channel mediates the connecting TGF (CTGF).[242] The afferent vasodilator effect of increased $E_{Na}C$ activity appears to be mediated by PGE2 acting on an EP4 receptor on the afferent arteriole.[243] An additional role of epoxyeicosatrienoic acid (EET) has also been suggested.[244] Furthermore, the presence of Ang II in the luminal fluid enhances the afferent arteriolar vasodilator effect caused by increases in luminal Na^+ concentration.[245] This CTGF mechanism is thought to mediate resetting of the macula densa TGF mechanism by partially reducing its sensitivity.[246] A modulating role of CTGF has been observed in experimental hypertension,[247] during high salt intake,[248] and in the renal vasodilatory response that occurs in the remaining kidney after unilateral nephrectomy.[249]

ENDOTHELIAL FACTORS AND GASEOUS TRANSMITTERS CONTROLLING RENAL HEMODYNAMICS AND GLOMERULAR FILTRATION RATE

A particularly intriguing and growing area of research involves the paracrine interactions between endothelial cells and the underlying smooth muscle cells. As shown in Fig. 3.21, endothelial cells respond to various physical and chemical stimuli, including pressure, flow, shear stress, and circumferential strain, as well as vasoactive factors normally present in the blood. Under normal conditions, an increase in shear stress may activate endothelial cells to produce NO and the prostanoids PGE2 and PGI2, which help adjust vascular tone to accommodate the increased load. However, during conditions of tissue injury or inflammation, the endothelial cells may also be stimulated to produce endothelin, thromboxane,

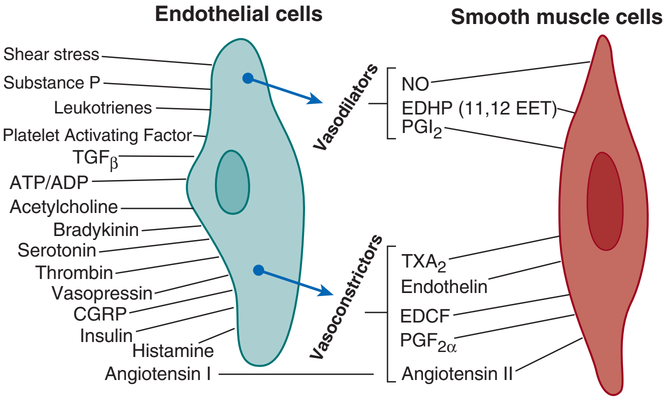

Fig. 3.21 Interaction of endothelial cells with smooth muscle or mesangial cells. As indicated in the text, several agents interact with endothelial cells to produce the vasodilator nitric oxide *(NO)*. Others yield vasoconstriction (see text and other sources). Angiotensin-converting enzyme converts angiotensin I to the potent vasoconstrictor angiotensin II. *ATP–ADP*, Adenosine trisphosphate–adenosine diphosphate; *CGRP,* calcitonin gene-related peptide; *EDCF,* endothelial-derived constricting factor; *EDHP,* endothelial-derived hyperpolarizing factor; *EET,* epoxyeicosatrienoic acid; *$PGF_{2\alpha}$,* prostaglandin $F_{2\alpha}$; *PGI_2,* prostaglandin I2; *TGF_β,* transforming growth factor beta; *TXA_2,* thromboxane A2. (Modified from Arendshorst W, Navar LG. Renal circulation and glomerular hemodyanamics. In: Schrier RW, Coffman TM, Falk RJ, Molitoris BA, Neilson EG, eds. *Schrier's diseases of the kidney*. Philadelphia: Lippincott Williams & Wilkins; 2013:74–131; Navar LG, Arendshorst WJ, Pallone TL, Inscho EW, Imig JD, Bell PD. The renal microcirculation. In: Tuma RF, Duran WN, Ley K, eds. *Handbook of physiology: Microcirculation.* Vol 2: Academic Press; 2008:550–683; Maddox, DA, Deen WL, and Brenner BM. Glomerular Filtration; *Handbook of Physiology: Section 8 Renal Physiology*. American Physiological Society, New York: Oxford University Press; 1992, 545–638; Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412.)

certain growth and profibrotic factors that elicit vasoconstriction, and/or additional paracrine factors associated with tissue injury and fibrosis. Several of these factors associated with regulation of the renal microcirculation are gaseous physiologic transmitters, called "gasotransmitters," which have been identified over the last 2 decades. NO is the first such gasotransmitter discovered, but carbon monoxide (CO) and hydrogen sulfide (H_2S) have also been shown to influence the renal microcirculation.

Nitric Oxide and Nitric Oxide Synthases

In 1980, Furchgott and Zawadzki[250] demonstrated that the vasodilatory action of acetylcholine requires the presence of an intact endothelium. Acetylcholine binds to receptors on endothelial cells, leading to the formation and release of an "endothelial-derived relaxing factor," now known to be NO.[251,252] Many cell types, including the endothelium, produce NO from the amino acid L-arginine[35,83,253] by a family of nitric oxide synthases (NOSs) that are present in many cell types, including vascular endothelial cells, macrophages, neurons, glomerular mesangial cells, macula densa, and renal tubular cells.[35,254–256] Three main NOS isoforms have been isolated. Neuronal NOS, also termed "NOS I" or "nNOS," and endothelial NOS, also called "NOS III" or "eNOS," are constitutively present in the kidney. A third NOS, iNOS or NOS II, is inducible, is expressed after transcriptional induction, and remains active for prolonged periods.[35,83] All three isoforms of NOS are found in the kidney. The arcuate and interlobular arteries, as well as the afferent and efferent arterioles, all produce NO, which regulates basal vascular tone, as indicated by the constriction that occurs in response to inhibition of endogenous NO production.[35,83]

Once released by the endothelium, NO diffuses into adjacent and downstream vascular smooth muscle cells,[257] where it stimulates the activity of soluble guanylate cyclase and increases cGMP formation.[83,258–263] cGMP reduces calcium influx, intracellular calcium release, and intracellular calcium concentration. This occurs, in part, through a cGMP-dependent protein kinase (PKG)-mediated phosphorylation of targets, which include inositol trisphosphate (IP3) receptors, calcium channels, and phospholipase A2,[264] thereby reducing the amount of free calcium available for contraction, hence promoting relaxation.[265]

In addition to stimulation by acetylcholine, NO formation in the vascular endothelium increases in response to bradykinin,[262,266–269] thrombin,[270] platelet-activating factor,[271] endothelin,[272] and calcitonin gene-related peptide.[267,273–276] Elevation of blood flow through vessels with intact endothelium or across cultured endothelial cells results in increased shear stress and increased NO release. Both the pulse frequency and pulse pressure modulate flow-induced NO release.[259,266,268,277–281] Elevated perfusion pressure and shear stress also increase NO release from afferent arterioles.[282]

NO plays a major role in modulation of renal hemodynamics, regulation of medullary perfusion, modulation of the sensitivity of the TGF mechanism, inhibition of tubular sodium reabsorption, modulation of renal sympathetic neural activity, and mediation of pressure natriuresis.[35,283–287] NO dominates integrated renal hyperemic responses to acetylcholine and bradykinin, and renal endothelium-dependent vasodilation is diminished in diabetes due to impaired NO function.[288] Pressure natriuresis in experimental models using stepwise increases of both renal perfusion pressure and medullary blood flow involve increased NO release, which can exert direct tubular effects to promote sodium and water excretion.[289,290] Tubular epithelial cells are capable of releasing NO but, during increased medullary flow, the vasa recta may be a primary source of the NO, as suggested by the fact that flow-dependent increases of NO also occur, even during microperfusion of isolated outer medullary vasa recta.[291]

There are important interactions among NO, Ang II, and renal nerves in the control of renal function and blood pressure.[292] Nonselective NOS inhibition using competitive inhibitors of NO results in decreases in RPF, increases in mean arterial blood pressure (AP), and generally a reduction in GFR.[165,293,294] These effects are largely prevented by the simultaneous administration of excess L-arginine, the NOS substrate.[293] Selective inhibition of neuronal NOS (nNOS or type I NOS), which is found in the thick ascending limb of the loop of Henle, the macula densa, and efferent arterioles,[256,285] decreases GFR without affecting blood pressure or RBF.[295] Because eNOS is found in the endothelium of renal blood vessels, including both the afferent and efferent arterioles and glomerular capillary endothelial cells,[256] differences in the effects of generalized NOS inhibition versus specific inhibition of nNOS on NO formation and RBF appear to be related to the distinct distribution of eNOS versus nNOS in the kidney. Both acute and chronic inhibition of

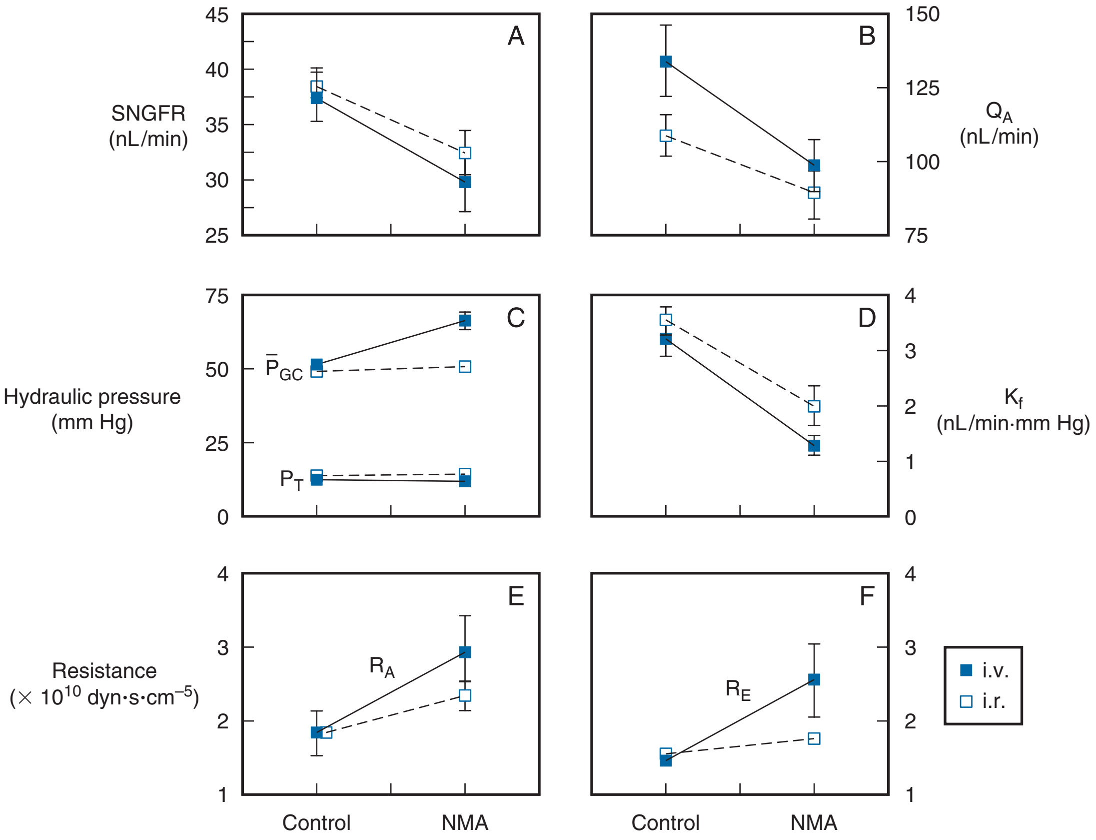

Fig. 3.22 (A–F) Role of nitric oxide in the control of glomerular filtration dynamics. Studies were performed in euvolemic Munich-Wistar rats receiving intravenous pressor doses of the nonselective nitric oxide synthase (NOS) blocker, *N*-monomethyl-L-arginine (*NMA;* i.v., *filled squares*) or nonpressor doses of NMA at the origin of the renal artery (i.r., *open squares*). K_f, Filtration coefficient; P_{GC}, pressure in the glomerular capillaries; P_T, pressure in the tubules; Q_A, glomerular plasma flow rate; R_A, preglomerular arteriolar resistance; R_E, efferent arteriolar resistance; *SNGFR,* single-nephron glomerular filtration rate. (Data [mean ± SE] obtained from Deng A, Baylis C: Locally produced NO controls preglomerular resistance and filtration coefficient. *Am J Physiol*. 1993;264:F212–F215.)

NO production result in systemic and glomerular capillary hypertension, an increase in preglomerular (R_A) and efferent arteriolar (R_E) resistance, a decrease in K_f, and decreases in both single-nephron plasma flow and GFR.[296–300]

As shown in Fig. 3.22, acute administration of pressor doses of a blocker of NO production results in a decline in SNGFR, Q_A, and K_f and increases in both preglomerular and efferent arteriolar resistances. Administration of nonpressor doses of the inhibitor of NO formation through the renal artery yielded an increase in preglomerular resistance and a decrease in SNGFR and K_f but no effect on efferent resistance[297] These studies suggest that the cortical afferent, but not efferent, arterioles are under tonic control by NO. However, others have found that the renal artery, arcuate and interlobular arteries, and afferent and efferent arterioles all produce NO and constrict in response to inhibition of endogenous NO production.[144,169,234,257,301–304] In agreement with this finding, investigators[31,302] have reported that NO dilates both efferent and afferent arterioles in perfused juxtamedullary nephrons. Interestingly, the modulatory influence of nNOS on afferent arteriolar tone is dependent on the maintenance of distal tubular fluid, indicative of a critical interaction with the TGF mechanism.[304]

Controversy exists regarding the role of the RAS in the genesis of the increase in vascular resistance that follows blockade of NOS. Studies of in vitro perfused nephrons[302] and of anesthetized rats in vivo[305] suggest that the increase in renal vascular resistance that follows NOS blockade is blunted when Ang II formation or receptor binding is blocked. NO inhibits renin release, whereas acute Ang II infusion increases cortical NOS activity and protein expression, and chronic Ang II infusion increases mRNA levels for eNOS and nNOS.[305,306] Ang II increases NO production in isolated perfused afferent arterioles via activation of the AT1 Ang II receptors.[307] In contrast, nonselective NOS inhibition increases renal oxygen consumption, independently of Ang II.[308] Additionally, inhibition of NOS in conscious rats had similar effects on renal hemodynamics in the intact and Ang II–blocked state.[309] This suggests that the vasoconstrictor response to NOS blockade is not mediated by Ang II. Further studies[310] have shown that when the Ang II levels are acutely raised by the infusion of Ang II, acute NO blockade amplifies the renal vasoconstrictor actions of Ang II. An observation in agreement with this finding is that intrarenal inhibition of NO enhances Ang II–induced afferent, but not efferent, arteriolar vasoconstriction.[144,311] In the juxtaglomerular nephron, however, blockade of nNOS enhanced efferent but not afferent arteriolar responsiveness to Ang II.[304] These data suggest that NO modulates the vasoconstrictor effects of Ang II on glomerular arterioles in vivo, perhaps blunting

Ang II's vasoconstrictor response in the afferent arteriole, with some results showing similar responses on the efferent arteriole.

Effects of Heme Oxygenase and Carbon Monoxide on Renal Function

Heme is degraded by heme oxygenase (HO) enzymes (HO-1 and HO-2) producing carbon monoxide (CO), biliverdin, and bilirubin and by the release of free iron.[2,35,312–315] Induction of HO-1 with hemin in anesthetized rats resulted in significant increases in RBF and GFR, a lower renal vascular resistance, and an increase in sodium excretion in untreated control rats without affecting blood pressure. Furthermore, autoregulatory responses to acute Ang II infusion were blunted, and these studies suggested a vasodilatory influence of HO-1 induction and hence CO production.[313] When a heme oxygenase inhibitor was administered either alone or to rats receiving an NOS inhibitor—N(ω)-nitro-L-arginine methyl ester (L-NAME)—for 4 days, blockade of HO in control rats decreased CO, HO-1 levels, urine volume, and sodium excretion, but did not affect AP, RBF, or GFR. In rats undergoing NOS inhibition with L-NAME, blockade of HO decreased endogenous CO and renal HO-1 levels, urine volume, and sodium excretion, but again had no effect on AP, RBF, or GFR. An increase in plasma renin activity was observed in untreated rats but not in L-NAME–treated rats, indicating that the effects on urine volume and sodium excretion are associated, even when NO was inhibited. This suggests that inhibition of HO promotes water and sodium excretion by a direct tubular action. independently of renal hemodynamics or the NO system.[314]

Inhibition of renal medullary HO activity and CO production decreases medullary blood flow and sodium excretion; the abundance of both the HO-1 and HO-2 isoforms of HO are higher in the inner medulla and lower in the cortex.[316] Inhibition of HO significantly reduces renal medullary cGMP concentrations when infused into the renal medullary interstitial space. These results suggest that both HO-1 and HO-2 are highly expressed in the renal medulla that HO, and its products play a key role in maintaining the constancy of blood flow to the renal medulla; cGMP may mediate the vasodilator effect of HO and CO in the renal medullary circulation.[316] In anesthetized rats, increases in renal perfusion pressure were found to increase CO concentrations in the renal medulla. An HO inhibitor reduced HO activity and pressure-dependent increases in CO in the medulla and blunted pressure natriuresis.[317] In conscious rats fed a normal-sodium diet, chronic infusion of an HO inhibitor into the renal medulla increased mean AP. When rats were placed on a high-salt diet, inhibition of HO activity caused a further increase in AP. Thus, renal medullary HO activity plays a key role in the control of arterial blood pressure and the control of pressure natriuresis.[317]

Using a CO-releasing molecule (CORM-A1), Ryan and coworkers demonstrated that increases in CO in the mouse increase RBF, with comparable results obtained from infusion of the vasodilator acetylcholine.[318] Pretreatment with an inhibitor of guanylate cyclase to block acetylcholine reduced the increase in RBF by CORM-A1. In isolated vasoconstricted renal interlobular arteries, CORM-A1–induced vasodilation was attenuated with the guanylate cyclase inhibitor, as observed in vivo. Inhibition of calcium-activated potassium channels (K_{Ca}) with iberiotoxin completely blocked CORM-A1 vasodilation. Thus, CO released from CORM-AI increases RBF and decreases vascular resistance by activating guanylate cyclase and opening K_{Ca} channels.[318]

The vascular effects of HO may be related to CO synthesis and are affected by NO release linked to the HO-CO system. Administration of a CO donor into the renal artery of rats increased RBF, GFR urinary cGMP excretion, and blood carboxyhemoglobin levels.[319] Inhibition of HO induced acute renal failure, with decreases in RBF, GFR, and cGMP excretion. These effects were nearly eliminated by the addition of a CO donor, which also decreased renal cortical NO concentration, urinary excretion of nitrates and nitrites, and urinary cGMP excretion and increased blood carboxyhemoglobin levels. Inhibition of renal HO resulted in acute renal failure, characterized by a large drop (–77%) in RBF and GFR (–93%). Supplementing HO inhibition with CO donor administration reversed the effects of HO inhibition on RBF and GFR, suggesting that the deleterious effects of HO on RBF and GFR were caused by the inhibition of CO. HO inhibition also decreased cortical NO concentration and increased urinary nitrate and/or nitrite excretion of the HO-CO system, whereas a CO donor increased renal NO levels and decreased nitrate and nitrite excretion. These results have suggested that changes in NO release contribute to the renal effects of the HO-CO system.[319]

Administration of heme decreases vascular resistance and increases RBF and sodium excretion, excretion of 6-keto-PGF1α, and the concentration of CO in renal cortical microdialysate. Pretreatment with an inhibitor of HO blunted heme-induced renal vasodilation and increased RBF. Pretreatment with sodium meclofenamate blunted the renal vasodilatory effect of heme, suggesting that heme-induced renal vasodilation is cyclooxygenase-dependent, yielding increased synthesis of PGI2.

Hydrogen Sulfide

A growing body of evidence has shown that H_2S, an endogenous bioactive gas synthesized in nearly all organs, plays an important role in the regulation of kidney function. H_2S generation by kidney cells is reduced in acute and chronic disease states, and H_2S donors ameliorate injury[320] but, under some conditions, H_2S may lead to kidney injury.[321] H_2S is produced by cystathionine beta-synthase (CBS) and cystathionine gamma-lyase (CGL) by the transsulfuration of homocysteine.[322,323] Incubation of renal tissue homogenates with L-cysteine as a substrate yields H_2S. This response was prevented by inhibitors of both CBS and CGL in combination, whereas either inhibitor alone induced only a small decrease in H_2S.[322]

H_2S plays a role in renal hemodynamics, as shown by the effects of intrarenal infusion of a donor of H_2S (NaHS), which increased renal blood flow and GFR, as well as urinary sodium and potassium excretion. Infusion of L-cysteine also increases endogenous H_2S production.[322] Simultaneous infusion of both an inhibitor of CBS and CGL to decrease H_2S production decreased GFR and sodium and potassium excretion, but either inhibitor alone did not affect these renal functions.[322] H_2S causes endothelium-dependent/cytochrome P450– dependent vasodilation and vascular smooth muscle hyperpolarization of small arterial vessels, increasing ryanodine-mediated Ca^{2+} release through the activation of large conductance calcium activated potassium channels, causing membrane hyperpolarization and vasodilation.[324] The involvement of cGMP-dependent protein kinase-I in H_2S-induced vasorelaxation was shown in preconstricted aortic rings, with or without intact endothelium.[325] Treatment of the aortic rings with NaHS (an H_2S donor) indicated that a cGMP-dependent protein kinase (PKG) is activated by H_2S. Incubation with a PKG-1 inhibitor blocked NaHS-stimulated vasodilation.[325] NaHS-induced vasorelaxation was reduced by removal of the endothelium and by inhibitors of either NO or cGMP production.[326] H_2S also relaxes smooth muscle by activating ATP-sensitive potassium channels.[327]

Increases in TGF-β1 are associated with the development of tubulointerstitial fibrosis and glomerular sclerosis in other renal diseases and are mediated, at least in part, by Ang II. Ang II- and transforming growth factor beta 1 (TGF-β1)–induced renal tubular epithelial-mesenchymal transition (EMT) plays a pivotal role leading to renal sclerosis. One study has demonstrated that Ang II stimulates EMT in renal tubular epithelial cells by increasing the level of α-smooth actin and decreasing E-cadherin.[328] This effect was blocked by a TGF-β receptor kinase inhibitor. Ang II stimulated TGF-β activation and exogenous TGF-β1–induced EMT. The H_2S donor NaHS blocked the promotion of EMT by Ang II and TGF-β1 and reduced TGF-β activity. H_2S cleaves the disulfide bond in dimeric active TGF-β1, promoting the formation of inactive TGF- β1 monomer.[328] These results have suggested the potential to treat sclerosis in the kidney (both glomerular sclerosis and tubulointerstitial fibrosis), as well as other diseases associated with fibrosis and TGF-β1 (e.g., pulmonary fibrosis) by stimulating H_2S or using another means to form the inactive TGF-β1 monomer.

REACTIVE OXYGEN SPECIES

Reactive oxygen species (ROS) are products of a one-electron reduction of dioxygen (oxygen gas, O_2) to form the anionic form of O_2, superoxide, O_2^- . Superoxide is generated by the catalytic actions of oxidative enzymes, such as nicotinamide adenine dinucleotide (NADH)/reduced NADPH oxidase (NOX) and cytochrome oxidase.[2,35] Superoxide is toxic and organisms creating O_2^- have developed isoforms of superoxide dismutase (SOD), which catalyzes the conversion of superoxide to hydrogen peroxide (H_2O_2). Other enzymes degrade H_2O_2, protecting against the deleterious actions of this ROS.[2,35] ROS are produced in the kidney by endothelial cells, epithelial cells, vascular smooth muscle cells, mesangial cells, podocytes, and other cell types and have effects on the kidney vasculature.[329]

Normally, oxidative and antioxidative enzymes in the kidney yield a balanced production of NO and the superoxide anion. ROS are formed in the arteries, arterioles, glomeruli, and juxtaglomerular apparatus and other nephron segments, and the oxidases NOX 1, 2, and 4, NOS, and COX are also found in the kidney.[2,35] In the renal vasculature, NOX 1 and NOX 2 produce O_2^-, whereas NOX 4 in epithelial cells produces H_2O_2. Superoxide dismutase converts superoxide to H_2O_2; catalase and glutathione peroxidase degrade H_2O_2.

Stimulants of O_2^- production include Ang II,[330–332] endothelin,[333] norepinephrine,[2] TGF-β1,[334] and stretch of vascular walls by increased intravascular pressure.[335,336] Ang II activates NADPH oxidases in afferent arterioles to form O_2^-, leading to calcium release from intracellular stores by activation of the inositol trisphosphate receptor (IP3R).[330] Superoxide also enhances calcium entry pathways into afferent arterioles through L-type channels by membrane depolarization.[337] NOX2 NADPH oxidase is activated by Ang II, promoting the generation of ROS, which scavenge NO and cause subsequent NO-deficiency.[331] O_2^- and H_2O_2 activate different signaling pathways in vascular smooth muscle cells linked to discrete membrane channels, with opposite effects on membrane potential and voltage-operated Ca^{2+} channels, and therefore have opposite effects on myogenic contractions.[335]

Increased perfusion pressure and Ang II increase O_2^- production and increase myogenic responses in arterioles of superoxide gene-deleted mice compared with controls.[338] In the macula densa of blood-perfused juxtamedullary nephrons, O_2^- was undetectable in control normotensive mice, but was markedly elevated in Ang II-induced hypertensive animals. NO was found in the macula densa of control mice but was undetectable in the macula densa of hypertensive animals.[339] These data suggest that under normal conditions, NO generated in the macula densa reduces tubuloglomerular feedback sensitivity, but in Ang II-induced hypertension, the TGF response is augmented by O_2^- generated by the macula densa.[339] Increased perfusion pressure causes vascular O_2^- production from NADPH oxidase, enhancing myogenic contractions independently of NO, whereas H_2O_2 impairs pressure-induced contractions but is not involved in the normal myogenic response.[340]

In the remnant kidney model, COX-2 is induced, leading to activation of thromboxane prostanoid receptors (TP-Rs), which enhance ET-1, ROS generation, and contractions.[333] Compared with controls, diabetic mouse afferent arterioles also have increased production of O_2^- and H_2O_2 and enhanced responses to ET-1. These responses are accompanied by reduced protein expression and activities for catalase and superoxide dismutase-2. ET-1 further increases O_2^-, whereas H_2O_2 is unchanged by ET-1. Increased ROS in diabetes (notably H_2O_2) contributes to the enhanced arteriolar responses to ET-1.[341] TGF-β1, a growth factor involved in glomerular and tubular injury in diabetes, blocks autoregulation of afferent arterioles, an effect prevented with an ROS scavenger or an NADPH oxidase inhibitor. In smooth muscle cells, TGF-β1 stimulated ROS formation that was inhibited by NADPH oxidase inhibitors.[334]

In afferent arterioles from rats with spontaneous hypertension, pressure-induced increases in ROS were four times greater in SHR than in WKY rats. Both a scavenger of O_2^- and an NOX2-based (NADPH oxidase) inhibitor attenuated pressure-induced constriction in SHR vessels but not in WKY. Thus, NOX2-derived O_2^- may contribute to an enhanced myogenic response in SHR afferent arterioles.[342] Of note, arterioles from rats with ischemia-reperfusion injury had a 38% increase in H_2O_2, which could act to buffer the effect of Ang II and be a protective mechanism.[343]

Endothelin

Endothelin is a potent vasoconstrictor agent derived primarily from vascular endothelial cells.[344] There are three distinct genes for endothelin, each encoding distinct 21–amino acid isopeptides, ET-1, ET-2, and ET-3.[344–346] Proteolytic cleavage of a 212–amino acid preproendothelin by furin yields a 38- to 40-amino acid proendothelin, which in turn is cleaved by endothelin-converting enzyme to yield endothelin peptides.[347,348] ET-1, the primary endothelin produced in the kidney, is formed in arcuate arteries and veins, interlobular arteries, afferent and efferent arterioles, glomerular capillary endothelial cells, glomerular epithelial cells, and glomerular mesangial cells.[349–360] ET-1 acts in an autocrine or paracrine fashion, or both,[361] to alter a variety of biologic processes in these cells. Endothelins are potent vasoconstrictors, and the renal vasculature is highly sensitive to these agents.[362] Once released from endothelial cells, endothelins bind to specific receptors on vascular smooth muscle, the ET_A receptors that bind both ET-1 and ET-2.[361,363–366] ET_B receptors are expressed in the glomerulus on mesangial cells and podocytes and have equal affinity for ET-1, ET-2, and ET-3.[365,366] There are two subtypes of ET_B receptors, the ET_{B1} linked to vasodilation and the ET_{B2} linked to vasoconstriction.[367] An endothelin-specific protease modulates endothelin levels in the kidney.[368–370]

Endothelin production is stimulated by physical factors, including shear stress and vascular stretch.[371,372] A variety of hormones, growth factors, and vasoactive peptides increase endothelin production, including TGF-β, platelet-derived growth factor, tumor necrosis factor-α, Ang II, arginine vasopressin, insulin, bradykinin, thromboxane A2, and thrombin.[349,353,355,359,373–376] Endothelin production is inhibited by atrial and brain natriuretic peptides acting through a cGMP-dependent process[368,377] and by factors that increase intracellular cAMP and protein kinase A activation, such as β-adrenergic agonists.[355] ATP-binding renal purinergic (P2) receptors may regulate ET-1 production.[378]

Intravenous infusion of ET-1 induces a marked, prolonged pressor response[344,379] accompanied by increases in preglomerular and efferent arteriolar resistances and a decrease in RBF and GFR, without changes in fractional Na excretion.[379] As shown in Fig. 3.23, infusion of subpressor doses of ET-1 decreases SNGFR, Q_A, and whole-kidney RBF and GFR,[380–384] accompanied by increases in preglomerular and postglomerular resistances and filtration fraction.[380,382,385] Vasoconstriction of afferent and efferent arterioles by endothelin has been confirmed in the split hydronephrotic rat kidney preparation[386,387] and in isolated perfused arterioles.[301,388,389] Endothelin also causes mesangial cell contraction.[346,390] The vasoconstrictor effects of the endothelins can be modulated by several factors,[364,391] including NO,[301,392] bradykinin,[393] prostaglandin E2,[394] and prostacyclin.[394,395] The endothelin pathways demonstrate significant sexual dimorphisms that may affect the progression of renal disease and treatment choices.[396]

The ET_A and ET_B receptors have been cloned and characterized.[366,397,398] ET_A receptors are abundant on vascular smooth muscle, have a high affinity for ET-1, and play a prominent role in the pressor response to endothelin.[399] ET_B receptors are present on endothelial cells, where they may mediate NO release and relaxation.[398] Both ET_A and ET_B receptors are expressed in the media of interlobular arteries and afferent and efferent arterioles. Only ET_A receptors are present on vascular smooth muscle cells of interlobar and arcuate arteries.[400] There is strong labeling of ET_B receptors on peritubular and glomerular capillaries, as well as the vasa

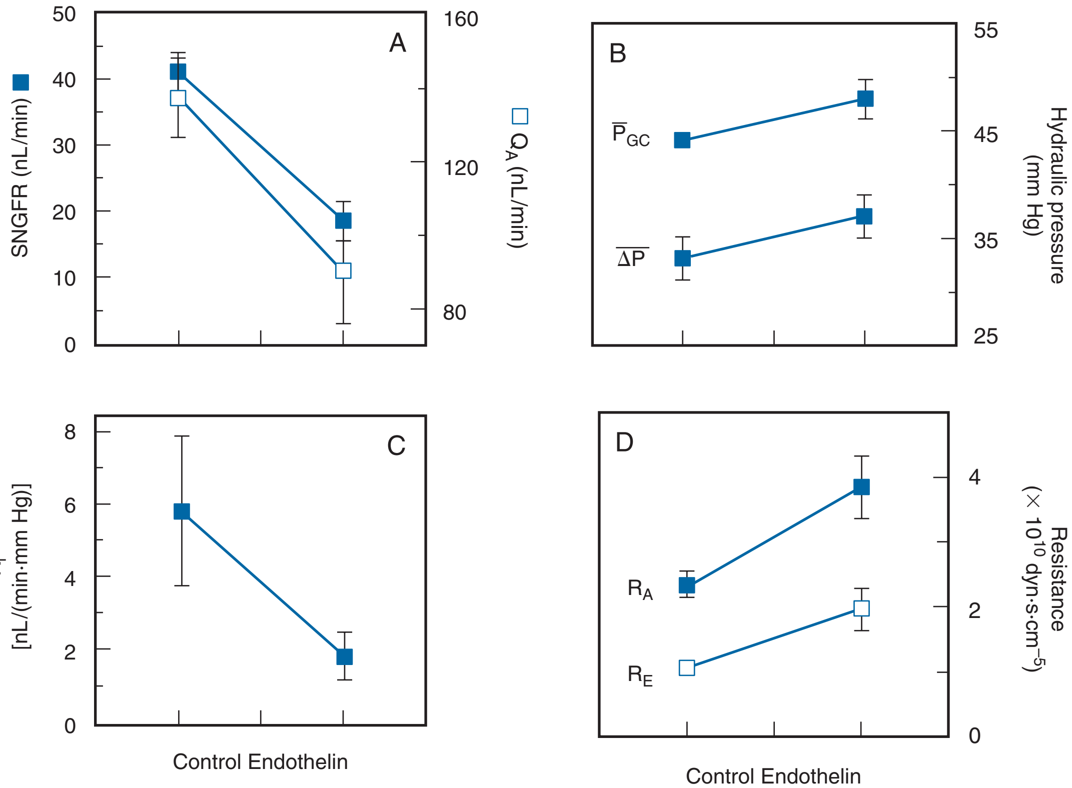

Fig. 3.23 (A–D) Effects of intravenous administration of endothelin (subpressor dose) on glomerular dynamics. K_f, Filtration coefficient; ΔP, mean transcapillary pressure gradient; P_{GC}, mean glomerular capillary pressure; Q_A, single-nephron glomerular plasma flow; R_A, afferent arteriolar resistance; R_E, efferent arteriolar resistance; *SNGFR,* single-nephron glomerular filtration rate. (Data [mean ± SE] obtained in Munich-Wistar rats from Badr KF, et al. Mesangial cell, glomerular and renal vascular responses to endothelin in the rat kidney. Elucidation of signal transduction pathways. *J Clin Invest*. 1989;83(1):336–342.)

recta endothelium.[400] ET_A receptors are evident on glomerular mesangial cells and pericytes of descending vasa recta bundles.[400] Endogenous endothelin may actually dilate the afferent arteriole and lower K_f via ET_B receptors.[401]

ET_A receptor antagonists may be useful for patients with diabetic nephropathy.[402] These ET_A receptor antagonists have been shown to reduce albuminuria in diabetic nephropathy, although the albuminuria returns on cessation of the drug. It has been used alone and in conjunction with renin-angiotensin blockade.[403,404] However, ET_A receptor antagonists have been associated with edema and heart failure, so patients already manifesting these conditions should be excluded from this type of medication.[405]

Endothelin stimulates the production of vasodilatory prostaglandins,[383,392,395,406,407] yielding a feedback loop to dampen the vasoconstrictor effects of endothelin. ET-1, ET-2, and ET-3 also stimulate NO production in the arterioles and glomerular mesangium via activation of the ET_B receptor.[270,272,301,392,408] Resistance in the renal and systemic vasculature is markedly increased during inhibition of NO production. There is a dynamic interrelationship between NO and endothelin effects, so that ET_A blockade or inhibition of endothelin-converting enzyme leads to increased renal resistance caused by NO inhibition.[409,410] The vasoconstrictive effects of Ang II may be mediated, in part, by stimulation of ET-1 production, which acts on ET_A receptors to produce vasoconstriction.[373,376] Chronic administration of Ang II reduces renal blood flow, an effect reduced by a mixed ET_A-ET_B receptor antagonist, suggesting that endothelin contributes to the renal vasoconstrictive effects of Ang II.[373]

Blood flow through the renal medulla is influenced by ET-1 as well as by Ang II, norepinephrine, nitric oxide, and vasodilatory prostaglandins. Medullary vasodilation measured by laser Doppler techniques was seen to occur at low doses of endothelin when cortical blood flow was decreased.[411] An ET_A receptor antagonist blocked cortical vasoconstriction by ET-1 but failed to prevent medullary dilation. The endothelin-induced medullary vasodilation was blocked by an $ET_{A/B}$ receptor antagonist and was mimicked by an ET_B receptor agonist. Inhibition of NO completely blocked the endothelin-induced vasodilation of medullary blood flow, and inhibition of prostaglandins attenuated the response. These results have indicated that endothelin causes cortical vasoconstriction mediated by ET_A receptors, whereas activation of ET_B receptors causes medullary vasodilation mediated by the release of NO.[411]

Medullary blood flow occurs through vasa recta capillaries, and most regions of the vasa recta are covered by pericytes capable of vasoconstriction. The role of these pericytes in controlling blood flow through the vasa recta has been examined with confocal microscopy and pericyte-mediated vasoconstriction and vasodilation were visualized. Ang II, endothelin, and norepinephrine all caused vasoconstriction at pericyte locations.[412] These effects were attenuated by an NO donor and enhanced with inhibitors of NO production or inhibition of prostaglandin release by nonselective cyclooxygenase inhibition with indomethacin. Because of the narrow diameter of the vasa recta (normally, ~10 μm), constriction of pericytes can cause impairment of the movement of red cells and hence blood flow through the medulla. These results suggest an important role for pericytes in the control of the medullary circulation.[412]

ROLE OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM IN THE CONTROL OF RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

The RAS exerts major autocrine, paracrine, and endocrine functions regulating RBF and GFR. As presented in detail in several recent reviews,[2,35,55,83] renin is a proteolytic enzyme synthesized, stored, and released from the kidney, and also synthesized in the liver. In the kidney, it is synthesized and secreted primarily by the granulated epithelioid cells of the JGA adjacent to the terminal portion of the afferent arteriole; renin is also formed in the proximal tubules and in the principal cells of the connecting tubule and collecting duct.[2,35,413–415] Renin release from the kidneys is stimulated by a decrease in sodium intake, a reduction in extracellular fluid volume and blood volume, a decrease in arterial blood pressure, and increased sympathetic nerve activity.[416] Renin cleaves a decapeptide, angiotensin I (Ang I) from angiotensinogen, a glycoprotein formed in the proximal tubules of the kidney[415] and the liver. Circulating angiotensinogen is present in the α_2-globulin fraction of plasma. Subsequent conversion of Ang I by angiotensin-converting enzyme (ACE), identical to kininase II, yields the octapeptide Ang II. ACE is present in many tissues, including lung. In the kidney, it is bound to the luminal sides of endothelial cells of blood vessels and tubular cells, including the brush border of the proximal tubule. All components needed for the production and degradation (the latter by angiotensinase A) of Ang II are present in the immediate region of the juxtaglomerular region of the nephron, allowing direct local regulation of glomerular blood flow and filtration rate.[35,55]

Ang II is a potent vasoconstrictor, and numerous studies have demonstrated that preglomerular vessels, including the arcuate arteries, interlobular arteries, and afferent arterioles, as well as the postglomerular efferent arterioles, constrict in response to exogenous and endogenous Ang II.[144,174,303,389,417,418]

Some studies have indicated that efferent arterioles have a greater sensitivity to Ang II, whereas others have shown similar effects on both afferent and efferent arterioles.[35,144,303,389,418] Fig. 3.24 shows the effects of Ang II on diameters in these vessels. As shown in Fig. 3.24, both L-type and T-type Ca^{++} channels are involved in the afferent arteriolar responses to Ang II while the T-type Ca^{++} channels are predominate at the efferent arterioles.[522,523] In addition to constricting vascular smooth muscle cells, Ang II increases myocardial contractility, stimulates aldosterone release, increases salt appetite and thirst, and helps regulate sodium transport by the kidney tubules and intestine.[419] The overall effect of Ang II is to minimize renal fluid and sodium losses and maintain extracellular fluid volume (ECFV) and arterial blood pressure.[416]

There are two major classes of Ang II receptors, AT1 and AT2, but the hypertensinogenic and vasoconstrictive actions of Ang II are primarily due to the AT1 receptor, which is widely distributed throughout all segments of the renal microvasculature in the cortical and medullary circulatory beds and are also present in the glomeruli, including mesangial cells, glomerular capillary endothelial cells, and podocytes.[35,83,420] In rodents, afferent arterioles have both AT1a and AT1b receptors, whereas efferent arterioles have only AT1a receptors.[420,421] AT1 receptors are also present in the proximal and distal tubules, loop of Henle and macula densa

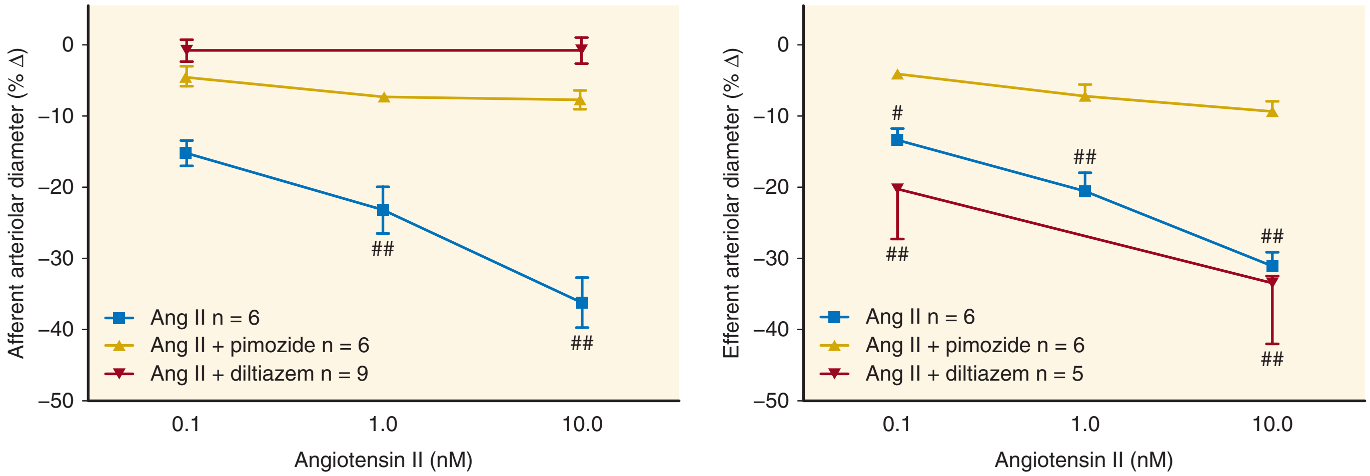

Fig. 3.24 Role of L- and T-type Ca^{2+} channels in angiotensin II (Ang II)–mediated afferent and efferent arteriolar vasoconstriction. Ang II constricts both afferent and efferent arterioles. The vasoconstrictor effects of Ang II on afferent arterioles are blocked by both L-type and T-type Ca^{2+} channel blockers. In contrast, the efferent vasoconstrictor effects of Ang II are not blocked by L-type Ca^{2+} channel blockers, but are blocked by T-type Ca^{2+} channel blockers.[35,522,523] (From Carmines PK, Navar LG. Disparate effects of Ca channel blockade on afferent and efferent arteriolar responses to Ang II. *Am J Physiol Renal Physiol*. 1989;256(6 Pt 2):F1015–F1020; Feng MG, Navar LG. Angiotensin II-mediated constriction of afferent and efferent arterioles involves T-type Ca^{2+} channel activation. *Am J Nephrol.* 2004;24(6):641–648.)

cells, cortical and medullary collecting ducts, glomerular podocytes, and mesangial cells.[419]

The vasoconstrictor effects of Ang II are blunted by the endogenous production of vasodilators including NO, cyclooxygenase, and cytochrome P450 epoxygenase metabolites in the afferent but not the efferent arterioles.[140,144,303,311,422–425] Ang II-simulated release of NO in the afferent arterioles occurs through activation of the AT1 receptors.[307,426] Ang II increases the production of prostaglandins (both PGE2 and PGI2) in afferent arteriolar smooth muscle cells, and PGE2, PGI2, and cAMP all blunt Ang II-induced calcium entry into these cells,[423] potentially explaining, at least in part, the different effects of Ang II on vasoconstriction of the afferent and efferent arterioles.[422,423] In contrast to effects on the afferent arteriole, PGE2 had no effect on Ang II-induced vasoconstriction of the efferent arteriole.[140] The effects of PGE2 on Ang II-induced vasoconstriction of the afferent arteriole are concentration-dependent, with low concentrations acting as a vasodilator via interaction with prostaglandin EP4 receptors and high concentrations of PGE2 acting on prostaglandin EP3 receptors to restore the Ang II effects in that segment.[140] Ang II infusion alone produces decreases in renal blood flow, with lesser effects on GFR resulting in an increase in filtration fraction.[35] However, when combined with cyclooxygenase inhibition, Ang II causes marked reductions in SNGFR and Q_A, suggesting an important role for endogenous vasodilatory prostaglandins in ameliorating the vasoconstrictor effects of Ang II.[427] Because Ang II increases renal production of vasodilatory prostaglandins, this serves as a feedback loop to modulate the vasoconstrictor effects on Ang II under chronic conditions when the RAAS is stimulated.[83]

Ang II decreases K_f[427] and contracts mesangial cells.[428] One possible cause for the changes in K_f is that contraction of the mesangial cells reduces effective filtration area by blocking flow through some glomerular capillaries, but no direct evidence has been obtained to support this hypothesis. Alternatively, Ang II might decrease hydraulic conductivity, rather than or as well as reducing the surface area available for filtration, thereby reducing K_f.[429] Glomerular epithelial cells possess both AT1 and AT2 receptors and respond to Ang II by increasing cAMP production, suggesting a possible role for these cells in reducing K_f.[430] Alterations in epithelial structure or the size of the filtration slits, however, have not been detected following infusion of Ang II at a dose sufficient to decrease GFR and K_f.[431]

The vasoconstrictive effect of Ang II on glomerular mesangial cells is markedly reduced by NO. Release of NO from endothelial cells stimulated by bradykinin increases cGMP production in co-incubated mesangial cells. Comparable results were obtained when mesangial cells were incubated with NO alone.[262] Ang II alone caused constriction of the mesangial cells, but this effect was largely eliminated when the cells were co-incubated with both Ang II and NO. These studies suggest that local NO production from endothelial cells can modify the effects of Ang II on glomerular mesangial cells.[262] Glomerular epithelial cells contain Ang II receptors and, together with mesangial cells, may play an important role in Ang II-mediated control of the glomerular filtration barrier.[430]

Blockade of the AT1 receptors caused a dose-dependent dilation when Ang II was added to the lumen or bath of isolated afferent arterioles preconstricted by norepinephrine. This effect was blocked by pretreatment with an AT2 receptor antagonist, suggesting that activation of the AT2 causes vasodilation of afferent arterioles. Disruption of the endothelium or simultaneous inhibition of the cytochrome P450 pathway also abolished the vasodilation by Ang II. Thus, activation of AT2 receptors may cause endothelium-dependent vasodilation via a cytochrome P450 pathway, counteracting the vasoconstrictor effects of Ang II at AT1 receptors.[432,433]

As indicated, Ang II produces prohypertensive and renal vasoconstrictor effects via the activation of AT1 receptors, whereas activation of AT2 receptors results in modest vasodilation.[35,433] Several metabolic fragments of the octapeptide

Ang II, once believed to be inactive, have now been shown to have physiologic effects on the kidney, often opposing the actions of Ang II. The related peptide, Ang 1-7, has been shown to induce vasodilation of preconstricted renal arterioles.[434] These effects of Ang 1-7 occur independently of binding to AT1 or AT2 receptors and appear to involve activation of the G protein–coupled Mas receptor,[435] which has been shown to be expressed on the afferent arteriole.[436] In addition, an isoform of ACE, known as "ACE2,"[437,438] is involved in the formation of Ang 1-7.[439] Ang 1-7, ACE2, and the Mas receptor have all been detected in the kidney. The balance between opposing actions of the vasoconstrictor peptide, Ang II, and the vasodilator peptide, Ang 1-7, may be influenced by the ratio of ACE to ACE2 and AT1 to Mas receptor content in specific vascular regions (and tubular segments) of the kidney. Cardiovascular and renal diseases may involve an imbalance of these peptides, enzymes, or receptors.[436]

Additional mechanisms involved in Ang II-induced hypertension have been proposed. The principal cells of the connecting tubule and collecting duct express prorenin receptors and produce renin, which, coupled with delivery of angiotensinogen from the proximal tubule,[414] allows for the local formation of Ang I.[2,413,433] Luminal ACE in the collecting duct is then available to produce Ang II, which in turn can increase sodium reabsorption by enhancing activity of $E_{Na}C$ and other transporters in the collecting duct, an effect that may be important in the progression of diabetes and Ang II-induced hypertension.[413,414,433,440]

ROLE OF ANGIOTENSIN II IN GLOMERULAR AND TISSUE INJURY

Chronic renal failure is characterized by a progressive decline in renal function, with glomerular and tubular injury and systemic hypertension. An important demonstration of an elevation of glomerular hydraulic pressure in kidney disease was in a model of nephrotoxic serum nephritis (NSN),[441] where Ang II is thought to have a key role in the renal complications that develop. Intravenous infusion of Ang II increases P_{GC} and ΔP in rats with reduced renal mass.[442] In the remnant kidney model of renal failure, inhibition of Ang II production by an ACE inhibitor lowered P_{GC} and ΔP toward normal and prevented proteinuria and glomerular injury.[443] Over the last 30 years, the efficacy of blockade of the RAAS with ACE inhibitors and/or Ang II type 1 receptor antagonists (angiotensin receptor blockers [ARBs]) in limiting or slowing the progression of some types of kidney disease has been established.[444] ACE inhibitors and ARBs also are effective in the treatment of spontaneous glomerulosclerosis associated with aging.[445,446] Both ACE inhibitors and ARBs prevented hypertension, limited proteinuria, and ameliorated sclerosis in several experimental models of glomerulonephritis.[447] In one study, ACE inhibition was suggested to promote regeneration of new renal tissue.[448] Similarly, in the remnant kidney model of renal disease, there is evidence that chronic treatment with an ACE inhibitor causes reversal of renal injury.[449]

Accumulation of extracellular matrix proteins, including collagen and fibronectin, leads to glomerular and tubular injury.[450] Enhanced production of TGF-β1 in the remnant kidney increases matrix formation by stimulating the production of TGF-β1. Administration of a neutralizing antibody or an Ang II receptor antagonist reduces plasma TGF-β1 levels and blood pressure toward normal.[451] ACE inhibition in the obese Zucker rat has significantly reduced proteinuria compared with untreated obese animals, downregulated expression of nephrin, and decreased expression of TGB-β1, collagen IV, and fibronectin.[452] Similarly, in an animal model of myocardial infarction (MI), rats with MI had significant renal injury, and treatment by ACE inhibition exerted beneficial effects.[453]

Ang II induces upregulation and activation of a transcription factor called "sterol-responsive element-binding protein" (SREBP-1), leading to upregulation of TGF-β1 and increased production of matrix collagen and fibronectin.[450] An ARB prevented the increase in TGF-β1 formation and the induction of SREBP-1. Inhibition of SREBP-1 prevented Ang II-induced glomerular upregulation of TGF-β1 and matrix synthesis, demonstrating a significant role of the transcription factor in glomerular injury and indicating that TGF-β1 plays a role in the stimulation of matrix production.[450] SREBP-1 upregulation is observed in diabetic kidneys; SREBP-1 is activated by high glucose levels and mediates profibrogenic responses in primary rat mesangial cells.[454] In these cells, high glucose levels activated SREBP-1, which binds to the TGF-β1 promoter, resulting in TGF-β1 upregulation.[454]

Treatment with an ARB in rats with NSN protected against interstitial inflammation, tubular degeneration, and segmental glomerulosclerosis.[455] NSN rats were treated with pirfenidone (an antifibrotic, antiinflammatory compound first used to treat pulmonary fibrosis) or with pirfenidone plus an ARB. Pirfenidone alone yielded a decrease in proteinuria and the ARB a slightly greater decrease; the two together had additive effects on proteinuria, interstitial inflammation, tubular degeneration, and segmental glomerulosclerosis.[455] Transgenic mice overexpressing active renin from the liver develop progressive pulmonary fibrosis, much like that seen in humans with pulmonary fibrosis. There was increased extracellular matrix deposition of fibronectin and collagens and increased production of TGF-β1 and connective tissue growth factor (CTGF).[456] Two weeks of treatment with a renin inhibitor or an ARB reduced production of TGF-β1 and CTGF and decreased deposition of matrix proteins.

ARACHIDONIC ACID METABOLITES REGULATING RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

The processing of the essential polyunsaturated fatty acid, linoleic acid, in the liver yields the polyunsaturated fatty acid arachidonic acid, which is stored in membrane phospholipids. Biologically active eicosanoids, C20 metabolites of arachidonic acid, are produced by three primary enzymatic pathways, including the following: cyclooxygenase-generating prostaglandins (PGs) and thromboxanes (TBXs); lipoxygenase (OX) yielding leukotrienes (LTs) and HETEs; and cytochrome P450 pathways synthesizing HETEs and EETs. Arachidonic acid (AA) is released from cell membranes, predominantly by phospholipase A2 (PLA2).[457–461] These eicosanoids regulate the renal microcirculation, in part by activating G protein–coupled receptors in the endothelium and vascular smooth muscles.[458,462]

Prostaglandins

Following the interaction of various hormones and vasoactive substances with their membrane receptors, PLA2 is activated, resulting in the release of AA from the cell membranes. This allows the enzymatic action of cyclooxygenase to process AA into prostaglandins—PGG2 and subsequently PGH2. PGH2 is then converted into a number of biologically active prostaglandins, including PGE2, prostacyclin (PGI2), PGF2α, PGE1, PGD2, and thromboxane (TxA2). Prostaglandins play important roles in healthy and diseased kidneys.[93,463–466] PGE2 receptors are the most abundant prostaglandin receptors in the kidney. The effects of PGE2 on the kidneys depend on the location of the receptor subtype and interaction with other active compounds. PGE2 activates at least four receptors, EP1 to EP4. Prostaglandin EP1 receptor activation results in contractile effects in smooth muscle through the Gq alpha protein linkage, whereas EP2 and EP4 are relaxant via the Gs alpha subunit. EP3 is an inhibitory receptor (Gi alpha), decreasing cAMP, which causes vasoconstriction.[35,463,464]

In the kidneys, PGE1, PGI2, and PGE2 are vasodilator prostaglandins that generally increase renal plasma flow yet produce little or no increase in GFR and SNGFR, in part due to a decline in K_f.[93,464–466] TxA2 generally results in contraction of the glomerular mesangial cells via the Gq alpha subunit. During blockade of endogenous prostaglandin production, infusion of PGE2 or PGI2 decreases SNGFR and Q_A, accompanied by an increase in renal vascular resistance (particularly R_E), increases in P_{GC} and ΔP, and a decline in K_f.[467] Additional blockade of Ang II receptors during cyclooxygenase inhibition results in vasodilation in response to PGE2 or PGI2, resulting in a return of SNGFR and Q_A equal to or more than control values, a fall in P_{GC} below control values, and a return of K_f to normal.[467] Thus, the renal vasoconstriction induced by exogenous PGE2 or PGI2 appears to be mediated by induction of renin and Ang II production.

Vasodilation at the whole-kidney level resulting from PGI2 infusion during cyclooxygenase and Ang II inhibition has not always been observed.[468] Topical application (but not luminal) of PGE2 to the afferent arteriole increased the vasoconstrictive effect of Ang II and norepinephrine, whereas PGI2 only attenuated norepinephrine-induced vasoconstriction.[469] PGE2 also constricted interlobular arteries but neither prostaglandin produced vasodilation of vessels preconstricted by Ang II.[469] Indomethacin alone induced vasoconstriction of preglomerular and postglomerular resistance vessels of superficial and juxtamedullary nephrons, indicating that vasodilatory prostaglandins normally modulate endogenous vasoconstrictors.[470] However, there appears to be gender-dependent differences in mechanisms, because indomethacin treatment results in net renal vasodilation in female rats.[471] The combination of cyclooxygenase inhibition with an ACE inhibitor caused vasodilation of preglomerular but not postglomerular vessels of the cortical nephrons due to the effects of continued NO production in the preglomerular vessels.[470] Additionally, the PGE2 EP3 receptor regulates COX-2 through a negative feedback loop in the thick ascending limb of the loop of Henle.[472] These data, taken together, indicate that differences in the response to vasoactive prostaglandins between superficial and deep nephrons may be due to prostaglandin receptor subtypes and cyclooxygenase activity.

Leukotrienes and Lipoxins

Leukotrienes are a class of lipid products formed from AA via the lipoxygenase pathway. Leukotrienes that affect glomerular filtration and renal blood flow are leukotriene C4 (LTC4), leukotriene D4 (LTD4), and leukotriene B4 (LTB4). LTC4 and LTD4 are potent vasoconstrictors,[473] whereas LTB4 produces moderate renal vasodilation and an increase in RBF, with no change in GFR in the normal rat.[474] Intravenous infusion of LTC4 increases renal vascular resistance, leading to a decrease in RBF and GFR, as well as a decrease in plasma volume and cardiac output.[475,476] The decline in RBF is diminished by saralasin, an Ang II receptor antagonist, and indomethacin, an inhibitor of cyclooxygenase, indicating the involvement of Ang II and cyclooxygenase products in the response to LTC4, as well as a direct effect of LTC4 on the renal resistance vessels.[475] Similarly, LTD4 decreased K_f, but increased renal vascular resistance, particularly, R_E, a fall in Q_A and SNGFR, and a rise in P_{GC} and ΔP during blockade of Ang II and control of renal perfusion pressure, demonstrating a direct effect of this leukotriene on renal hemodynamics.[477] Leukotrienes often mediate drug-associated nephrotoxicity in some models of kidney disease.[478]

Inflammatory injury also activates the 5-, 12-, and 15-lipoxygenase pathways in neutrophils and platelets to form acyclic eicosanoids called lipoxins (LXs), of which there are two main types, LXA4 and LXB4.[479] LXB4 and 7-*cis*-11-*trans*-LXA4 produce renal vasoconstriction.[478,480] By contrast, intrarenal infusion of LXA4 induces a reduction in preglomerular resistance (R_A) without affecting R_E, thereby resulting in an increase in P_{GC} and ΔP.[477] The specific vasodilation of the preglomerular vessels by LXA4 was blocked by cyclooxygenase inhibition, indicating that vasodilatory prostaglandins are responsible for this effect.[477,480] Unique to this compound, LXA4 produced vasodilation while simultaneously causing a reduction in K_f.[480] Because P_{GC}, ΔP, and Q_A were increased, SNGFR also increased.[477] Furthermore, transfection of rat kidney with the 15-lipoxygenase gene suppressed inflammation and preserved function in experimental glomerulonephritis.[481] The lipoxins have also been used to treat acute renal failure in mice.[482,483]

Cytochrome P450

A rapidly growing field of study is that of eicosanoids formed through the cytochrome P 450 (CYP450) monooxygenase pathway. AA is metabolized via the CYP450 enzymes to other active metabolites, including EETs, dihydrooxyeicosatetraenoic acids (DiHETEs), and HETEs, which act locally in a paracrine manner.[35] Early researchers demonstrated the role of one metabolite, 20-HETE, involved in regulating renal hemodynamics through direct actions, as well as exerting modulatory actions on the TGF mechanism.[460] In addition to its vascular effects, 20-HETE inhibits sodium transport in the proximal tubule and the thick ascending loop of Henle. 20-HETE elicits an enhancement of myogenic tone and vasoconstricts afferent arterioles, which may be partially mediated through the augmentation of the TGF mechanism.[484] Blockade of 20-HETE formation attenuated vascular responses to Ang II, endothelin, norepinephrine, NO, and CO.[222,484] NO inhibits the production of 20-HETE, which contributes to the vasodilatory effect of NO.[222,460,463,485–487]

In contrast to the contractile responses of 20-HETE, some of the EETs mentioned also formed via the CYP450 pathway are potent vasodilators.[35,488] EETs are formed locally and serve as both autocrine and paracrine factors. EETs are endothelium-derived hyperpolarizing factors that elicit potent vasodilation of the afferent arteriole.[488] These actions are mediated in part by activating large conductance potassium channels.

NEURAL REGULATION OF RENAL CIRCULATION

The renal vasculature, including the afferent and efferent arterioles, macula densa cells of the distal tubule, and glomerular mesangium are richly innervated.[83,489] Innervation primarily includes renal efferent sympathetic adrenergic nerves[489,490] and renal afferent sensory fibers containing peptides, including calcitonin gene-related peptide (CGRP) and substance P.[83,489] Sympathetic efferent nerves are found in all segments of the vascular tree, from the main renal artery to the afferent arteriole (including the renin-containing juxtaglomerular cells) and efferent arteriole.[489,490] They play an important role in the regulation of renal hemodynamics, sodium transport, and renin secretion.[491] Afferent nerves containing CGRP and substance P are localized primarily in the main renal artery and interlobar arteries, with some innervation also observed in the arcuate artery, interlobular artery, afferent arteriole, and juxtaglomerular apparatus.[489,490] Peptidergic nerve fibers immunoreactive for neuropeptide Y (NPY), neurotensin, vasoactive intestinal polypeptide, and somatostatin are also found in the kidney.[55] Neuronal NOS-immunoreactive neurons have been identified in the kidney.[489] The NOS-containing neuronal stomata are seen in the wall of the renal pelvis, at the renal hilus close to the renal artery, along the interlobar arteries, the arcuate arteries, and extending to the afferent arteriole, supporting their role in the control of RBF.[489] They are also present in nerve bundles that have vasomotor and sensory fibers, suggesting that they modulate renal neural function.[489]

In micropuncture studies of the effects of renal nerve stimulation (RNS), RNS alone increased R_A and R_E, resulting in a decrease in plasma flow and SNGFR, without any effect on K_f.[492] When prostaglandin production was inhibited by indomethacin, however, the same level of RNS produced even greater increases in R_A and R_E, accompanied by very large declines in plasma flow and SNGFR and decreases in K_f, P_{GC}, and ΔP.[492] When saralasin was administered as a competitive inhibitor of endogenous Ang II in conjunction with indomethacin, RNS had no effect on K_f, but both R_A and R_E were still increased, and ΔP was slightly reduced.[492] The release of norepinephrine by RNS enhances Ang II production and results in arteriolar vasoconstriction and reductions in K_f. The increase in Ang II production may then enhance vasodilator prostaglandin production,[492,493] which partially ameliorates the constriction.

Continued vasoconstriction by RNS during the blockade of endogenous prostaglandins and Ang II has indicated that norepinephrine has vasoconstrictive properties by itself. The findings that norepinephrine causes constriction of preglomerular vessels support the direct effects of norepinephrine on renal microvessels.[422] Inhibition of NOS results in a decline in SNGFR in normal rats but not in rats with surgical renal denervation, suggesting that NO normally modulates the effects of renal adrenergic activity.[494] However, this modulation does not appear to be related to the sympathetic modulation of renin secretion.[495] Under conditions of renal injury and inflammation, the effects of increased renal nerve activity are exacerbated.[496] The actions of increased sympathetic activity to decrease renal cortical perfusion are attenuated by temporally mediated reductions in ROS.[497]

Renal denervation in animals undergoing acute water deprivation (48-hour duration) or with congestive heart failure produces increases in SNGFR, plasma flow, and K_f.[498] This indicates that the natural activity of the renal nerves in these settings plays an important role in the constriction of the arterioles and reductions in K_f that were observed with water deprivation and in congestive heart failure.[498] The vasoconstrictive effects of the renal nerves in both settings were mediated in part by a stimulatory effect on Ang II release, together with direct vasoconstrictive effects on the preglomerular and postglomerular blood vessels.[498] These studies have demonstrated the important role of the renal nerves in pathophysiologic settings.

Clinical trials have focused on whether catheter-based renal artery denervation reduces blood pressure in patients with resistant hypertension. Some studies have demonstrated sustained reduction in blood pressure in patients who underwent renal denervation; however, the blood pressure response did not differ from the sham-treated patients at 6 months after treatment.[499] More recent clinical trials using improved catheters have provided more promising long-term effects.[500]

GLOMERULAR HEMODYNAMICS IN THE AGING KIDNEY

Aging is commonly associated with the progressive development of chronic renal disease characterized by global glomerulosclerosis, tubular atrophy, interstitial fibrosis, and arteriosclerosis. Over 75 years ago, Davies and Shock demonstrated that beyond 40 to 50 years of age, inulin clearance in humans declined approximately 8 mL/min/1.73 m^2 in each subsequent decade of age.[501] In healthy adult living kidney donors, the prevalence of glomerulosclerosis, followed by local tubular atrophy and interstitial fibrosis, was found to increase progressively from 2.7% for patients aged 18 to 29 years to 73% for patients aged 70 to 77 years.[502] Despite these changes, the age-related decline in GFR was not fully explained by these age-related histologic changes.[502,503] Fig. 3.25 shows the declines in GFR beyond the ages of 40 to 50 years[83,504–506] Kidney disease associated with aging is generally accompanied by declines in GFR, RPF, and RBF associated with an increase in filtration fraction. In one study, the calculated filtration coefficient, K_f, was lower in older than in younger subjects (4.9 ± 1.7 nL/[min·mm Hg] in older subjects vs 7.0 ± 2.9 nL/[min·mm Hg] in younger subjects).[507] However, age-related declines in GFR and RBF do not necessarily occur in everyone.[502,506,508] As shown in Fig. 3.25, associated with the decline in GFR and RBF is the fact that systolic blood pressure increases with age,[507,509] which may contribute to a further decline in renal function. Nevertheless, in disease-free older individuals, GFR is generally adequate to sustain quality of life (see Fig. 3.25).

Factors likely involved in the decline in renal function associated with aging include eating high-protein meals, which may lead to episodes of hyperfiltration and glomerular capillary hypertension, development of diabetes, with associated

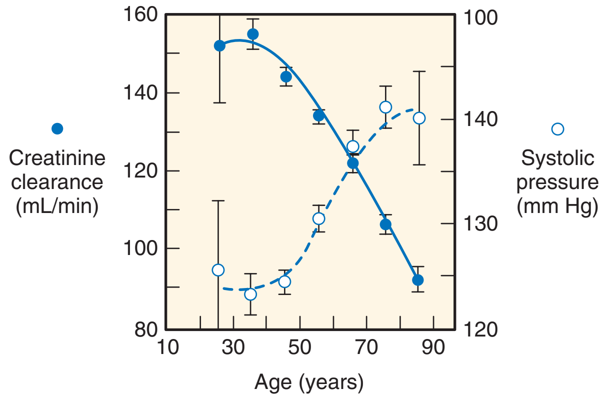

Fig. 3.25 Correlation of aging with systolic blood pressure and creatinine clearance (C_{Cr}) in normal adults. (From Lindeman R, Tobin J, Shock N. Association between blood pressure and the rate of decline in renal function with age. *Kidney Int.* 1984;26: 861–868.)

advanced glycosylated end products, obesity, oxidative stress due to chronic exposure to oxygen free radicals, dyslipidemia, excessive exposure to some drugs that reduce renal function, infections, atherosclerotic disease, male gender, genetic and racial differences, and environmental factors.[506,510–512] Sclerotic glomeruli seen with aging are smaller than functional glomeruli, leading to a decline in average glomerular size. With the progressive loss of filtering glomeruli, unaffected glomeruli undergo compensatory hypertrophy, which helps preserve GFR.[502] As has been shown in studies of the rat, following removal of 50% to 80% of total renal mass, the remaining nephrons undergo compensatory hypertrophy and increases in SNGFR and glomerular capillary hydraulic pressures.[14,513,514] In the aging human, progressive loss of functioning nephrons and hypertrophy of remaining filtering units would also result in glomerular capillary hypertension and damage to these remaining nephrons. Because these alterations use renal reserve capacity, there is a reduced capability of the renal vasculature in older individuals to respond to amino acid infusion, so that there is only an increase in GFR and filtration fraction and RPF remains unchanged.[508,515] In contrast, the vasoconstrictive response to Ang II is identical in younger and older people.[516] These data suggest that there is a blunted renal vasodilatory response but a vasoconstrictive response is maintained in older subjects, indicating that the aged kidney has used reserve capacity and is in a state of compensatory renal vasodilation to compensate for the chronic glomerular injury and loss of functional glomeruli.[515]

Complete reference list available at ExpertConsult.com.

KEY REFERENCES

2. Arendshorst W, Navar LG. Renal circulation and glomerular hemodyanamics. In: Schrier RW, Coffman TM, Falk RJ, et al, eds. *Schrier's Diseases of the Kidney*. Philadelphia: Lippincott Williams & Wilkins; 2013:74–131.
4. Navar LG, Bell PD, Evan AP. The regulation of glomerular filtration rate in mammalian kidneys. In: Andreoli TE, Hoffman J, Fanestil D, et al, eds. *Physiology of Membrane Disorders*. New York: Plenum Medical Book Co.; 1986:637–667.
5. Smith HW. *The Kidney: Structure and Function in Health and Disease*. Vol 4. New York: Oxford University Press; 1951.
25. Evans RG, Harrop GK, Ngo JP, et al. Basal renal O2 consumption and the efficiency of O2 utilization for Na+ reabsorption. *Am J Physiol Renal Physiol*. 2014;306(5):F551–F560.
26. Mimura I, Nangaku M. The suffocating kidney: tubulointerstitial hypoxia in end-stage renal disease. *Nat Rev Nephrol*. 2010;6(11): 667–678.
35. Navar LG, Arendshorst WJ, Pallone TL, et al. The renal microcirculation. In: Tuma RF, Duran WN, Ley K, eds. *Handbook of Physiology: Microcirculation*. Vol 2. Academic Press; 2008:550–683.
41. Beeuwkes R 3rd. The vascular organization of the kidney. *Annu Rev Physiol*. 1980;42:531–542.
47. Neal CR, Arkill K, Bell JS, et al. Novel haemodynamic structures in the human glomerulus. *Am J Physiol Renal Physiol*. 2018;315:F1370–F1384.
60. Lawrence MG, Altenburg MK, Sanford R, et al. Permeation of macromolecules into the renal glomerular basement membrane and capture by the tubules. *Proc Natl Acad Sci USA*. 2017;114(11): 2958–2963.
62. Haraldsson B, Nystrom J, Deen WM. Properties of the glomerular barrier and mechanisms of proteinuria. *Physiol Rev*. 2008;88(2): 451–487.
67. Suh JH, Miner JH. The glomerular basement membrane as a barrier to albumin. *Nat Rev Nephrol*. 2013;9(8):470–477.
74. Arendshorst WJ, Gottschalk CW. Glomerular ultrafiltration dynamics: historical perspective. *Am J Physiol Renal Physiol*. 1985;248(2 Pt 2): F163–F174.
75. Oken DE. An analysis of glomerular dynamics in rat, dog, and man. *Kidney Int*. 1982;22(2):136–145.
89. Lenihan CR, Busque S, Derby G, et al. Longitudinal study of living kidney donor glomerular dynamics after nephrectomy. *J Clin Invest*. 2015;125(3):1311–1318.
102. Pallone TL, Zhang Z, Rhinehart K. Physiology of the renal medullary microcirculation. *Am J Physiol Renal Physiol*. 2003;284(2):F253–F266.
127. Peppiatt-Wildman CM. The evolving role of renal pericytes. *Curr Opin Nephrol Hypertens*. 2013;22(1):10–16.
132. Pannabecker TL, Dantzler WH. Three-dimensional architecture of inner medullary vasa recta. *Am J Physiol Renal Physiol*. 2006; 290(6):F1355–F1366.
154. Peti-Peterdi J, Morishima S, Bell PD, et al. Two-photon excitation fluorescence imaging of the living juxtaglomerular apparatus. *Am J Physiol Renal Physiol*. 2002;283(1):F197–F201.
163. Griffin KA, Hacioglu R, Abu-Amarah I, et al. Effects of calcium channel blockers on “dynamic” and “steady-state step” renal autoregulation. *Am J Physiol Renal Physiol*. 2004;286(6):F1136–F1143.
191. Hashimoto S, Huang Y, Briggs J, et al. Reduced autoregulatory effectiveness in adenosine 1 receptor-deficient mice. *Am J Physiol Renal Physiol*. 2006;290(4):F888–F891.
200. Loutzenhiser R, Griffin K, Williamson G, et al. Renal autoregulation: new perspectives regarding the protective and regulatory roles of the underlying mechanisms. *Am J Physiol Regul Integr Comp Physiol*. 2006;290(5):R1153–R1167.
202. Schnermann J, Briggs JP. Tubuloglomerular feedback: mechanistic insights from gene-manipulated mice. *Kidney Int*. 2008;74(4):418–426.
217. Feng MG, Navar LG. Afferent arteriolar vasodilator effect of adenosine predominantly involves adenosine A2B receptor activation. *Am J Physiol Renal Physiol*. 2010;299:F310–F315.
220. Fan F, Roman RJ. Effect of cytochrome P450 metabolites of arachidonic acid in nephrology. *J Am Soc Nephrol*. 2017;28(10): 2845–2855.
222. Schnermann J. Concurrent activation of multiple vasoactive signaling pathways in vasoconstriction caused by tubuloglomerular feedback: a quantitative assessment. *Annu Rev Physiol*. 2015;77:301–322.
239. Kovacs G, Komlosi P, Fuson A, et al. Neuronal nitric oxide synthase: its role and regulation in macula densa cells. *J Am Soc Nephrol*. 2003;14(10):2475–2483.
313. Botros FT, Dobrowolski L, Navar LG. Renal heme oxygenase-1 induction with hemin augments renal hemodynamics, renal autoregulation, and excretory function. *Int J Hypertens*. 2012;2012:189512.
320. Kasinath BS, Feliers D, Lee HJ. Hydrogen sulfide as a regulatory factor in kidney health and disease. *Biochem Pharmacol*. 2018;149:29–41.
330. Fellner SK, Arendshorst WJ. Angiotensin II, reactive oxygen species, and Ca2+ signaling in afferent arterioles. *Am J Physiol Renal Physiol*. 2005;289(5):F1012–F1019.
336. Li L, Lai EY, Luo Z, et al. Superoxide and hydrogen peroxide counterregulate myogenic contractions in renal afferent arterioles from a mouse model of chronic kidney disease. *Kidney Int*. 2017;92(3):625–633.

337. Vogel PA, Yang X, Moss NG, et al. Superoxide enhances Ca2+ entry through L-type channels in the renal afferent arteriole. *Hypertension*. 2015;66(2):374–381.
339. Song J, Lu Y, Lai EY, et al. Oxidative status in the macula densa modulates tubuloglomerular feedback responsiveness in angiotensin II-induced hypertension. *Acta Physiol (Oxf)*. 2015;213(1):249–258.
341. Zhang S, Huang Q, Wang Q, et al. Enhanced renal afferent arteriolar reactive oxygen species and contractility to endothelin-1 are associated with canonical wnt signaling in diabetic mice. *Kidney Blood Press Res*. 2018;43(3):860–871.
370. De Miguel C, Speed JS, Kasztan M, et al. Endothelin-1 and the kidney: new perspectives and recent findings. *Curr Opin Nephrol Hypertens*. 2016;25(1):35–41.
378. Gohar EY, Kasztan M, Pollock DM. Interplay between renal endothelin and purinergic signaling systems. *Am J Physiol Renal Physiol*. 2017;313(3):F666–F668.
413. Prieto MC, Gonzalez AA, Navar LG. Evolving concepts on regulation and function of renin in distal nephron. *Pflugers Arch*. 2013;465(1):121–132.
414. Ramkumar N, Kohan DE. Role of collecting duct renin in blood pressure regulation. *Am J Physiol Regul Integr Comp Physiol*. 2013;305(2):R92–R94.
420. Navar LG. Intrarenal renin-angiotensin system in regulation of glomerular function. *Curr Opin Nephrol Hypertens*. 2014;23(1):38–45.
440. Gonzalez AA, Lara LS, Prieto MC. Role of Collecting Duct Renin in the Pathogenesis of Hypertension. *Curr Hypertens Rep*. 2017;19(8):62.
450. Wang TN, Chen X, Li R, et al. SREBP-1 mediates angiotensin II-induced TGF-beta1 upregulation and glomerular fibrosis. *J Am Soc Nephrol*. 2015;26(8):1839–1854.
457. Navar LG, Inscho EW, Majid SA, et al. Paracrine regulation of the renal microcirculation. *Physiol Rev*. 1996;76(2):425–536.
464. Carlstrom M, Wilcox CS, Arendshorst WJ. Renal autoregulation in health and disease. *Physiol Rev*. 2015;95(2):405–511.
472. Vio CP, Quiroz-Munoz M, Cuevas CA, et al. Prostaglandin E2 EP3 receptor regulates cyclooxygenase-2 expression in the kidney. *Am J Physiol Renal Physiol*. 2012;303(3):F449–F457.
478. Rubinstein M, Dvash E. Leukotrienes and kidney diseases. *Curr Opin Nephrol Hypertens*. 2018;27(1):42–48.
483. Chandrasekharan JA, Sharma-Walia N. Lipoxins: nature's way to resolve inflammation. *J Inflamm Res*. 2015;8:181–192.
484. Roman RJ, Fan F. 20-HETE: hypertension and beyond. *Hypertension*. 2018;72(1):12–18.
488. Imig JD. Epoxyeicosatrienoic acids, hypertension, and kidney injury. *Hypertension*. 2015;65(3):476–482.
499. Bhatt DL, Kandzari DE, O'Neill WW, et al. A controlled trial of renal denervation for resistant hypertension. *N Engl J Med*. 2014;370(15):1393–1401.
502. Glassock RJ, Rule AD. The implications of anatomical and functional changes of the aging kidney: with an emphasis on the glomeruli. *Kidney Int*. 2012;82(3):270–277.
505. Maddox DA, Alavi FK, Zawada ET Jr. The kidney and aging. In: Massry SG, Glassock RJ, eds. *Textbook of Nephrology*. Baltimore: Lippincott Williams & Wilkins; 2001:1094–1105.

REFERENCES

1. Stein JH, Fadem SZ. The renal circulation. *JAMA*. 1978;239(13):1308–1312.
2. Arendshorst W, Navar LG. Renal circulation and glomerular hemodyanamics. In: Schrier RW, Coffman TM, Falk RJ, et al, eds. *Schrier's Diseases of the Kidney*. Philadelphia: Lippincott Williams & Wilkins; 2013:74–131.
3. Navar LG, Evan AP, Rosivall L. Microcirculation of the kidneys. In: Mortillaro N, ed. *The Physiology and Pharmacology of the Microcirculation*. New York: Academic Press; 1983:397–488.
4. Navar LG, Bell PD, Evan AP. The regulation of glomerular filtration rate in mammalian kidneys. In: Andreoli TE, Hoffman J, Fanestil D, et al, eds. *Physiology of Membrane Disorders*. New York: Plenum Medical Book Co.; 1986:637–667.
5. Smith HW. *The Kidney: Structure and Function in Health and Disease*. Vol 4. New York: Oxford University Press; 1951.
6. Dworkin LD, Brenner BM. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004.
7. Maddox DA, Alavi FK, Santella RN, et al. Prevention of obesity-linked renal disease: age-dependent effects of dietary food restriction. *Kidney Int*. 2002;62(1):208–219.
8. Kost CK Jr, Li P, Williams DS, et al. Renal vascular responses to angiotensin II in conscious spontaneously hypertensive and normotensive rats. *J Cardiovasc Pharmacol*. 1998;31(6):854–861.
9. Matsuda H, Hayashi K, Arakawa K, et al. Zonal heterogeneity in action of angiotensin-converting enzyme inhibitor on renal microcirculation: role of intrarenal bradykinin. *J Am Soc Nephrol*. 1999;10(11):2272–2282.
10. Sommer G, Corrigan G, Fredrickson J, et al. Renal blood flow: measurement in vivo with rapid spiral MR imaging. *Radiology*. 1998;208(3):729–734.
11. Sommer G, Noorbehesht B, Pelc N, et al. Normal renal blood flow measurement using phase-contrast cine magnetic resonance imaging. *Invest Radiol*. 1992;27(6):465–470.
12. Krier JD, Ritman EL, Bajzer Z, et al. Noninvasive measurement of concurrent single-kidney perfusion, glomerular filtration, and tubular function. *Am J Physiol Renal Physiol*. 2001;281(4):F630–F638.
13. Deng A, Tang T, Singh P, et al. Regulation of oxygen utilization by angiotensin II in chronic kidney disease. *Kidney Int*. 2009;75(2):197–204.
14. Maddox DA, Horn JF, Famiano FC, et al. Load dependence of proximal tubular fluid and bicarbonate reabsorption in the remnant kidney of the Munich-Wistar rat. *J Clin Invest*. 1986;77(5):1639–1649.
15. Correa-Rotter R, Hostetter TH, Manivel JC, et al. Renin expression in renal ablation. *Hypertension*. 1992;20(4):483–490.
16. Sykes D. The correlation between renal vascularisation and lobulation of the kidney. *Br J Urol*. 1964;36:549–555.
17. Sykes D. The arterial supply of the human kidney with special reference to accessory renal arteries. *Br J Surg*. 1963;50:368–374.
18. Boijsen E. Angiographic studies of the anatomy of single and multiple renal arteries. *Acta Radiol Suppl*. 1959;183:1–135.
19. Graves F. *The Arterial Anatomy of the Kidney*. Williams and Wilkins; 1971.
20. Beeuwkes R 3rd. Efferent vascular patterns and early vascular-tubular relations in the dog kidney. *Am J Physiol*. 1971;221(5):1361–1374.
21. Kosinski H. Variation of the structure and course of the interlobular arteries in human kidney. *Folia Morphol (Warsz)*. 1997;56(4):249–252.
22. Moffat DB, Fourman J. A vascular pattern of the rat kidney. 1963. *J Am Soc Nephrol*. 2001;12(3):624–632.
23. Gardiner BS, Thompson SL, Ngo JP, et al. Diffusive oxygen shunting between vessels in the preglomerular renal vasculature: anatomic observations and computational modeling. *Am J Physiol Renal Physiol*. 2012;303(5):F605–F618.
24. Welch WJ, Baumgartl H, Lubbers D, et al. Nephron pO2 and renal oxygen usage in the hypertensive rat kidney. *Kidney Int*. 2001;59(1):230–237.
25. Evans RG, Harrop GK, Ngo JP, et al. Basal renal O2 consumption and the efficiency of O2 utilization for Na+ reabsorption. *Am J Physiol Renal Physiol*. 2014;306(5):F551–F560.
26. Mimura I, Nangaku M. The suffocating kidney: tubulointerstitial hypoxia in end-stage renal disease. *Nat Rev Nephrol*. 2010;6(11):667–678.
27. Evans RG, Gardiner BS, Smith DW, et al. Intrarenal oxygenation: unique challenges and the biophysical basis of homeostasis. *Am J Physiol Renal Physiol*. 2008;295(5):F1259–F1270.
28. Palm F, Onozato M, Welch WJ, et al. Blood pressure, blood flow, and oxygenation in the clipped kidney of chronic 2-kidney, 1-clip rats: effects of tempol and Angiotensin blockade. *Hypertension*. 2010;55(2):298–304.
29. Ngo JP, Ow CP, Gardiner BS, et al. Diffusive shunting of gases and other molecules in the renal vasculature: physiological and evolutionary significance. *Am J Physiol Regul Integr Comp Physiol*. 2016;311(5):R797–R810.
30. Casellas D, Navar LG. In vitro perfusion of juxtamedullary nephrons in rats. *Am J Physiol*. 1984;246:F349–F358.
31. Imig JD, Roman RJ. Nitric oxide modulates vascular tone in preglomerular arterioles. *Hypertension*. 1992;19(6 Pt 2):770–774.
32. Peti-Peterdi J. Multiphoton imaging of renal tissues in vitro. *Am J Physiol Renal Physiol*. 2005;288(6):F1079–F1083.
33. Peti-Peterdi J, Toma I, Sipos A, et al. Multiphoton imaging of renal regulatory mechanisms. *Physiology (Bethesda)*. 2009;24:88–96.
34. Evans RG, Eppel GA, Anderson WP, et al. Mechanisms underlying the differential control of blood flow in the renal medulla and cortex. *J Hypertens*. 2004;22(8):1439–1451.
35. Navar LG, Arendshorst WJ, Pallone TL, et al. The renal microcirculation. In: Tuma RF, Duran WN, Ley K, eds. *Handbook of Physiology: Microcirculation*. Vol 2. Academic Press; 2008:550–683.
36. Cowley AW Jr. Renal medullary oxidative stress, pressure-natriuresis, and hypertension. *Hypertension*. 2008;52(5):777–786.
37. Pallone TL, Robertson CR, Jamison RL. Renal medullary microcirculation. *Physiol Rev*. 1990;70(3):885–920.
38. Beeuwkes R 3rd, Bonventre JV. The organization and vascular perfusion of canine renal tubules. *Physiologist*. 1973;16:264.
39. Beeuwkes R 3rd, Bonventre JV. Tubular organization and vascular-tubular relations in the dog kidney. *Am J Physiol*. 1975;229(3):695–713.
40. Briggs JP, Wright FS. Feedback control of glomerular filtration rate: site of the effector mechanism. *Am J Physiol*. 1979;236(1):F40–F47.
41. Beeuwkes R 3rd. The vascular organization of the kidney. *Annu Rev Physiol*. 1980;42:531–542.
42. Steinhausen M. Further information on the cortical countercurrent system in rat kidney. *Yale J Biol Med*. 1972;45(3–4):451–456.
43. Steinhausen M, Eisenbach GM, Galaske R. Countercurrent system in the renal cortex of rats. *Science*. 1970;167(925):1631–1633.
44. Weinstein SW, Szyjewicz J. Superficial nephron tubular-vascular relationships in the rat kidney. *Am J Physiol*. 1978;234(3):F207–F214.
45. Elger M, Sakai T, Kriz W. The vascular pole of the renal glomerulus of rat. *Adv Anat Embryol Cell Biol*. 1998;139:1–98.
46. Rosivall L, Peti-Peterdi J. Heterogeneity of the afferent arteriole–correlations between morphology and function. *Nephrol Dial Transplant*. 2006;21(10):2703–2707.
47. Neal CR, Arkill K, Bell JS, et al. Novel haemodynamic structures in the human glomerulus. *Am J Physiol Renal Physiol*. 2018;315:F1370–F1384.
48. Navar LG, Bell PD. Romancing the macula densa at UAB. *Kidney Int*. 2004;66(91):S34–S40.
49. Kriz W, Elger M, Mundel P, et al. Structure-stabilizing forces in the glomerular tuft. *J Am Soc Nephrol*. 1995;5(10):1731–1739.
50. Sraer JD, Adida C, Peraldi MN, et al. Species-specific properties of the glomerular mesangium. *J Am Soc Nephrol*. 1993;3(7):1342–1350.
51. Feng Z, Wei C, Chen X, et al. Essential role of Ca2+ release channels in angiotensin II-induced Ca2+ oscillations and mesangial cell contraction. *Kidney Int*. 2006;70(1):130–138.
52. Inkyo-Hayasaka K, Sakai T, Kobayashi N, et al. Three-dimensional analysis of the whole mesangium in the rat. *Kidney Int*. 1996;50(2):672–683.
53. Schlondorff D, Wyatt CM, Campbell KN. Revisiting the determinants of the glomerular filtration barrier: what goes round must come round. *Kidney Int*. 2017;92(3):533–536.
54. Brenner BM, Troy JL. Postglomerular vascular protein concentration: evidence for a causal role in governing fluid reabsorption and glomerulotublar balance by the renal proximal tubule. *J Clin Invest*. 1971;50(2):336–349.
55. Maddox DA, Deen WM, Brenner BM. *Handbook of Physiology: Section 8: Renal Physiology*. Vol 1. New York: Oxford University Press; 1992.
56. Oliver JD 3rd, Anderson S, Troy JL, et al. Determination of glomerular size-selectivity in the normal rat with Ficoll. *J Am Soc Nephrol*. 1992;3(2):214–228.
57. Scandling JD, Myers BD. Glomerular size-selectivity and microalbuminuria in early diabetic glomerular disease. *Kidney Int*. 1992;41(4):840–846.

58. Walker AM, Bott PA, Oliver JJ, et al. The collection and analysis of fluid from single nephrons of the mammalian kidney. *Am J Physiol.* 1941;134:580–595.
59. Wearn JT, Richards AN. Observations on the composition of glomerular urine, with particular reference to the problem of reabsorption in the renal tubule. *Am J Physiol.* 1924;71:209–227.
60. Lawrence MG, Altenburg MK, Sanford R, et al. Permeation of macromolecules into the renal glomerular basement membrane and capture by the tubules. *Proc Natl Acad Sci USA.* 2017;114(11): 2958–2963.
61. Bohrer MP, Baylis C, Humes HD, et al. Permselectivity of the glomerular capillary wall. Facilitated filtration of circulating polycations. *J Clin Invest.* 1978;61(1):72–78.
62. Haraldsson B, Nystrom J, Deen WM. Properties of the glomerular barrier and mechanisms of proteinuria. *Physiol Rev.* 2008;88(2): 451–487.
63. Chang RL, Ueki IF, Troy JL, et al. Permselectivity of the glomerular capillary wall to macromolecules. II. Experimental studies in rats using neutral dextran. *Biophys J.* 1975;15(9):887–906.
64. Whiteside C, Silverman M. Determination of glomerular permselectivity to neutral dextrans in the dog. *Am J Physiol.* 1983;245(4): F485–F495.
65. Guasch A, Deen WM, Myers BD. Charge selectivity of the glomerular filtration barrier in healthy and nephrotic humans. *J Clin Invest.* 1993;92(5):2274–2282.
66. Deen WM, Lazzara MJ, Myers BD. Structural determinants of glomerular permeability. *Am J Physiol Renal Physiol.* 2001;281(4):F579–F596.
67. Suh JH, Miner JH. The glomerular basement membrane as a barrier to albumin. *Nat Rev Nephrol.* 2013;9(8):470–477.
68. Salmon AH, Satchell SC. Endothelial glycocalyx dysfunction in disease: albuminuria and increased microvascular permeability. *J Pathol.* 2012;226(4):562–574.
69. Deen WM. What determines glomerular capillary permeability? *J Clin Invest.* 2004;114(10):1412–1414.
70. Drumond MC, Deen WM. Structural determinants of glomerular hydraulic permeability. *Am J Physiol.* 1994;266(1 Pt 2):F1–F12.
71. Garg P, Rabelink T. Glomerular proteinuria: a complex interplay between unique players. *Adv Chronic Kidney Dis.* 2011;18(4):233–242.
72. Liu G, Kaw B, Kurfis J, et al. Neph1 and nephrin interaction in the slit diaphragm is an important determinant of glomerular permeability. *J Clin Invest.* 2003;112(2):209–221.
73. Drumond MC, Kristal B, Myers BD, et al. Structural basis for reduced glomerular filtration capacity in nephrotic humans. *J Clin Invest.* 1994;94(3):1187–1195.
74. Arendshorst WJ, Gottschalk CW. Glomerular ultrafiltration dynamics: historical perspective. *Am J Physiol Renal Physiol.* 1985;248(2 Pt 2):F163–F174.
75. Oken DE. An analysis of glomerular dynamics in rat, dog, and man. *Kidney Int.* 1982;22(2):136–145.
76. Gertz KH, Mangos JA, Braun G, et al. Pressure in the glomerular capillaries of the rat kidney and its relation to arterial blood pressure. *Pflugers Arch Gesamte Physiol Menschen Tiere.* 1966;288(4): 369–374.
77. Brenner BM, Troy JL, Daugharty TM. The dynamics of glomerular ultrafiltration in the rat. *J Clin Invest.* 1971;50(8):1776–1780.
78. Maddox DA, Deen WM, Brenner BM. Dynamics of glomerular ultrafiltration: VI. Studies in the primate. *Kidney Int.* 1974;5:271–278.
79. Maddox DA, Price DC, Rector FC Jr. Effect of surgery on plasma volume and salt and water excretion in rats. *Am J Physiol.* 1977;233:F600–F606.
80. Deen WM, Robertson CR, Brenner BM. A model of glomerular ultrafiltration in the rat. *Am J Physiol.* 1972;223(5):1178–1183.
81. Thomas CE, Bell PD, Navar LG. Glomerular filtration dynamics in the dog during elevated plasma colloid osmotic pressure. *Kidney Int.* 1979;15:502–512.
82. Pinnick RV, Savin VJ. Filtration by superficial and deep glomeruli of normovolemic and volume-depleted rats. *Am J Physiol.* 1986;250:F86–F91.
83. Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney.* 7th ed. Philadelphia: Saunders; 2004:353–412.
84. Lowenstein J, Beranbaum ER, Chasis H, et al. Intrarenal pressure and exaggerated natriuresis in essential hypertension. *Clin Sci.* 1970;38(3):359–374.
85. Willassen Y, Ofstad J. Renal sodium excretion and the peritubular capillary physical factors in essential hypertension. *Hypertension.* 1980;2(6):771–779.
86. Lambert PP, Verniory A, Gassee JP, et al. Sieving equations and effective glomerular filtration pressure. *Kidney Int.* 1972;2(3):131–146.
87. Andersen S, Blouch K, Bialek J, et al. Glomerular permselectivity in early stages of overt diabetic nephropathy. *Kidney Int.* 2000; 58(5):2129–2137.
88. Tan JC, Busque S, Workeneh B, et al. Effects of aging on glomerular function and number in living kidney donors. *Kidney Int.* 2010;78(7):686–692.
89. Lenihan CR, Busque S, Derby G, et al. Longitudinal study of living kidney donor glomerular dynamics after nephrectomy. *J Clin Invest.* 2015;125(3):1311–1318.
90. Brenner BM, Troy JL, Daugharty TM, et al. Dynamics of glomerular ultrafiltration in the rat. II. Plasma-flow dependence of GFR. *Am J Physiol.* 1972;223(5):1184–1190.
91. Daniels BS, Hauser EB, Deen WM, et al. Glomerular basement membrane: in vitro studies of water and protein permeability. *Am J Physiol.* 1992;262(6 Pt 2):F919–F926.
92. Blantz RC, Rector FC Jr, Seldin DW. Effect of hyperoncotic albumin expansion upon glomerular ultrafiltration in the rat. *Kidney Int.* 1974;6(4):209–221.
93. Baylis C, Deen WM, Myers BD, et al. Effects of some vasodilator drugs on transcapillary fluid exchange in renal cortex. *Am J Physiol.* 1976;230(4):1148–1158.
94. Charonis AS, Wissig SL. Anionic sites in basement membranes. Differences in their electrostatic properties in continuous and fenestrated capillaries. *Microvasc Res.* 1983;25(3):265–285.
95. Kriz W, Napiwotzky P. Structural and functional aspects of the renal interstitium. *Contrib Nephrol.* 1979;16:104–108.
96. Aukland K, Bogusky RT, Renkin EM. Renal cortical interstitium and fluid absorption by peritubular capillaries. *Am J Physiol.* 1994;266(2 Pt 2):F175–F184.
97. Whiteside C, Silverman M. Postglomerular capillary solute flux restricted by shape and charge in the dog. *Am J Physiol.* 1987;253(3 Pt 2):F500–F512.
98. Venkatachalam MA, Karnovsky MJ. Extravascular protein in the kidney. An ultrastructural study of its relation to renal peritubular capillary permeability using protein tracers. *Lab Invest.* 1972;27(5): 435–444.
99. Deen WM, Ueki IF, Brenner BM. Permeability of renal peritubular capillaries to neutral dextrans dextrans and endogenous albumin. *Am J Physiol.* 1976;231(2):283–291.
100. Zimmerhackl B, Robertson CR, Jamison RL. The microcirculation of the renal medulla. *Circ Res.* 1985;57(5):657–667.
101. Rosivall L, Posch E, Simon G, et al. Intrarenal distribution of renal blood flow in the rat. *Acta Physiol Acad Sci Hung.* 1979;53(4):389–397.
102. Pallone TL, Zhang Z, Rhinehart K. Physiology of the renal medullary microcirculation. *Am J Physiol Renal Physiol.* 2003;284(2):F253–F266.
103. Edwards JG. Efferent arterioles of glomeruli in the juxtamedullary zone of the human kidney. *Anat Rec.* 1956;125(3):521–529.
104. Fourman J. Structural aspects of the kidney. *J Endocrinol.* 1971; 50(3):iv–v.
105. Kriz W, Koepsell H. The structural organization of the mouse kidney. *Z Anat Entwicklungsgesch.* 1974;144(2):137–163.
106. Kriz W. Structural organization of renal medullary circulation. *Nephron.* 1982;31(4):290–295.
107. Dieterich HJ. [Structure of blood vessels in the kidney]. *Norm Pathol Anat (Stuttg).* 1978;35:1–108.
108. Inoue M, Maeda M, Takao S. Regional differentiation of blood flow responses to microinjection of sodium nitroprusside into the nucleus tractus solitarius of anesthetized rats. *J Auton Nerv Syst.* 1997;63(3):172–178.
109. Lim SW, Han KH, Jung JY, et al. Ultrastructural localization of UT-A and UT-B in rat kidneys with different hydration status. *Am J Physiol Regul Integr Comp Physiol.* 2006;290(2):R479–R492.
110. Edwards A, Silldforff EP, Pallone TL. The renal medullary microcirculation. *Front Biosci.* 2000;5:E36–E52.
111. Zimmerhackl B, Robertson CR, Jamison RL. Effect of arginine vasopressin on renal medullary blood flow. A videomicroscopic study in the rat. *J Clin Invest.* 1985;76(2):770–778.
112. Fadem SZ, Hernandez-Llamas G, Patak RV, et al. Studies on the mechanism of sodium excretion during drug-induced vasodilatation in the dog. *J Clin Invest.* 1982;69(3):604–610.
113. Ganguli M, Tobian L, Ferris T, et al. Acute prostaglandin reduction with indomethacin and chronic prostaglandin reduction with an essential fatty acid deficient diet both decrease plasma flow to the renal papilla in the rat. *Prostaglandins.* 1989;38(1):3–19.

114. Nafz B, Berger K, Rosler C, et al. Kinins modulate the sodium-dependent autoregulation of renal medullary blood flow. *Cardiovasc Res.* 1998;40(3):573–579.
115. Zou AP, Nithipatikom K, Li PL, et al. Role of renal medullary adenosine in the control of blood flow and sodium excretion. *Am J Physiol.* 1999;276(3 Pt 2):R790–R798.
116. Dunn BR, Ichikawa I, Pfeffer JM, et al. Renal and systemic hemodynamic effects of synthetic atrial natriuretic peptide in the anesthetized rat. *Circ Res.* 1986;59(3):237–246.
117. Pallone TL, Mattson DL. Role of nitric oxide in regulation of the renal medulla in normal and hypertensive kidneys. *Curr Opin Nephrol Hypertens.* 2002;11(1):93–98.
118. Oliver JJ, Rajapakse NW, Evans RG. Effects of indomethacin on responses of regional kidney perfusion to vasoactive agents in rabbits. *Clin Exp Pharmacol Physiol.* 2002;29(10):873–879.
119. Badzynska B, Grzelec-Mojzesowicz M, Dobrowolski L, et al. Differential effect of angiotensin II on blood circulation in the renal medulla and cortex of anaesthetised rats. *J Physiol.* 2002;538(Pt 1):159–166.
120. Sarkis A, Liu KL, Lo M, et al. Angiotensin II and renal medullary blood flow in Lyon rats. *Am J Physiol Renal Physiol.* 2003;284(2):F365–F372.
121. Omoro SA, Majid DS, El Dahr SS, et al. Roles of Ang II and bradykinin in the renal regional blood flow responses to ACE inhibition in sodium-depleted dogs. *Am J Physiol Renal Physiol.* 2000;279(2):F289–F293.
122. Kiberd B, Robertson CR, Larson T, et al. Effect of V2-receptor-mediated changes on inner medullary blood flow induced by AVP. *Am J Physiol.* 1987;253(3 Pt 2):F576–F581.
123. Mattson DL. Importance of the renal medullary circulation in the control of sodium excretion and blood pressure. *Am J Physiol Regul Integr Comp Physiol.* 2003;284(1):R13–R27.
124. Daniel PM, Peabody CN, Prichard MM. Cortical ischaemia of the kidney with maintained blood flow through the medulla. *Q J Exp Physiol Cogn Med Sci.* 1952;37(1):11–18.
125. Kriz W, Schnermann J, Koepsell H. The position of short and long loops of Henle in the rat kidney. *Z Anat Entwicklungsgesch.* 1972;138(3):301–319.
126. Park F, Mattson DL, Roberts LA, et al. Evidence for the presence of smooth muscle alpha-actin within pericytes of the renal medulla. *Am J Physiol.* 1997;273(5 Pt 2):R1742–R1748.
127. Peppiatt-Wildman CM. The evolving role of renal pericytes. *Curr Opin Nephrol Hypertens.* 2013;22(1):10–16.
128. Kriz W, Barrett JM, Peter S. The renal vasculature: anatomical-functional aspects. *Int Rev Physiol.* 1976;11:1–21.
129. Schwartz MM, Karnovsky MJ, Vehkatachalam MA. Ultrastructural differences between rat inner medullary descending and ascending vasa recta. *Lab Invest.* 1976;35(2):161–170.
130. Kriz W, Kaissling B. *Structural Organization of the Mammalian Kidney.* 3rd ed. Lippincott Williams & Wilkins; 2000.
131. Marsh DJ, Segel LA. Analysis of countercurrent diffusion exchange in blood vessels of the renal medulla. *Am J Physiol.* 1971;221(3):817–828.
132. Pannabecker TL, Dantzler WH. Three-dimensional architecture of inner medullary vasa recta. *Am J Physiol Renal Physiol.* 2006;290(6):F1355–F1366.
133. Imai M. Functional heterogeneity of the descending limbs of Henle's loop. II. Interspecies differences among rabbits, rats, and hamsters. *Pflugers Arch.* 1984;402(4):393–401.
134. Haley DP, Sarrafian M, Bulger RE, et al. Structural and functional correlates of effects of angiotensin-induced changes in rat glomerulus. *Am J Physiol.* 1987;253(1 Pt 2):F111–F119.
135. Zimmerhackl B, Parekh N, Kücherer H, et al. Influence of systemically applied angiotensin II on the microcirculation of glomerular capillaries in the rat. *Kidney Int.* 1985;27:17–24.
136. Andrews PM, Coffey AK. Cytoplasmic contractile elements in glomerular cells. *Fed Proc.* 1983;42(14):3046–3052.
137. Click RL, Joyner WL, Gilmore JP. Reactivity of gomerular afferent and efferent arterioles in renal hypertension. *Kidney Int.* 1979;15(2):109–115.
138. Steinhausen M, Sterzel RB, Fleming JT, et al. Acute and chronic effects of angiotensin II on the vessels of the split hydronephrotic kidney. *Kidney Int Suppl.* 1987;20:S64–S73.
139. Loutzenhiser R, Bidani A, Chilton L. Renal myogenic response: kinetic attributes and physiological role. *Circ Res.* 2002;90(12):1316–1324.
140. Tang L, Loutzenhiser K, Loutzenhiser R. Biphasic actions of prostaglandin E(2) on the renal afferent arteriole : role of EP(3) and EP(4) receptors. *Circ Res.* 2000;86(6):663–670.
141. Tang L, Parker M, Fei Q, et al. Afferent arteriolar adenosine A2a receptors are coupled to KATP in in vitro perfused hydronephrotic rat kidney. *Am J Physiol.* 1999;277(6 Pt 2):F926–F933.
142. Carmines PK, Morrison TK, Navar LG. Angiotensin II effects on microvascular diameters of in vitro blood-perfused juxtamedullary nephrons. *Am J Physiol.* 1986;251(4 Pt 2):F610–F618.
143. Edwards RM. Segmental effects of norepinephrine and angiotensin II on isolated renal microvessels. *Am J Physiol.* 1983;244(5):F526–F534.
144. Ito S, Johnson CS, Carretero OA. Modulation of angiotensin II-induced vasoconstriction by endothelium-derived relaxing factor in the isolated microperfused rabbit afferent arteriole. *J Clin Invest.* 1991;87(5):1656–1663.
145. Robertson CR, Deen WM, Troy JL, et al. Dynamics of glomerular ultrafiltration in the rat. III. Hemodynamics and autoregulation. *Am J Physiol.* 1972;223:1191–1200.
146. Navar LG, Bell PD, Burke TJ. Autoregulatory responses of superficial nephrons and their association with sodium excretion during arterial pressure alterations in the dog. *Circ Res.* 1977;41(4):487–496.
147. Eppel GA, Bergstrom G, Anderson WP, et al. Autoregulation of renal medullary blood flow in rabbits. *Am J Physiol Regul Integr Comp Physiol.* 2003;284(1):R233–R244.
148. Pallone TL, Silldorff EP, Turner MR. Intrarenal blood flow: microvascular anatomy and the regulation of medullary perfusion. *Clin Exp Pharmacol Physiol.* 1998;25(6):383–392.
149. Majid DS, Navar LG. Medullary blood flow responses to changes in arterial pressure in canine kidney. *Am J Physiol.* 1996;270(5 Pt 2):F833–F838.
150. Steinhausen M, Blum M, Fleming JT, et al. Visualization of renal autoregulation in the split hydronephrotic kidney of rats. *Kidney Int.* 1989;35(5):1151–1160.
151. Heyeraas KJ, Aukland K. Interlobular arterial resistance: influence of renal arterial pressure and angiotensin II. *Kidney Int.* 1987;31(6):1291–1298.
152. Carmines PK, Inscho EW, Gensure RC. Arterial pressure effects on preglomerular microvasculature of juxtamedullary nephrons. *Am J Physiol.* 1990;258(1 Pt 2):F94–F102.
153. Takenaka T, Harrison-Bernard LM, Inscho EW, et al. Autoregulation of afferent arteriolar blood flow in juxtamedullary nephrons. *Am J Physiol.* 1994;267(5 Pt 2):F879–F887.
154. Peti-Peterdi J, Morishima S, Bell PD, et al. Two-photon excitation fluorescence imaging of the living juxtaglomerular apparatus. *Am J Physiol Renal Physiol.* 2002;283(1):F197–F201.
155. Roman RJ, Cowley AW Jr, Garcia-Estan J, et al. Pressure-diuresis in volume-expanded rats. Cortical and medullary hemodynamics. *Hypertension.* 1988;12(2):168–176.
156. Takenaka T, Suzuki H, Okada H, et al. Mechanosensitive cation channels mediate afferent arteriolar myogenic constriction in the isolated rat kidney. *J Physiol.* 1998;511(Pt 1):245–253.
157. Davis MJ, Hill MA. Signaling mechanisms underlying the vascular myogenic response. *Physiol Rev.* 1999;79(2):387–423.
158. Hayashi K, Epstein M, Loutzenhiser R. Enhanced myogenic responsiveness of renal interlobular arteries in spontaneously hypertensive rats. *Hypertension.* 1992;19(2):153–160.
159. Hayashi K, Epstein M, Loutzenhiser R. Determinants of renal actions of atrial natriuretic peptide. Lack of effect of atrial natriuretic peptide on pressure-induced vasoconstriction. *Circ Res.* 1990;67(1):1–10.
160. Wagner AJ, Holstein-Rathlou NH, Marsh DJ. Endothelial Ca2+ in afferent arterioles during myogenic activity. *Am J Physiol.* 1996;270(1 Pt 2):F170–F178.
161. Yip KP, Marsh DJ. [Ca2+]i in rat afferent arteriole during constriction measured with confocal fluorescence microscopy. *Am J Physiol.* 1996;271(5 Pt 2):F1004–F1011.
162. Navar LG, Inscho EW, Imig JD, et al. Heterogeneous activation mechanisms in the renal microvasculature. *Kidney Int Suppl.* 1998;67:S17–S21.
163. Griffin KA, Hacioglu R, Abu-Amarah I, et al. Effects of calcium channel blockers on "dynamic" and "steady-state step" renal autoregulation. *Am J Physiol Renal Physiol.* 2004;286(6):F1136–F1143.
164. Imig JD, Falck JR, Inscho EW. Contribution of cytochrome P450 epoxygenase and hydroxylase pathways to afferent arteriolar autoregulatory responsiveness. *Br J Pharmacol.* 1999;127(6):1399–1405.
165. Baumann JE, Persson PB, Ehmke H, et al. Role of endothelium-derived relaxing factor in renal autoregulation in conscious dogs. *Am J Physiol.* 1992;263(2 Pt 2):F208–F213.

166. Beierwaltes WH, Sigmon DH, Carretero OA. Endothelium modulates renal blood flow but not autoregulation. *Am J Physiol.* 1992;262(6 Pt 2):F943–F949.
167. Majid DS, Inscho EW, Navar LG. P2 purinoceptor saturation by adenosine triphosphate impairs renal autoregulation in dogs. *J Am Soc Nephrol.* 1999;10(3):492–498.
168. Majid DS, Navar LG. Suppression of blood flow autoregulation plateau during nitric oxide blockade in canine kidney. *Am J Physiol.* 1992;262(1 Pt 2):F40–F46.
169. Hoffend J, Cavarape A, Endlich K, et al. Influence of endothelium-derived relaxing factor on renal microvessels and pressure-dependent vasodilation. *Am J Physiol.* 1993;265(2 Pt 2):F285–F292.
170. Navar LG. Integrating multiple paracrine regulators of renal microvascular dynamics. *Am J Physiol.* 1998;274:F433–F444.
171. Bayliss W. On the local reactions of the artrial wall to changes in internal pressure. *J Physiol.* 1902;28:220.
172. Holstein-Rathlou NH, Marsh DJ. A dynamic model of renal blood flow autoregulation. *Bull Math Biol.* 1994;56:411–429.
173. Just A. Mechanisms of renal blood flow autoregulation: dynamics and contributions. *Am J Physiol Regul Integr Comp Physiol.* 2007; 292(1):R1–R17.
174. Just A, Ehmke H, Toktomambetova L, et al. Dynamic characteristics and underlying mechanisms of renal blood flow autoregulation in the conscious dog. *Am J Physiol Renal Physiol.* 2001;280(6):F1062–F1071.
175. Walker MI, Harrison-Bernard LM, Cook AK, et al. Dynamic interaction between myogenic and TGF mechanisms in afferent arteriolar blood flow regulation. *Am J Physiol.* 2000;279:F858–F865.
176. Casellas D, Moore LC. Autoregulation of intravascular pressure in preglomerular juxtamedullary vessels. *Am J Physiol.* 1993;264: F315–F321.
177. Casellas D, Bouriquet N, Moore LC. Branching patterns and autoregulatory responses of juxtamedullary afferent arterioles. *Am J Physiol.* 1997;272(3 Pt 2):F416–F421.
178. Hayashi K, Epstein M, Loutzenhiser R, et al. Impaired myogenic responsiveness of the afferent arteriole in streptozotocin-induced diabetic rats: role of eicosanoid derangements. *J Am Soc Nephrol.* 1992;2(11):1578–1586.
179. Heller J, Horacek V. Autoregulation of superficial nephron function in the alloperfused dog kidney. *Pflugers Arch.* 1979;382(1):99–104.
180. Pelayo JC, Westcott JY. Impaired autoregulation of glomerular capillary hydrostatic pressure in the rat remnant nephron. *J Clin Invest.* 1991;88(1):101–105.
181. Hayashi K, Epstein M, Loutzenhiser R. Pressure-induced vasoconstriction of renal microvessels in normotensive and hypertensive rats. *Circ Res.* 1989;65:1475–1484.
182. Takenaka T, Forster H, De Micheli A, et al. Impaired myogenic responsiveness of renal microvessels in Dahl salt-sensitive rats. *Circ Res.* 1992;71(2):471–480.
183. Cangiotti AM, Lorenzi T, Zingaretti MC, et al. Polarized ends of human macula densa cells: ultrastructural investigation and morphofunctional correlations. *Anat Rec (Hoboken).* 2018;301(5):922–931.
184. Kriz W, Sakai T. Morphological aspects of glomerular function. In: Davison AM, ed. *Nephrology:Proceedings of the Tenth International Congress of Nephrology.* Vol 1988. London: Bailliere-Tindall; 1987:3–23.
185. Schnermann J, Wright FS, Davis JM, et al. Regulation of superficial nephron filtration rate by tubulo-glomerular feedback. *Pflugers Arch.* 1970;318(2):147–175.
186. Vallon V. Tubuloglomerular feedback and the control of glomerular filtration rate. *News Physiol Sci.* 2003;18:169–174.
187. Schnermann J. Localization, mediation and function of the glomerular vascular response to alterations of distal fluid delivery. *Fed Proc.* 1981;40(1):109–115.
188. Navar LG, Bell PD, Burke TJ. Role of a macula densa feedback mechanism as a mediator of renal autoregulation. *Kidney Int Suppl.* 1982;12:S157–S164.
189. Moore LC, Casellas D. Tubuloglomerular feedback dependence of autoregulation in rat juxtamedullary afferent arterioles. *Kidney Int.* 1990;37:1402–1408.
190. Schlatter E, Salomonsson M, Persson AE, et al. Macula densa cells sense luminal NaCl concentration via furosemide sensitive Na+2Cl-K+ cotransport. *Pflugers Arch.* 1989;414(3):286–290.
191. Hashimoto S, Huang Y, Briggs J, et al. Reduced autoregulatory effectiveness in adenosine 1 receptor-deficient mice. *Am J Physiol Renal Physiol.* 2006;290(4):F888–F891.
192. Holstein-Rathlou NH. Oscillations and chaos in renal blood flow control. *J Am Soc Nephrol.* 1993;4(6):1275–1287.
193. Leyssac PP, Baumbach L. An oscillating intratubular pressure response to alterations in Henle loop flow in the rat kidney. *Acta Physiol Scand.* 1983;117(3):415–419.
194. Leyssac PP, Holstein-Rathlou NH. Effects of various transport inhibitors on oscillating TGF pressure responses in the rat. *Pflugers Arch.* 1986;407(3):285–291.
195. Holstein-Rathlou NH, Wagner AJ, Marsh DJ. Tubuloglomerular feedback dynamics and renal blood flow autoregulation in rats. *Am J Physiol.* 1991;260(1 Pt 2):F53–F68.
196. Flemming B, Arenz N, Seeliger E, et al. Time-dependent autoregulation of renal blood flow in conscious rats. *J Am Soc Nephrol.* 2001;12(11):2253–2262.
197. Bell PD, Navar LG. Cytoplasmic calcium in the mediation of macula densa tubulo-glomerular feedback responses. *Science.* 1982; 215(4533):670–673.
198. Gotshall R, Hess T, Mills T. Efficiency of canine renal blood flow autoregulation in kidneys with or without glomerular filtration. *Blood Vessels.* 1985;22(1):25–31.
199. Aukland K, Oien AH. Renal autoregulation: models combining tubuloglomerular feedback and myogenic response. *Am J Physiol.* 1987;252(4 Pt 2):F768–F783.
200. Loutzenhiser R, Griffin K, Williamson G, et al. Renal autoregulation: new perspectives regarding the protective and regulatory roles of the underlying mechanisms. *Am J Physiol Regul Integr Comp Physiol.* 2006;290(5):R1153–R1167.
201. Singh P, Thomson SC. Renal homeostasis and tubuloglomerular feedback. *Curr Opin Nephrol Hypertens.* 2010;19(1):59–64.
202. Schnermann J, Briggs JP. Tubuloglomerular feedback: mechanistic insights from gene-manipulated mice. *Kidney Int.* 2008;74(4):418–426.
203. Gutsche HU, Brunkhorst R, Muller-Ott K, et al. Effect of diuretics on the tubuloglomerular feedback response. *Can J Physiol Pharmacol.* 1984;62(4):412–417.
204. Bell PD, Navar LG. Relationship between tubulo-glomerular feedback responses and perfusate hypotonicity at constant low chloride concentration. *Kidney Int.* 1982;22:234–239.
205. Bell PD, Navar LG, Ploth DW, et al. Tubuloglomerular feedback responses during perfusion with nonelectrolyte solutions in the rat. *Kidney Int.* 1980;18:460–471.
206. Sipos A, Vargas S, Peti-Peterdi J. Direct demonstration of tubular fluid flow sensing by macula densa cells. *Am J Physiol Renal Physiol.* 2010;299(5):F1087–F1093.
207. Bell PD, Lapointe JY, Sabirov R, et al. Macula densa cell signaling involves ATP release through a maxi anion channel. *Proc Natl Acad Sci USA.* 2003;100(7):4322–4327.
208. Inscho EW, Ohishi K, Cook AK, et al. Calcium activation mechanisms in the renal microvascular response to extracellular ATP. *Am J Physiol.* 1995;268(5 Pt 2):F876–F884.
209. Peti-Peterdi J, Bell PD. Cytosolic [Ca2+] signaling pathway in macula densa cells. *Am J Physiol.* 1999;277(3 Pt 2):F472–F476.
210. Hansen PB, Friis UG, Uhrenholt TR, et al. Intracellular signalling pathways in the vasoconstrictor response of mouse afferent arterioles to adenosine. *Acta Physiol (Oxf).* 2007;191(2):89–97.
211. Franco M, Bell PD, Navar LG. Effect of adenosine A1 analogue on tubuloglomerular feedback mechanism. *Am J Physiol.* 1989;257(2 Pt 2):F231–F236.
212. Brown R, Ollerstam A, Johansson B, et al. Abolished tubuloglomerular feedback and increased plasma renin in adenosine A1 receptor-deficient mice. *Am J Physiol Regul Integr Comp Physiol.* 2001;281(5):R1362–R1367.
213. Sun D, Samuelson LC, Yang T, et al. Mediation of tubuloglomerular feedback by adenosine: evidence from mice lacking adenosine 1 receptors. *Proc Natl Acad Sci USA.* 2001;98(17):9983–9988.
214. Thomson S, Bao D, Deng A, et al. Adenosine formed by 5'-nucleotidase mediates tubuloglomerular feedback. *J Clin Invest.* 2000;106(2):289–298.
215. Ren Y, Arima S, Carretero OA, et al. Possible role of adenosine in macula densa control of glomerular hemodynamics. *Kidney Int.* 2002;61(1):169–176.
216. Feng MG, Navar LG. Adenosine A2 receptor activation attenuates afferent arteriolar autoregulation during adenosine receptor saturation in rats. *Hypertension.* 2007;50(4):744–749.
217. Feng MG, Navar LG. Afferent arteriolar vasodilator effect of adenosine predominantly involves adenosine A2B receptor activation. *Am J Physiol Renal Physiol.* 2010;299:F310–F315.
218. Ren Y, Garvin JL, Carretero OA. Efferent arteriole tubuloglomerular feedback in the renal nephron. *Kidney Int.* 2001;59(1):222–229.

219. Ren YL, Garvin JL, Carretero OA. Role of macula densa nitric oxide and cGMP in the regulation of tubuloglomerular feedback. *Kidney Int.* 2000;58:2053–2060.
220. Fan F, Roman RJ. Effect of cytochrome P450 metabolites of arachidonic acid in nephrology. *J Am Soc Nephrol.* 2017;28(10):2845–2855.
221. Schnermann JB, Traynor T, Yang T, et al. Absence of tubuloglomerular feedback responses in AT1A receptor-deficient mice. *Am J Physiol.* 1997;273(2 Pt 2):F315–F320.
222. Schnermann J. Concurrent activation of multiple vasoactive signaling pathways in vasoconstriction caused by tubuloglomerular feedback: a quantitative assessment. *Annu Rev Physiol.* 2015;77:301–322.
223. Traynor T, Yang T, Huang YG, et al. Tubuloglomerular feedback in ACE-deficient mice. *Am J Physiol.* 1999;276(5 Pt 2):F751–F757.
224. Traynor TR, Schnermann J. Renin-angiotensin system dependence of renal hemodynamics in mice. *J Am Soc Nephrol.* 1999;10(suppl 11):S184–S188.
225. Schnermann J, Traynor T, Yang T, et al. Tubuloglomerular feedback: new concepts and developments. *Kidney Int Suppl.* 1998;67:S40–S45.
226. Welch WJ, Wilcox CS. Feedback responses during sequential inhibition of angiotensin and thromboxane. *Am J Physiol.* 1990;258(3 Pt 2):F457–F466.
227. Vallon V. Tubuloglomerular feedback in the kidney: insights from gene-targeted mice. *Pflugers Arch.* 2003;445:470–476.
228. Ploth DW, Roy RN. Renin-angiotensin influences on tubuloglomerular feedback activity in the rat. *Kidney Int.* 1982;22(suppl 12):S114–S121.
229. Wang H, Garvin JL, Carretero OA. Angiotensin II enhances tubuloglomerular feedback via luminal AT(1) receptors on the macula densa. *Kidney Int.* 2001;60(5):1851–1857.
230. Munger KA, Jackson EK. Effects of selective A1 receptor blockade on glomerular hemodynamics: involvement of renin-angiotensin system. *Am J Physiol.* 1994;267(5 Pt 2):F783–F790.
231. Wilcox CS, Welch WJ, Murad F, et al. Nitric oxide synthase in macula densa regulates glomerular capillary pressure. *Proc Natl Acad Sci USA.* 1992;89(24):11993–11997.
232. Ichihara A, Imig JD, Inscho EW, et al. Cyclooxygenase-2 participates in tubular flow-dependent afferent arteriolar tone: interaction with neuronal NOS. *Am J Physiol.* 1998;275(4 Pt 2):F605–F612.
233. Liu R, Carretero OA, Ren Y, et al. Increased intracellular pH at the macula densa activates nNOS during tubuloglomerular feedback. *Kidney Int.* 2005;67(5):1837–1843.
234. Ito S, Ren Y. Evidence for the role of nitric oxide in macula densa control of glomerular hemodynamics. *J Clin Invest.* 1993;92(2):1093–1098.
235. Thorup C, Erik A, Persson G. Macula densa derived nitric oxide in regulation of glomerular capillary pressure. *Kidney Int.* 1996;49(2):430–436.
236. Welch WJ, Wilcox CS. Macula densa arginine delivery and uptake in the rat regulates glomerular capillary pressure. Effects of salt intake. *J Clin Invest.* 1997;100(9):2235–2242.
237. Wilcox CS, Welch WJ. Macula densa nitric oxide synthase: expression, regulation, and function. *Kidney Int Suppl.* 1998;67:S53–S57.
238. Vidal MJ, Romero JC, Vanhoutte PM. Endothelium-derived relaxing factor inhibits renin release. *Eur J Pharmacol.* 1988;149(3):401–402.
239. Kovacs G, Komlosi P, Fuson A, et al. Neuronal nitric oxide synthase: its role and regulation in macula densa cells. *J Am Soc Nephrol.* 2003;14(10):2475–2483.
240. Maddox DA, Troy JL, Brenner BM. Autoregulation of filtration rate in the absence of macula densa-glomerulus feedback. *Am J Physiol.* 1974;227(1):123–131.
241. Dorup J, Morsing P, Rasch R. Tubule-tubule and tubule-arteriole contacts in rat kidney distal nephrons. *Lab Invest.* 1992;67:761–769.
242. Ren Y, Garvin JL, Liu R, et al. Crosstalk between the connecting tubule and the afferent arteriole regulates renal microcirculation. *Kidney Int.* 2007;71(11):1116–1121.
243. Ren Y, D'Ambrosio MA, Garvin JL, et al. Prostaglandin E2 mediates connecting tubule glomerular feedback. *Hypertension.* 2013;62(6):1123–1128.
244. Ren Y, D'Ambrosio MA, Garvin JL, et al. Possible mediators of connecting tubule glomerular feedback. *Hypertension.* 2009;53(2):319–323.
245. Ren Y, D'Ambrosio MA, Garvin JL, et al. Angiotensin II enhances connecting tubule glomerular feedback. *Hypertension.* 2010;56:636–642.
246. Wang H, D'Ambrosio MA, Garvin JL, et al. Connecting tubule glomerular feedback mediates acute tubuloglomerular feedback resetting. *Am J Physiol Renal Physiol.* 2012;302(10):F1300–F1304.
247. Wang H, D'Ambrosio MA, Garvin JL, et al. Connecting tubule glomerular feedback in hypertension. *Hypertension.* 2013;62(4):738–745.
248. Wang H, Romero CA, Masjoan Juncos JX, et al. Effect of salt intake on afferent arteriolar dilatation: role of connecting tubule glomerular feedback (CTGF). *Am J Physiol Renal Physiol.* 2017;313(6):F1209–F1215.
249. Monu SR, Ren Y, Masjoan-Juncos JX, et al. Connecting tubule glomerular feedback mediates tubuloglomerular feedback resetting after unilateral nephrectomy. *Am J Physiol Renal Physiol.* 2018;315(4):F806–F811.
250. Furchgott RF, Zawadzki JV. The obligatory role of endothelial cells in the relaxation of arterial smooth muscle by acetylcholine. *Nature.* 1980;288(5789):373–376.
251. Ignarro LJ, Buga GM, Wood KS, et al. Endothelium-derived relaxing factor produced and released from artery and vein is nitric oxide. *Proc Natl Acad Sci USA.* 1987;84(24):9265–9269.
252. Palmer RM, Ferrige AG, Moncada S. Nitric oxide release accounts for the biological activity of endothelium-derived relaxing factor. *Nature.* 1987;327(6122):524–526.
253. Ignarro LJ. Nitric oxide. A novel signal transduction mechanism for transcellular communication. *Hypertension.* 1990;16(5):477–483.
254. Romero JC, Lahera V, Salom MG, et al. Role of the endothelium-dependent relaxing factor nitric oxide on renal function. *J Am Soc Nephrol.* 1992;2(9):1371–1387.
255. Shultz PJ, Tayeh MA, Marletta MA, et al. Synthesis and action of nitric oxide in rat glomerular mesangial cells. *Am J Physiol.* 1991;261(4 Pt 2):F600–F606.
256. Bachmann S, Bosse HM, Mundel P. Topography of nitric oxide synthesis by localizing constitutive NO synthases in mammalian kidney. *Am J Physiol.* 1995;268(5 Pt 2):F885–F898.
257. Kon V, Harris RC, Ichikawa I. A regulatory role for large vessels in organ circulation. Endothelial cells of the main renal artery modulate intrarenal hemodynamics in the rat. *J Clin Invest.* 1990;85(6):1728–1733.
258. Greenberg SG, He XR, Schnermann JB, et al. Effect of nitric oxide on renin secretion. I. Studies in isolated juxtaglomerular granular cells. *Am J Physiol.* 1995;268(5 Pt 2):F948–F952.
259. Lamontagne D, Pohl U, Busse R. Mechanical deformation of vessel wall and shear stress determine the basal release of endothelium-derived relaxing factor in the intact rabbit coronary vascular bed. *Circ Res.* 1992;70(1):123–130.
260. Murphy ME, Brayden JE. Apamin-sensitive K+ channels mediate an endothelium-dependent hyperpolarization in rabbit mesenteric arteries. *J Physiol.* 1995;489(Pt 3):723–734.
261. Radermacher J, Forstermann U, Frolich JC. Endothelium-derived relaxing factor influences renal vascular resistance. *Am J Physiol.* 1990;259(1 Pt 2):F9–F17.
262. Shultz PJ, Schorer AE, Raij L. Effects of endothelium-derived relaxing factor and nitric oxide on rat mesangial cells. *Am J Physiol.* 1990;258(1 Pt 2):F162–F167.
263. Tolins JP, Palmer RM, Moncada S, et al. Role of endothelium-derived relaxing factor in regulation of renal hemodynamic responses. *Am J Physiol.* 1990;258(3 Pt 2):H655–H662.
264. Lucas KA, Pitari GM, Kazerounian S, et al. Guanylyl cyclases and signaling by cyclic GMP. *Pharmacol Rev.* 2000;52(3):375–414.
265. Rapoport RM. Cyclic guanosine monophosphate inhibition of contraction may be mediated through inhibition of phosphatidylinositol hydrolysis in rat aorta. *Circ Res.* 1986;58(3):407–410.
266. Buga GM, Gold ME, Fukuto JM, et al. Shear stress-induced release of nitric oxide from endothelial cells grown on beads. *Hypertension.* 1991;17(2):187–193.
267. Chin JH, Azhar S, Hoffman BB. Inactivation of endothelial derived relaxing factor by oxidized lipoproteins. *J Clin Invest.* 1992;89:10–18.
268. Cooke JP, Rossitch E Jr, Andon NA, et al. Flow activates an endothelial potassium channel to release an endogenous nitrovasodilator. *J Clin Invest.* 1991;88(5):1663–1671.
269. Luckhoff A, Busse R. Calcium influx into endothelial cells and formation of endothelium-derived relaxing factor is controlled by the membrane potential. *Pflugers Arch.* 1990;416(3):305–311.
270. Marsden PA, Brock TA, Ballermann BJ. Glomerular endothelial cells respond to calcium-mobilizing agonists with release of EDRF. *Am J Physiol.* 1990;258(5 Pt 2):F1295–F1303.
271. Handa RK, Strandhoy JW. Nitric oxide mediates the inhibitory action of platelet-activating factor on angiotensin II-induced renal vasoconstriction, in vivo. *J Pharmacol Exp Ther.* 1996;277(3):1486–1491.

272. Edwards RM, Pullen M, Nambi P. Activation of endothelin ETB receptors increases glomerular cGMP via an L-arginine-dependent pathway. *Am J Physiol*. 1992;263(6 Pt 2):F1020–F1025.
273. Fiscus RR, Zhou HL, Wang X, et al. Calcitonin gene-related peptide (CGRP)-induced cyclic AMP, cyclic GMP and vasorelaxant responses in rat thoracic aorta are antagonized by blockers of endothelium-derived relaxant factor (EDRF). *Neuropeptides*. 1991;20(2):133–143.
274. Samuelson UE, Jernbeck J. Calcitonin gene-related peptide relaxes porcine arteries via one endothelium-dependent and one endothelium-independent mechanism. *Acta Physiol Scand*. 1991; 141(2):281–282.
275. Gray DW, Marshall I. Nitric oxide synthesis inhibitors attenuate calcitonin gene-related peptide endothelium-dependent vasorelaxation in rat aorta. *Eur J Pharmacol*. 1992;212(1):37–42.
276. Gray DW, Marshall I. Human alpha-calcitonin gene-related peptide stimulates adenylate cyclase and guanylate cyclase and relaxes rat thoracic aorta by releasing nitric oxide. *Br J Pharmacol*. 1992;107(3):691–696.
277. Hutcheson IR, Griffith TM. Release of endothelium-derived relaxing factor is modulated both by frequency and amplitude of pulsatile flow. *Am J Physiol*. 1991;261(1 Pt 2):H257–H262.
278. Koller A, Kaley G. Endothelial regulation of wall shear stress and blood flow in skeletal muscle microcirculation. *Am J Physiol*. 1991;260(3 Pt 2):H862–H868.
279. Nollert MU, Eskin SG, McIntire LV. Shear stress increases inositol trisphosphate levels in human endothelial cells. *Biochem Biophys Res Commun*. 1990;170(1):281–287.
280. O'Neill WC. Flow-mediated NO release from endothelial cells is independent of K+ channel activation or intracellular Ca2+. *Am J Physiol*. 1995;269(4 Pt 1):C863–C869.
281. Pohl U, Herlan K, Huang A, et al. EDRF-mediated shear-induced dilation opposes myogenic vasoconstriction in small rabbit arteries. *Am J Physiol*. 1991;261(6 Pt 2):H2016–H2023.
282. Pittner J, Wolgast M, Casellas D, et al. Increased shear stress-released NO and decreased endothelial calcium in rat isolated perfused juxtamedullary nephrons. *Kidney Int*. 2005;67(1):227–236.
283. Blantz RC, Deng A, Lortie M, et al. The complex role of nitric oxide in the regulation of glomerular ultrafiltration. *Kidney Int*. 2002;61(3):782–785.
284. Mount PF, Power DA. Nitric oxide in the kidney: functions and regulation of synthesis. *Acta Physiol (Oxf)*. 2006;187(4):433–446.
285. Welch WJ, Tojo A, Lee JU, et al. Nitric oxide synthase in the JGA of the SHR: expression and role in tubuloglomerular feedback. *Am J Physiol*. 1999;277(1 Pt 2):F130–F138.
286. Majid DSA, Chin S, Omoro S, et al. Direct assessment of intrarenal nitric oxide (NO) levels during changes in renal arterial pressure (RAP) in anesthetized dogs. *FASEB J*. 1996;10:A40.
287. Navar LG, Majid DSA. Contribution of intrarenal nitric oxide to the mechanism of pressure natriuresis. *J Physiol Biochem*. 1997; 53:34.
288. Edgley AJ, Tare M, Evans RG, et al. In vivo regulation of endothelium-dependent vasodilation in the rat renal circulation and the effect of streptozotocin-induced diabetes. *Am J Physiol Regul Integr Comp Physiol*. 2008;295(3):R829–R839.
289. Jin C, Hu C, Polichnowski A, et al. Effects of renal perfusion pressure on renal medullary hydrogen peroxide and nitric oxide production. *Hypertension*. 2009;53(6):1048–1053.
290. Majid DSA, Navar LG. Nitric oxide in the control of renal hemodynamics and excretory function. *Am J Hypertens*. 2001;14: 74S–82S.
291. Zhang Z, Pallone TL. Response of descending vasa recta to luminal pressure. *Am J Physiol Renal Physiol*. 2004;287(3):F535–F542.
292. Gabbai FB, Blantz RC. Role of nitric oxide in renal hemodynamics. *Semin Nephrol*. 1999;19(3):242–250.
293. Baylis C, Harton P, Engels K. Endothelial derived relaxing factor controls renal hemodynamics in the normal rat kidney. *J Am Soc Nephrol*. 1990;1(6):875–881.
294. Treeck B, Aukland K. Effect of L-NAME on glomerular filtration rate in deep and superficial layers of rat kidneys. *Am J Physiol*. 1997;272(3 Pt 2):F312–F318.
295. Sigmon DH, Beierwaltes WH. Influence of nitric oxide derived from neuronal nitric oxide synthase on glomerular filtration. *Gen Pharmacol*. 2000;34(2):95–100.
296. Baylis C, Mitruka B, Deng A. Chronic blockade of nitric oxide synthesis in the rat produces systemic hypertension and glomerular damage. *J Clin Invest*. 1992;90(1):278–281.
297. Deng A, Baylis C. Locally produced EDRF controls preglomerular resistance and ultrafiltration coefficient. *Am J Physiol*. 1993;264(2 Pt 2):F212–F215.
298. Gonzalez JD, Llinas MT, Nava E, et al. Role of nitric oxide and prostaglandins in the long-term control of renal function. *Hypertension*. 1998;32(1):33–38.
299. Qiu C, Baylis C. Endothelin and angiotensin mediate most glomerular responses to nitric oxide inhibition. *Kidney Int*. 1999;55(6): 2390–2396.
300. Zatz R, de Nucci G. Effects of acute nitric oxide inhibition on rat glomerular microcirculation. *Am J Physiol*. 1991;261(2 Pt 2): F360–F363.
301. Ito S, Juncos LA, Nushiro N, et al. Endothelium-derived relaxing factor modulates endothelin action in afferent arterioles. *Hypertension*. 1991;17(6 Pt 2):1052–1056.
302. Ohishi K, Carmines PK, Inscho EW, et al. EDRF-angiotensin II interactions in rat juxtamedullary afferent and efferent arterioles. *Am J Physiol*. 1992;263(5 Pt 2):F900–F906.
303. Ito S, Arima S, Ren YL, et al. Endothelium-derived relaxing factor/ nitric oxide modulates angiotensin II action in the isolated microperfused rabbit afferent but not efferent arteriole. *J Clin Invest*. 1993;91(5):2012–2019.
304. Ichihara A, Inscho EW, Imig JD, et al. Neuronal nitric oxide synthase modulates rat renal microvascular function. *Am J Physiol Renal Physiol*. 1998;274:F516–F524.
305. Sigmon DH, Carretero OA, Beierwaltes WH. Endothelium-derived relaxing factor regulates renin release in vivo. *Am J Physiol*. 1992;263(2 Pt 2):F256–F261.
306. Moreno C, Lopez A, Llinas MT, et al. Changes in NOS activity and protein expression during acute and prolonged Ang II administration. *Am J Physiol Regul Integr Comp Physiol*. 2002;282(1):R31–R37.
307. Patzak A, Lai EY, Mrowka R, et al. AT1 receptors mediate angiotensin II-induced release of nitric oxide in afferent arterioles. *Kidney Int*. 2004;66(5):1949–1958.
308. Deng A, Miracle CM, Suarez JM, et al. Oxygen consumption in the kidney: effects of nitric oxide synthase isoforms and angiotensin II. *Kidney Int*. 2005;68(2):723–730.
309. Baylis C, Engels K, Samsell L, et al. Renal effects of acute endothelial-derived relaxing factor blockade are not mediated by angiotensin II. *Am J Physiol*. 1993;264(1 Pt 2):F74–F78.
310. Baylis C, Harvey J, Engels K. Acute nitric oxide blockade amplifies the renal vasoconstrictor actions of angiotension II. *J Am Soc Nephrol*. 1994;5(2):211–214.
311. Schnackenberg CG, Wilkins FC, Granger JP. Role of nitric oxide in modulating the vasoconstrictor actions of angiotensin II in preglomerular and postglomerular vessels in dogs. *Hypertension*. 1995;26(6 Pt 2):1024–1029.
312. Stec DE, Vera T, Storm MV, et al. Blood pressure and renal blow flow responses in heme oxygenase-2 knockout mice. *Am J Physiol Regul Integr Comp Physiol*. 2009;297(6):R1822–R1828.
313. Botros FT, Dobrowolski L, Navar LG. Renal heme oxygenase-1 induction with hemin augments renal hemodynamics, renal autoregulation, and excretory function. *Int J Hypertens*. 2012;2012:189512.
314. Jackson KE, Jackson DW, Quadri S, et al. Inhibition of heme oxygenase augments tubular sodium reabsorption. *Am J Physiol Renal Physiol*. 2011;300(4):F941–F946.
315. Abraham NG, Kappas A. Heme oxygenase and the cardiovascular-renal system. *Free Radic Biol Med*. 2005;39(1):1–25.
316. Zou AP, Billington H, Su N, et al. Expression and actions of heme oxygenase in the renal medulla of rats. *Hypertension*. 2000;35(1 Pt 2): 342–347.
317. Li N, Yi F, dos Santos EA, et al. Role of renal medullary heme oxygenase in the regulation of pressure natriuresis and arterial blood pressure. *Hypertension*. 2007;49(1):148–154.
318. Ryan MJ, Jernigan NL, Drummond HA, et al. Renal vascular responses to CORM-A1 in the mouse. *Pharmacol Res*. 2006;54(1):24–29.
319. Arregui B, Lopez B, Garcia Salom M, et al. Acute renal hemodynamic effects of dimanganese decacarbonyl and cobalt protoporphyrin. *Kidney Int*. 2004;65(2):564–574.
320. Kasinath BS, Feliers D, Lee HJ. Hydrogen sulfide as a regulatory factor in kidney health and disease. *Biochem Pharmacol*. 2018;149:29–41.
321. Feliers D, Lee HJ, Kasinath BS. Hydrogen sulfide in renal physiology and disease. *Antioxid Redox Signal*. 2016;25(13):720–731.
322. Xia M, Chen L, Muh RW, et al. Production and actions of hydrogen sulfide, a novel gaseous bioactive substance, in the kidneys. *J Pharmacol Exp Ther*. 2009;329(3):1056–1062.

323. Jung KJ, Jang HS, Kim JI, et al. Involvement of hydrogen sulfide and homocysteine transsulfuration pathway in the progression of kidney fibrosis after ureteral obstruction. *Biochim Biophys Acta*. 2013;1832(12):1989–1997.
324. Jackson-Weaver O, Osmond JM, Riddle MA, et al. Hydrogen sulfide dilates rat mesenteric arteries by activating endothelial large-conductance Ca(2)(+)-activated K(+) channels and smooth muscle Ca(2)(+) sparks. *Am J Physiol Heart Circ Physiol*. 2013;304(11):H1446–H1454.
325. Bucci M, Papapetropoulos A, Vellecco V, et al. cGMP-dependent protein kinase contributes to hydrogen sulfide-stimulated vasorelaxation. *PLoS ONE*. 2012;7(12):e53319.
326. Materazzi S, Zagli G, Nassini R, et al. Vasodilator activity of hydrogen sulfide (H2S) in human mesenteric arteries. *Microvasc Res*. 2017;109:38–44.
327. Li Y, Zang Y, Fu S, et al. H2S relaxes vas deferens smooth muscle by modulating the large conductance Ca2+ -activated K+ (BKCa) channels via a redox mechanism. *J Sex Med*. 2012;9(11):2806–2813.
328. Huang Y, Zhang Z, Huang Y, et al. Induction of inactive TGF-beta1 monomer formation by hydrogen sulfide contributes to its suppressive effects on Ang II- and TGF-beta1-induced EMT in renal tubular epithelial cells. *Biochem Biophys Res Commun*. 2018;501(2):534–540.
329. Gill PS, Wilcox CS. NADPH oxidases in the kidney. *Antioxid Redox Signal*. 2006;8(9–10):1597–1607.
330. Fellner SK, Arendshorst WJ. Angiotensin II, reactive oxygen species, and Ca2+ signaling in afferent arterioles. *Am J Physiol Renal Physiol*. 2005;289(5):F1012–F1019.
331. Carlstrom M, Lai EY, Ma Z, et al. Role of NOX2 in the regulation of afferent arteriole responsiveness. *Am J Physiol Regul Integr Comp Physiol*. 2009;296(1):R72–R79.
332. Lai EY, Solis G, Luo Z, et al. p47(phox) is required for afferent arteriolar contractile responses to angiotensin II and perfusion pressure in mice. *Hypertension*. 2012;59(2):415–420.
333. Wang C, Luo Z, Kohan D, et al. Thromboxane prostanoid receptors enhance contractions, endothelin-1, and oxidative stress in microvessels from mice with chronic kidney disease. *Hypertension*. 2015;65(5):1055–1063.
334. Sharma K, Cook A, Smith M, et al. TGF-beta impairs renal autoregulation via generation of ROS. *Am J Physiol Renal Physiol*. 2005;288(5):F1069–F1077.
335. Li L, Lai EY, Wellstein A, et al. Differential effects of superoxide and hydrogen peroxide on myogenic signaling, membrane potential, and contractions of mouse renal afferent arterioles. *Am J Physiol Renal Physiol*. 2016;310(11):F1197–F1205.
336. Li L, Lai EY, Luo Z, et al. Superoxide and hydrogen peroxide counterregulate myogenic contractions in renal afferent arterioles from a mouse model of chronic kidney disease. *Kidney Int*. 2017;92(3):625–633.
337. Vogel PA, Yang X, Moss NG, et al. Superoxide enhances Ca2+ entry through L-type channels in the renal afferent arteriole. *Hypertension*. 2015;66(2):374–381.
338. Li L, Feng D, Luo Z, et al. Remodeling of afferent arterioles from mice with oxidative stress does not account for increased contractility but does limit excessive wall stress. *Hypertension*. 2015;66(3):550–556.
339. Song J, Lu Y, Lai EY, et al. Oxidative status in the macula densa modulates tubuloglomerular feedback responsiveness in angiotensin II-induced hypertension. *Acta Physiol (Oxf)*. 2015;213(1):249–258.
340. Lai EY, Wellstein A, Welch WJ, et al. Superoxide modulates myogenic contractions of mouse afferent arterioles. *Hypertension*. 2011;58(4):650–656.
341. Zhang S, Huang Q, Wang Q, et al. Enhanced renal afferent arteriolar reactive oxygen species and contractility to endothelin-1 are associated with canonical wnt signaling in diabetic mice. *Kidney Blood Press Res*. 2018;43(3):860–871.
342. Ren Y, D'Ambrosio MA, Liu R, et al. Enhanced myogenic response in the afferent arteriole of spontaneously hypertensive rats. *Am J Physiol Heart Circ Physiol*. 2010;298(6):H1769–H1775.
343. Huang Q, Wang Q, Zhang S, et al. Increased hydrogen peroxide impairs angiotensin II contractions of afferent arterioles in mice after renal ischaemia-reperfusion injury. *Acta Physiol (Oxf)*. 2016;218(2):136–145.
344. Yanagisawa M, Kurihara H, Kimura S, et al. A novel potent vasoconstrictor peptide produced by vascular endothelial cells. *Nature*. 1988;332(6163):411–415.
345. Goraca A. New views on the role of endothelin (minireview). *Endocr Regul*. 2002;36(4):161–167.
346. Simonson MS, Dunn MJ. Endothelin-1 stimulates contraction of rat glomerular mesangial cells and potentiates beta-adrenergic-mediated cyclic adenosine monophosphate accumulation. *J Clin Invest*. 1990;85(3):790–797.
347. Barnes K, Brown C, Turner AJ. Endothelin-converting enzyme: ultrastructural localization and its recycling from the cell surface. *Hypertension*. 1998;31(1):3–9.
348. Barnes K, Murphy LJ, Takahashi M, et al. Localization and biochemical characterization of endothelin-converting enzyme. *J Cardiovasc Pharmacol*. 1995;26(suppl 3):S37–S39.
349. Bakris GL, Fairbanks R, Traish AM. Arginine vasopressin stimulates human mesangial cell production of endothelin. *J Clin Invest*. 1991;87(4):1158–1164.
350. Herman WH, Emancipator SN, Rhoten RL, et al. Vascular and glomerular expression of endothelin-1 in normal human kidney. *Am J Physiol*. 1998;275(1 Pt 2):F8–F17.
351. Karet FE, Davenport AP. Localization of endothelin peptides in human kidney. *Kidney Int*. 1996;49(2):382–387.
352. Kasinath BS, Fried TA, Davalath S, et al. Glomerular epithelial cells synthesize endothelin peptides. *Am J Pathol*. 1992;141(2):279–283.
353. Kohan DE. Production of endothelin-1 by rat mesangial cells: regulation by tumor necrosis factor. *J Lab Clin Med*. 1992;119(5):477–484.
354. Marsden PA, Dorfman DM, Collins T, et al. Regulated expression of endothelin 1 in glomerular capillary endothelial cells. *Am J Physiol*. 1991;261(1 Pt 2):F117–F125.
355. Sakamoto H, Sasaki S, Nakamura Y, et al. Regulation of endothelin-1 production in cultured rat mesangial cells. *Kidney Int*. 1992;41(2):350–355.
356. Sakamoto H, Sasaki S, Hirata Y, et al. Production of endothelin-1 by rat cultured mesangial cells. *Biochem Biophys Res Commun*. 1990;169(2):462–468.
357. Ujiie K, Terada Y, Nonoguchi H, et al. Messenger RNA expression and synthesis of endothelin-1 along rat nephron segments. *J Clin Invest*. 1992;90(3):1043–1048.
358. Wilkes BM, Susin M, Mento PF, et al. Localization of endothelin-like immunoreactivity in rat kidneys. *Am J Physiol*. 1991;260(6 Pt 2):F913–F920.
359. Zoja C, Orisio S, Perico N, et al. Constitutive expression of endothelin gene in cultured human mesangial cells and its modulation by transforming growth factor-beta, thrombin, and a thromboxane A2 analogue. *Lab Invest*. 1991;64(1):16–20.
360. Badr KF, Munger KA, Sugiura M, et al. High and low affinity binding sites for endothelin on cultured rat glomerular mesangial cells. *Biochem Biophys Res Commun*. 1989;161(2):776–781.
361. Kohan DE. Endothelins in the normal and diseased kidney. *Am J Kidney Dis*. 1997;29(1):2–26.
362. Madeddu P, Troffa C, Glorioso N, et al. Effect of endothelin on regional hemodynamics and renal function in awake normotensive rats. *J Cardiovasc Pharmacol*. 1989;14(6):818–825.
363. Clozel M, Fischli W. Human cultured endothelial cells do secrete endothelin-1. *J Cardiovasc Pharmacol*. 1989;13(suppl 5):S229–S231.
364. Marsden PA, Danthuluri NR, Brenner BM, et al. Endothelin action on vascular smooth muscle involves inositol trisphosphate and calcium mobilization. *Biochem Biophys Res Commun*. 1989;158(1):86–93.
365. Martin ER, Brenner BM, Ballermann BJ. Heterogeneity of cell surface endothelin receptors. *J Biol Chem*. 1990;265(23):14044–14049.
366. Sakurai T, Yanagisawa M, Masaki T. Molecular characterization of endothelin receptors. *Trends Pharmacol Sci*. 1992;13(3):103–108.
367. Pollock DM, Keith TL, Highsmith RF. Endothelin receptors and calcium signaling. *FASEB J*. 1995;9(12):1196–1204.
368. Deng Y, Martin LL, DelGrande D, et al. A soluble protease identified from rat kidney degrades endothelin-1 but not proendothelin-1. *J Biochem*. 1992;112(1):168–172.
369. Maguire JJ, Davenport AP. Endothelin receptors and their antagonists. *Semin Nephrol*. 2015;35(2):125–136.
370. De Miguel C, Speed JS, Kasztan M, et al. Endothelin-1 and the kidney: new perspectives and recent findings. *Curr Opin Nephrol Hypertens*. 2016;25(1):35–41.
371. Katusic ZS, Shepherd JT, Vanhoutte PM. Endothelium-dependent contraction to stretch in canine basilar arteries. *Am J Physiol*. 1987;252(3 Pt 2):H671–H673.
372. Yoshizumi M, Kurihara H, Sugiyama T, et al. Hemodynamic shear stress stimulates endothelin production by cultured endothelial cells. *Biochem Biophys Res Commun*. 1989;161(2):859–864.

373. Herizi A, Jover B, Bouriquet N, et al. Prevention of the cardiovascular and renal effects of angiotensin II by endothelin blockade. *Hypertension*. 1998;31:10–14.
374. Kohno M, Horio T, Ikeda M, et al. Angiotensin II stimulates endothelin-1 secretion in cultured rat mesangial cells. *Kidney Int*. 1992;42(4):860–866.
375. Marsden PA, Brenner BM. Transcriptional regulation of the endothelin-1 gene by TNF-alpha. *Am J Physiol*. 1992;262(4 Pt 1):C854–C861.
376. Rajagopalan S, Laursen JB, Borthayre A, et al. Role for endothelin-1 in angiotensin II-mediated hypertension. *Hypertension*. 1997;30(1 Pt 1):29–34.
377. Munger KA, Sugiura M, Takahashi K, et al. A role for atrial natriuretic peptide in endothelin-induced natriuresis. *J Am Soc Nephrol*. 1991;1(12):1278–1283.
378. Gohar EY, Kasztan M, Pollock DM. Interplay between renal endothelin and purinergic signaling systems. *Am J Physiol Renal Physiol*. 2017;313(3):F666–F668.
379. King AJ, Brenner BM, Anderson S. Endothelin: a potent renal and systemic vasoconstrictor peptide. *Am J Physiol*. 1989;256(6 Pt 2):F1051–F1058.
380. Badr KF, Murray JJ, Breyer MD, et al. Mesangial cell, glomerular and renal vascular responses to endothelin in the rat kidney. Elucidation of signal transduction pathways. *J Clin Invest*. 1989;83(1):336–342.
381. Clavell AL, Stingo AJ, Margulies KB, et al. Role of endothelin receptor subtypes in the in vivo regulation of renal function. *Am J Physiol*. 1995;268(3 Pt 2):F455–F460.
382. Heller J, Kramer HJ, Horacek V. Action of endothelin-1 on glomerular haemodynamics in the dog: lack of direct effects on glomerular ultrafiltration coefficient. *Clin Sci*. 1996;90(5):385–391.
383. Perico N, Dadan J, Gabanelli M, et al. Cyclooxygenase products and atrial natriuretic peptide modulate renal response to endothelin. *J Pharmacol Exp Ther*. 1990;252(3):1213–1220.
384. Stacy DL, Scott JW, Granger JP. Control of renal function during intrarenal infusion of endothelin. *Am J Physiol*. 1990;258(5 Pt 2):F1232–F1236.
385. Kon V, Yoshioka T, Fogo A, et al. Glomerular actions of endothelin in vivo. *J Clin Invest*. 1989;83(5):1762–1767.
386. Fretschner M, Endlich K, Gulbins E, et al. Effects of endothelin on the renal microcirculation of the split hydronephrotic rat kidney. *Ren Physiol Biochem*. 1991;14(3):112–127.
387. Loutzenhiser R, Epstein M, Hayashi K, et al. Direct visualization of effects of endothelin on the renal microvasculature. *Am J Physiol*. 1990;258(1 Pt 2):F61–F68.
388. Edwards RM, Trizna W, Ohlstein EH. Renal microvascular effects of endothelin. *Am J Physiol*. 1990;259(2 Pt 2):F217–F221.
389. Lanese DM, Yuan BH, McMurtry IF, et al. Comparative sensitivities of isolated rat renal arterioles to endothelin. *Am J Physiol*. 1992;263(5 Pt 2):F894–F899.
390. Dlugosz JA, Munk S, Zhou X, et al. Endothelin-1-induced mesangial cell contraction involves activation of protein kinase C-alpha, -delta, and -epsilon. *Am J Physiol*. 1998;275(3 Pt 2):F423–F432.
391. Noll G, Wenzel RR, Luscher TF. Endothelin and endothelin antagonists: potential role in cardiovascular and renal disease. *Mol Cell Biochem*. 1996;157(1–2):259–267.
392. Lin H, Smith MJ Jr, Young DB. Roles of prostaglandins and nitric oxide in the effect of endothelin-1 on renal hemodynamics. *Hypertension*. 1996;28(3):372–378.
393. Momose N, Fukuo K, Morimoto S, et al. Captopril inhibits endothelin-1 secretion from endothelial cells through bradykinin. *Hypertension*. 1993;21(6 Pt 2):921–924.
394. Prins BA, Hu RM, Nazario B, et al. Prostaglandin E2 and prostacyclin inhibit the production and secretion of endothelin from cultured endothelial cells. *J Biol Chem*. 1994;269(16):11938–11944.
395. Chou SY, Dahhan A, Porush JG. Renal actions of endothelin: interaction with prostacyclin. *Am J Physiol*. 1990;259(4 Pt 2):F645–F652.
396. Gohar EY, Giachini FR, Pollock DM, et al. Role of the endothelin system in sexual dimorphism in cardiovascular and renal diseases. *Life Sci*. 2016;159:20–29.
397. Arai H, Hori S, Aramori I, et al. Cloning and expression of a cDNA encoding an endothelin receptor. *Nature*. 1990;348(6303):730–732.
398. Sakurai T, Yanagisawa M, Takuwa Y, et al. Cloning of a cDNA encoding a non-isopeptide-selective subtype of the endothelin receptor. *Nature*. 1990;348(6303):732–735.
399. Ihara M, Noguchi K, Saeki T, et al. Biological profiles of highly potent novel endothelin antagonists selective for the ETA receptor. *Life Sci*. 1992;50(4):247–255.
400. Wendel M, Knels L, Kummer W, et al. Distribution of endothelin receptor subtypes ETA and ETB in the rat kidney. *J Histochem Cytochem*. 2006;54(11):1193–1203.
401. Qiu C, Engels K, Baylis C. Endothelin modulates the pressor actions of acute systemic nitric oxide blockade. *J Am Soc Nephrol*. 1995;6(5):1476–1481.
402. Heerspink HJL, Andress DL, Bakris G, et al. Rationale and protocol of the Study Of diabetic Nephropathy with AtRasentan (SONAR) trial: a clinical trial design novel to diabetic nephropathy. *Diabetes Obes Metab*. 2018;20(6):1369–1376.
403. de Zeeuw D, Coll B, Andress D, et al. The endothelin antagonist atrasentan lowers residual albuminuria in patients with type 2 diabetic nephropathy. *J Am Soc Nephrol*. 2014;25(5):1083–1093.
404. Kohan DE, Pritchett Y, Molitch M, et al. Addition of atrasentan to renin-angiotensin system blockade reduces albuminuria in diabetic nephropathy. *J Am Soc Nephrol*. 2011;22(4):763–772.
405. Mann JF, Green D, Jamerson K, et al. Avosentan for overt diabetic nephropathy. *J Am Soc Nephrol*. 2010;21(3):527–535.
406. Oyekan A, Balazy M, McGiff JC. Renal oxygenases: differential contribution to vasoconstriction induced by ET-1 and Ang II. *Am J Physiol*. 1997;273(1 Pt 2):R293–R300.
407. Stier CT Jr, Quilley CP, McGiff JC. Endothelin-3 effects on renal function and prostanoid release in the rat isolated kidney. *J Pharmacol Exp Ther*. 1992;262(1):252–256.
408. Owada A, Tomita K, Terada Y, et al. Endothelin (ET)-3 stimulates cyclic guanosine 3',5'-monophosphate production via ETB receptor by producing nitric oxide in isolated rat glomerulus, and in cultured rat mesangial cells. *J Clin Invest*. 1994;93(2):556–563.
409. Filep JG. Endogenous endothelin modulates blood pressure, plasma volume, and albumin escape after systemic nitric oxide blockade. *Hypertension*. 1997;30(1 Pt 1):22–28.
410. Thompson A, Valeri CR, Lieberthal W. Endothelin receptor A blockade alters hemodynamic response to nitric oxide inhibition in rats. *Am J Physiol*. 1995;269(2 Pt 2):H743–H748.
411. Gurbanov K, Rubinstein I, Hoffman A, et al. Differential regulation of renal regional blood flow by endothelin-1. *Am J Physiol*. 1996;271(6 Pt 2):F1166–F1172.
412. Crawford C, Kennedy-Lydon T, Sprott C, et al. An intact kidney slice model to investigate vasa recta properties and function in situ. *Nephron Physiol*. 2012;120(3):p17–p31.
413. Prieto MC, Gonzalez AA, Navar LG. Evolving concepts on regulation and function of renin in distal nephron. *Pflugers Arch*. 2013;465(1):121–132.
414. Ramkumar N, Kohan DE. Role of collecting duct renin in blood pressure regulation. *Am J Physiol Regul Integr Comp Physiol*. 2013;305(2):R92–R94.
415. Rohrwasser A, Morgan T, Dillon HF, et al. Elements of a paracrine tubular renin-angiotensin system along the entire nephron. *Hypertension*. 1999;34(6):1265–1274.
416. Navar LG. Physiology: hemodynamics, endothelial function, renin-angiotensin-aldosterone system, sympathetic nervous system. *J Am Soc Hypertens*. 2014;8(7):519–524.
417. Denton KM, Anderson WP, Sinniah R. Effects of angiotensin II on regional afferent and efferent arteriole dimensions and the glomerular pole. *Am J Physiol Regul Integr Comp Physiol*. 2000;279(2):R629–R638.
418. Yuan BH, Robinette JB, Conger JD. Effect of angiotensin II and norepinephrine on isolated rat afferent and efferent arterioles. *Am J Physiol*. 1990;258(3 Pt 2):F741–F750.
419. Navar LG, Harrison-Bernard LM, Imig JD, et al. Renal responses to AT1 receptor blockade. *Am J Hypertens*. 2000;13(1 Pt 2):45S–54S.
420. Navar LG. Intrarenal renin-angiotensin system in regulation of glomerular function. *Curr Opin Nephrol Hypertens*. 2014;23(1):38–45.
421. Harrison-Bernard LM, Monjure CJ, Bivona BJ. Efferent arterioles exclusively express the subtype 1A angiotensin receptor: functional insights from genetic mouse models. *Am J Physiol Renal Physiol*. 2006;290(5):F1177–F1186.
422. Juncos LA, Ren Y, Arima S, et al. Angiotensin II action in isolated microperfused rabbit afferent arterioles is modulated by flow. *Kidney Int*. 1996;49(2):374–381.
423. Purdy KE, Arendshorst WJ. Prostaglandins buffer Ang II-mediated increases in cytosolic calcium in preglomerular VSMC. *Am J Physiol*. 1999;277(6 Pt 2):F850–F858.
424. Kohagura K, Endo Y, Ito O, et al. Endogenous nitric oxide and epoxyeicosatrienoic acids modulate angiotensin II-induced

constriction in the rabbit afferent arteriole. *Acta Physiol Scand*. 2000;168(1):107–112.
425. Patzak A, Kleinmann F, Lai EY, et al. Nitric oxide counteracts angiotensin II induced contraction in efferent arterioles in mice. *Acta Physiol Scand*. 2004;181(4):439–444.
426. Patzak A, Lai E, Persson PB, et al. Angiotensin II-nitric oxide interaction in glomerular arterioles. *Clin Exp Pharmacol Physiol*. 2005;32(5–6):410–414.
427. Baylis C, Brenner BM. Modulation by prostaglandin synthesis inhibitors of the action of exogenous angiotensin II on glomerular ultrafiltration in the rat. *Circ Res*. 1978;43(6):889–898.
428. Takeda K, Meyer-Lehnert H, Kim JK, et al. Effect of angiotensin II on Ca2+ kinetics and contraction in cultured rat glomerular mesangial cells. *Am J Physiol*. 1988;254(2 Pt 2):F254–F266.
429. Wiegmann TB, MacDougall ML, Savin VJ. Glomerular effects of angiotensin II require intrarenal factors. *Am J Physiol*. 1990;258(3 Pt 2):F717–F721.
430. Sharma M, Sharma R, Greene AS, et al. Documentation of angiotensin II receptors in glomerular epithelial cells. *Am J Physiol*. 1998;274(3 Pt 2):F623–F627.
431. Pagtalunan ME, Rasch R, Rennke HG, et al. Morphometric analysis of effects of angiotensin II on glomerular structure in rats. *Am J Physiol*. 1995;268(1 Pt 2):F82–F88.
432. Inishi Y, Okuda T, Arakawa T, et al. Insulin attenuates intracellular calcium responses and cell contraction caused by vasoactive agents. *Kidney Int*. 1994;45(5):1318–1325.
433. Arima S, Endo Y, Yaoita H, et al. Possible role of P-450 metabolite of arachidonic acid in vasodilator mechanism of angiotensin II type 2 receptor in the isolated microperfused rabbit afferent arteriole. *J Clin Invest*. 1997;100(11):2816–2823.
434. Ren Y, Garvin JL, Carretero OA. Vasodilator action of angiotensin-(1-7) on isolated rabbit afferent arterioles. *Hypertension*. 2002;39(3):799–802.
435. Santos RA, Simoes e Silva AC, Maric C, et al. Angiotensin-(1-7) is an endogenous ligand for the G protein-coupled receptor Mas. *Proc Natl Acad Sci USA*. 2003;100(14):8258–8263.
436. Chappell MC. Emerging evidence for a functional angiotensin-converting enzyme 2-angiotensin-(1-7)-MAS receptor axis: more than regulation of blood pressure? *Hypertension*. 2007;50(4):596–599.
437. Tipnis SR, Hooper NM, Hyde R, et al. A human homolog of angiotensin-converting enzyme. Cloning and functional expression as a captopril-insensitive carboxypeptidase. *J Biol Chem*. 2000;275(43):33238–33243.
438. Turner AJ, Tipnis SR, Guy JL, et al. ACEH/ACE2 is a novel mammalian metallocarboxypeptidase and a homologue of angiotensin-converting enzyme insensitive to ACE inhibitors. *Can J Physiol Pharmacol*. 2002;80(4):346–353.
439. Zisman LS, Meixell GE, Bristow MR, et al. Angiotensin-(1-7) formation in the intact human heart: in vivo dependence on angiotensin II as substrate. *Circulation*. 2003;108(14):1679–1681.
440. Gonzalez AA, Lara LS, Prieto MC. Role of Collecting Duct Renin in the Pathogenesis of Hypertension. *Curr Hypertens Rep*. 2017;19(8):62.
441. Maddox DA, Bennett CM, Deen WM, et al. Determinants of glomerular filtration in experimental glomerulonephritis in the rat. *J Clin Invest*. 1975;55(2):305–318.
442. Myers BD, Deen WM, Brenner B. Effects of norepinephrine and angiotensin II on the determinants of glomerular ultrafiltration and proximal tubule fluid reabsorption in the rat. *Circ Res*. 1975;37:101–110.
443. Anderson S, Rennke HG, Brenner BM. Therapeutic advantage of converting enzyme inhibitors in arresting progressive renal disease associated with systemic hypertension in the rat. *J Clin Invest*. 1986;77(6):1993–2000.
444. Ruster C, Wolf G. Renin-angiotensin-aldosterone system and progression of renal disease. *J Am Soc Nephrol*. 2006;17(11):2985–2991.
445. Remuzzi A, Puntorieri S, Battaglia C, et al. Angiotensin converting enzyme inhibition ameliorates glomerular filtration of macromolecules and water and lessens glomerular injury in the rat. *J Clin Invest*. 1990;85(2):541–549.
446. Remuzzi A, Malanchini B, Battaglia C, et al. Comparison of the effects of angiotensin-converting enzyme inhibition and angiotensin II receptor blockade on the evolution of spontaneous glomerular injury in male MWF/Ztm rats. *Exp Nephrol*. 1996;4(1):19–25.
447. Urushihara M, Kinoshita Y, Kondo S, et al. Involvement of the intrarenal renin-angiotensin system in experimental models of glomerulonephritis. *J Biomed Biotechnol*. 2012;2012:601786.
448. Remuzzi A, Gagliardini E, Sangalli F, et al. ACE inhibition reduces glomerulosclerosis and regenerates glomerular tissue in a model of progressive renal disease. *Kidney Int*. 2006;69(7):1124–1130.
449. Adamczak M, Gross ML, Amann K, et al. Reversal of glomerular lesions involves coordinated restructuring of glomerular microvasculature. *J Am Soc Nephrol*. 2004;15(12):3063–3072.
450. Wang TN, Chen X, Li R, et al. SREBP-1 mediates angiotensin II-induced TGF-beta1 upregulation and glomerular fibrosis. *J Am Soc Nephrol*. 2015;26(8):1839–1854.
451. Lavoie P, Robitaille G, Agharazii M, et al. Neutralization of transforming growth factor-beta attenuates hypertension and prevents renal injury in uremic rats. *J Hypertens*. 2005;23(10):1895–1903.
452. Sharma R, Sharma M, Reddy S, et al. Chronically increased intrarenal angiotensin II causes nephropathy in an animal model of type 2 diabetes. *Front Biosci*. 2006;11:968–976.
453. Ichikawa I, Pfeffer JM, Pfeffer MA, et al. Role of angiotensin II in the altered renal function of congestive heart failure. *Circ Res*. 1984;55(5):669–675.
454. Uttarwar L, Gao B, Ingram AJ, et al. SREBP-1 activation by glucose mediates TGF-beta upregulation in mesangial cells. *Am J Physiol Renal Physiol*. 2012;302(3):F329–F341.
455. Leh S, Vaagnes O, Margolin SB, et al. Pirfenidone and candesartan ameliorate morphological damage in mild chronic anti-GBM nephritis in rats. *Nephrol Dial Transplant*. 2005;20(1):71–82.
456. Wang J, Chen L, Chen B, et al. Chronic activation of the renin-angiotensin system induces lung fibrosis. *Sci Rep*. 2015;5:15561.
457. Navar LG, Inscho EW, Majid SA, et al. Paracrine regulation of the renal microcirculation. *Physiol Rev*. 1996;76(2):425–536.
458. Imig JD. Eicosanoid regulation of the renal vasculature. *Am J Physiol Renal Physiol*. 2000;279(6):F965–F981.
459. Hao CM, Breyer MD. Physiologic and pathophysiologic roles of lipid mediators in the kidney. *Kidney Int*. 2007;71(11):1105–1115.
460. Roman RJ. P-450 metabolites of arachidonic acid in the control of cardiovascular function. *Physiol Rev*. 2002;82(1):131–185.
461. Smith WL, Marnett LJ, DeWitt DL. Prostaglandin and thromboxane biosynthesis. *Pharmacol Ther*. 1991;49(3):153–179.
462. Breyer RM, Bagdassarian CK, Myers SA, et al. Prostanoid receptors: subtypes and signaling. *Annu Rev Pharmacol Toxicol*. 2001;41:661–690.
463. Li Y, Xia W, Zhao F, et al. Prostaglandins in the pathogenesis of kidney diseases. *Oncotarget*. 2018;9(41):26586–26602.
464. Carlstrom M, Wilcox CS, Arendshorst WJ. Renal autoregulation in health and disease. *Physiol Rev*. 2015;95(2):405–511.
465. Nielsen CB, Bech JN, Pedersen EB. Effects of prostacyclin on renal haemodynamics, renal tubular function and vasoactive hormones in healthy humans. A placebo-controlled dose-response study. *Br J Clin Pharmacol*. 1997;44(5):471–476.
466. Villa E, Garcia-Robles R, Haas J, et al. Comparative effect of PGE2 and PGI2 on renal function. *Hypertension*. 1997;30(3 Pt 2):664–666.
467. Schor N, Ichikawa I, Brenner B. Mechanisms of action of various hormones and vasoactive substances on glomerular ultrafiltration in the rat. *Kidney Int*. 1981;20:442–451.
468. Yoshioka T, Yared A, Miyazawa H, et al. In vivo influence of prostaglandin I2 on systemic and renal circulation in the rat. *Hypertension*. 1985;7(6 Pt 1):867–872.
469. Inscho EW, Carmines PK, Navar LG. Prostaglandin influences on afferent arteriolar responses to vasoconstrictor agonists. *Am J Physiol*. 1990;259(1 Pt 2):F157–F163.
470. Endlich K, Forssmann WG, Steinhausen M. Effects of urodilatin in the rat kidney: comparison with ANF and interaction with vasoactive substances. *Kidney Int*. 1995;47(6):1558–1568.
471. Munger KA, Blantz RC. Cyclooxygenase-dependent mediators of renal hemodynamic function in female rats. *Am J Physiol*. 1990;258(5 Pt 2):F1211–F1217.
472. Vio CP, Quiroz-Munoz M, Cuevas CA, et al. Prostaglandin E2 EP3 receptor regulates cyclooxygenase-2 expression in the kidney. *Am J Physiol Renal Physiol*. 2012;303(3):F449–F457.
473. Dahlen SE, Bjork J, Hedqvist P, et al. Leukotrienes promote plasma leakage and leukocyte adhesion in postcapillary venules: in vivo effects with relevance to the acute inflammatory response. *Proc Natl Acad Sci USA*. 1981;78(6):3887–3891.
474. Yared A, Albrightson-Winslow C, Griswold D, et al. Functional significance of leukotriene B4 in normal and glomerulonephritic kidneys. *J Am Soc Nephrol*. 1991;2(1):45–56.
475. Badr KF, Baylis C, Pfeffer JM, et al. Renal and systemic hemodynamic responses to intravenous infusion of leukotriene C4 in the rat. *Circ Res*. 1984;54(5):492–499.

476. Filep J, Rigter B, Frolich JC. Vascular and renal effects of leukotriene C4 in conscious rats. *Am J Physiol*. 1985;249(5 Pt 2):F739–F744.
477. Badr KF, Serhan CN, Nicolaou KC, et al. The action of lipoxin-A on glomerular microcirculatory dynamics in the rat. *Biochem Biophys Res Commun*. 1987;145(1):408–414.
478. Rubinstein M, Dvash E. Leukotrienes and kidney diseases. *Curr Opin Nephrol Hypertens*. 2018;27(1):42–48.
479. Serhan CN, Sheppard KA. Lipoxin formation during human neutrophil-platelet interactions. Evidence for the transformation of leukotriene A4 by platelet 12-lipoxygenase in vitro. *J Clin Invest*. 1990;85(3):772–780.
480. Katoh T, Takahashi K, DeBoer DK, et al. Renal hemodynamic actions of lipoxins in rats: a comparative physiological study. *Am J Physiol*. 1992;263(3 Pt 2):F436–F442.
481. Munger KA, Montero A, Fukunaga M, et al. Transfection of rat kidney with human 15-lipoxygenase suppresses inflammation and preserves function in experimental glomerulonephritis. *Proc Natl Acad Sci USA*. 1999;96(23):13375–13380.
482. Kieran NE, Doran PP, Connolly SB, et al. Modification of the transcriptomic response to renal ischemia/reperfusion injury by lipoxin analog. *Kidney Int*. 2003;64(2):480–492.
483. Chandrasekharan JA, Sharma-Walia N. Lipoxins: nature's way to resolve inflammation. *J Inflamm Res*. 2015;8:181–192.
484. Roman RJ, Fan F. 20-HETE: hypertension and beyond. *Hypertension*. 2018;72(1):12–18.
485. Dordea AC, Vandenwijngaert S, Garcia V, et al. Androgen-sensitive hypertension associated with soluble guanylate cyclase-alpha1 deficiency is mediated by 20-HETE. *Am J Physiol Heart Circ Physiol*. 2016;310(11):H1790–H1800.
486. Tunctan B, Sari AN, Kacan M, et al. NS-398 reverses hypotension in endotoxemic rats: contribution of eicosanoids, NO, and peroxynitrite. *Prostaglandins Other Lipid Mediat*. 2013;104-105:93–108.
487. Tunctan B, Korkmaz B, Cuez T, et al. Contribution of vasoactive eicosanoids and nitric oxide production to the effect of selective cyclooxygenase-2 inhibitor, NS-398, on endotoxin-induced hypotension in rats. *Basic Clin Pharmacol Toxicol*. 2010;107(5):877–882.
488. Imig JD. Epoxyeicosatrienoic acids, hypertension, and kidney injury. *Hypertension*. 2015;65(3):476–482.
489. Liu L, Liu GL, Barajas L. Distribution of nitric oxide synthase-containing ganglionic neuronal somata and postganglionic fibers in the rat kidney. *J Comp Neurol*. 1996;369(1):16–30.
490. Barajas L, Liu L, Powers K. Anatomy of the renal innervation: intrarenal aspects and ganglia of origin. *Can J Physiol Pharmacol*. 1992;70(5):735–749.
491. DiBona GF. Neural control of renal function in health and disease. *Clin Auton Res*. 1994;4:69–74.
492. Pelayo JC, Westcott JY. Renal adrenergic effector mechanisms: glomerular sites for prostaglandin interaction. *Am J Physiol*. 1988;254:F184–F190.
493. Pelayo JC, Ziegler MG, Blantz RC. Angiotensin II in adrenergic-induced alterations in glomerular hemodynamics. *Am J Physiol*. 1984;247:F799–F807.
494. Gabbai FB, Thomson SC, Peterson O, et al. Glomerular and tubular interactions between renal adrenergic activity and nitric oxide. *Am J Physiol*. 1995;268(6 Pt 2):F1004–F1008.
495. Beirwaltes WH. Sympathetic stimulation of renin is independent of direct regulation by renal nitric oxide. *Vascul Pharmacol*. 2003; 40:43–49.
496. Abdulla MH, Johns EJ. The innervation of the kidney in renal injury and inflammation: a cause and consequence of deranged cardiovascular control. *Acta Physiol (Oxf)*. 2017;220(4):404–416.
497. Ahmeda AF, Rae MG, Johns EJ. Effect of reactive oxygen species and nitric oxide in the neural control of intrarenal haemodynamics in anaesthetized normotensive rats. *Acta Physiol (Oxf)*. 2013;209(2):156–166.
498. Kon V, Yared A, Ichikawa I. Role of sympathetic nerves in mediating hypoperfusion of renal cortical microcirculation in experimental congestive heart failure and acute extracellular volume depletion. *J Clin Invest*. 1985;76:1913–1920.
499. Bhatt DL, Kandzari DE, O'Neill WW, et al. A controlled trial of renal denervation for resistant hypertension. *N Engl J Med*. 2014;370(15):1393–1401.
500. Esler MD. Fine tuning renal denervation. *J Hypertens*. 2018;36(12): 2312–2313.
501. Davies DF, Shock NW. Age changes in glomerular filtration rate, effective renal plasma flow, and tubular excretory capacity in adult males. *J Clin Invest*. 1950;29(5):496–507.
502. Glassock RJ, Rule AD. The implications of anatomical and functional changes of the aging kidney: with an emphasis on the glomeruli. *Kidney Int*. 2012;82(3):270–277.
503. Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med*. 2010;152(9):561–567.
504. Lindeman RD, Tobin J, Shock NW. Longitudinal studies on the rate of decline in renal function with age. *J Am Geriatr Soc*. 1985;33(4):278–285.
505. Maddox DA, Alavi FK, Zawada ET Jr. The kidney and aging. In: Massry SG, Glassock RJ, eds. *Textbook of Nephrology*. Baltimore: Lippincott Williams & Wilkins; 2001:1094–1105.
506. Elliott MJ, O'Hare AM, Hemmelgarn BR. Care of the Older Adult with Chronic Renal Disease. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's the Kidney*. Philadelphia: Elsevier; 2016:2586–2601.
507. Hoang K, Tan JC, Derby G, et al. Determinants of glomerular hypofiltration in aging humans. *Kidney Int*. 2003;64(4):1417–1424.
508. Fliser D, Zeier M, Nowack R, et al. Renal functional reserve in healthy elderly subjects. *J Am Soc Nephrol*. 1993;3(7):1371–1377.
509. Lindeman RD, Tobin JD, Shock NW. Association between blood pressure and the rate of decline in renal function with age. *Kidney Int*. 1984;26(6):861–868.
510. Zawada ET Jr, Alavi FK, Santella RN, et al. Influence of dietary macronutrients on glomerular senescence. In: *Current Nephrology*. Vol 20. Mosby Yearbook; 1997:1–47.
511. Baylis C, Corman B. The aging kidney: insights from experimental studies. *J Am Soc Nephrol*. 1998;9(4):699–709.
512. Weinstein JR, Anderson S. The aging kidney: physiological changes. *Adv Chronic Kidney Dis*. 2010;17(4):302–307.
513. Deen WM, Maddox DA, Robertson CR, et al. Dynamics of glomerular ultrafiltration in the rat. VII. Response to reduced renal mass. *Am J Physiol*. 1974;227(3):556–562.
514. Brenner BM. Nephron adaptation to renal injury or ablation. *Am J Physiol*. 1985;249(3 Pt 2):F324–F337.
515. Choudhury D, Palmer B, Levi M. Renal function and dysfunction in aging. In: Seldin DW, Giebisch G, eds. *Seldin and Giebisch's the Kidney, Physiology and Pathophysiology*. Philadelphia: Lippincott Williams & Wilkins; 2000:2571–2595.
516. Baylis C, Fredericks M, Wilson C, et al. Renal vasodilatory response to intravenous glycine in the aging rat kidney. *Am J Kidney Dis*. 1990;15(3):244–251.
517. Giordano M, DeFronzo RA. Acute effect of human recombinant insulin-like growth factor I on renal function in humans. *Nephron*. 1995;71:10–15.
518. Winetz JA, Golbetz HV, Spencer RJ, et al. Glomerular function in advanced human diabetic nephropathy. *Kidney Int*. 1982;21(5): 750–756.
519. Hostetter TH. Human renal response to meat meal. *Am J Physiol*. 1986;250(4 Pt 2):F613–F618.
520. Deen WM, Bridges CR, Brenner BM, et al. Heteroporous model of glomerular size selectivity: application to normal and nephrotic humans. *Am J Physiol*. 1985;249(3 Pt 2):F374–F389.
521. Chagnac A, Weinstein T, Korzets A, et al. Glomerular hemodynamics in severe obesity. *Am J Physiol Renal Physiol*. 2000;278(5):F817–F822.
522. Carmines PK, Navar LG. Disparate effects of Ca channel blockade on afferent and efferent arteriolar responses to Ang II. *Am J Physiol Renal Physiol*. 1989;256(6 Pt 2):F1015–F1020.
523. Feng MG, Navar LG. Angiotensin II-mediated constriction of afferent and efferent arterioles involves T-type Ca2+ channel activation. *Am J Nephrol*. 2004;24(6):641–648.

BOARD REVIEW QUESTIONS

1. A 21-year old woman runs a marathon and her extracellular fluid volume becomes severely decreased. Immediately following the race, she takes double the prescribed dose of ibuprofen (a non-steroidal anti-inflammatory drug) to alleviate the pain of her aching ankle and knee joints. Which one of the following adverse effects is likely to occur in the renal circulation?
   a. Increased prostaglandin levels in the kidney.
   b. Decreased efferent arteriolar resistance.
   c. Increased afferent arteriolar resistance.
   d. Increased urinary excretion of salt and water.
   e. Increased renal blood flow and GFR.

   Answer: c

   Rationale: The decreased extracellular fluid volume stimulates increased renal sympathetic activity as well as activation of the renin-angiotensin system. This causes direct renal vasoconstriction of both the afferent and efferent arterioles. This is normally counteracted by vasodilator prostanoids such as prostacyclin and PGE_2 that cause afferent vasodilation. However, NSAID drugs block cyclooxygenase, and therefore prevent afferent vasodilation, leading to impaired renal blood flow and GFR.

2. A 35-year old man is diagnosed with severe hypertension due to bilateral renal artery stenosis. An angiotensin receptor antagonist is administered in order to block the actions of angiotensin II (Ang II) on AT1 receptors. Although blood pressure returns to normal, GFR decreases markedly (by about 50%). In addition to the reduction in blood pressure, which one of the following mechanisms is most responsible for the decrease in GFR?
   a. Increased bradykinin levels in the kidney.
   b. Decreased glomerular ultrafiltration coefficient (K_f).
   c. Increased erythropoietin levels in the kidney.
   d. Decreased efferent arteriolar resistance.
   e. Decreased renin secretion by the juxtaglomerular cells.

   Answer: d

   Rationale: Bilateral renal artery stenosis reduces renal perfusion pressure beyond the stenosis leading to activation of the renin-angiotensin system and increased systemic and intrarenal Ang II levels. The increased Ang II levels increases afferent and efferent arteriolar resistance but the vasodilator influence of the tubuloglomerular feedback mechanism exerts vasodilator influence on the afferent arterioles, leaving the Ang II influence on the efferent arterioles unopposed. The reduction in blood pressure plus the efferent arteriolar vasodilation due to blockade of the AT_1 receptors cause profound decreases in glomerular pressure and the GFR.

3. Mr. H.I. Stress has not seen a doctor in several years but comes to see you because a screening at a mall showed a blood pressure of 190/100 mmHg. Records from 10 years previously show that he had a plasma creatinine of 1.0 mg/dl and normal blood pressure. The plasma creatinine concentration is now 4 mg/dL. Which one of the following changes in renal function is illustrated in this patient?
   a. His renal function has been stable.
   b. Cannot make conclusions based on data given.
   c. He has had a 75% reduction in glomerular function.
   d. He has had a 25% reduction in glomerular function.
   e. He has had a 50% reduction in GFR.

   Answer: c

   Rationale: Although not exact, there is a reciprocal relationship between plasma creatinine and GFR such that a 4-fold increase in plasma creatinine concentration reflects reduced glomerular functions to ¼ of his original value assuming that the daily production of creatinine has remained unchanged.

4. A 47-year old man is admitted to the intensive care unit because of severe alcohol withdrawal with symptoms and signs of excessive sympathetic stimulation. The effect of this on renal nerve activity would result in which one of the following:
   a. Decrease in renal blood flow but not GFR.
   b. Decrease in both renal blood flow and GFR.
   c. Decrease in renal blood flow but increase GFR.
   d. Increase in both renal blood flow and GFR due to the increase in arterial pressure.
   e. Cause no changes in renal blood flow and GFR because of renal autoregulation.

   Answer: b

   Rationale: The renal vasculature has abundant sympathetic innervation to all vascular elements. Thus, marked renal sympathetic activity will vasoconstrict both afferent and efferent arterioles causing marked decreases in renal blood flow. Because of the greater amount of preglomerular vascular smooth muscle cells, the preglomerular vasoconstriction will predominate with increased renal sympathetic activity leading to profound decreases in glomerular pressure and thus GFR.

5. A 70-year-old male is seen by his physician for a physical, his first in almost 10 years. He has gained 40 pounds and his BMI has increased from 29 to 34. His HbA1c is now 9.0, up from 6.0 previously. His serum creatinine has gone from 0.8 to 1.8 suggesting some loss of GFR due to type II diabetes and perhaps an age-related decline in renal function. His blood pressure has also increased from 120/80 to 150/90. The physician prescribes an angiotensin converting enzyme (ACE) inhibitor to inhibit the production of angiotensin II (Ang II). Which one of the following responses regarding the effects of Ang II on the kidney is most likely?
   a. The ACE inhibitor will result in a decrease in renal blood flow and GFR.
   b. The reduced intrarenal Ang II will increase sodium reabsorption in the cortical collecting duct.
   c. The ACE inhibitor will lead to reductions in intrarenal Ang II allowing maintenance of glomerular capillary hydraulic pressure and GFR.
   d. The reduced intrarenal Ang II will decrease production of proliferative factors.
   e. The ACE inhibition will increase intrarenal Ang II.

   Answer: c

   Rationale: Following the conversion of circulating and intrarenal angiotensinogen by renal renin, all components needed to synthesize and degrade Ang II are present in the immediate region of the juxtaglomerular region of the nephron thereby allowing direct local regulation of glomerular blood flow and filtration rate. Both ACE inhibitors and ARBs prevent hypertension, limit proteinuria and ameliorate sclerosis in several experimental models of glomerulonephritis.